

Fourier Analysis Methods for PDE's

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Introduction

Since the 80's, Fourier analysis methods have known a growing interest in the study of linear and nonlinear PDE's. In particular, techniques based on *Littlewood-Paley decomposition* and *paradifferential calculus* have proved to be very efficient. Littlewood-Paley decomposition has been introduced more than fifty years ago in harmonic analysis but its systematic use in the PDE's framework is rather recent. Paradifferential calculus, as for it, has been introduced in 1981 by J.-M. Bony for the study of the propagation of microlocal singularities in nonlinear hyperbolic PDE's (see [4]).

In the present notes, we aim at giving a survey of those techniques and a few examples of how they may be used to solve PDE's. We focus on two linear models: the heat equation and the transport equation. For each of them, an example of related nonlinear problem is given. Although most of the results we present here belong to the mathematical folklore, we want to point out that Fourier analysis methods are very efficient to tackle most of well-posedness problems for evolutionary PDE's in the whole space or in the torus.

The notes are structured as follows. The first chapter deals with Fourier analysis. We introduce Littlewood-Paley decomposition and show how it may be used to characterize functional spaces. We also give a (non so) short insight into the theory of *homogeneous spaces* which turn out to be well adapted to the study of many PDE's and are omitted in most of textbooks on functional analysis. Finally, we introduce some smatterings of paradifferential calculus and prove estimates for the product of two temperate distributions, when it makes sense.

In the second chapter, we focus on the heat equation. We state a priori estimates in Besov spaces with optimal gain of derivatives. As an application, we prove global well-posedness in Besov spaces with critical regularity for the incompressible Navier-Stokes equations with small data.

The third chapter is devoted to the study of transport equations associated to Lipschitz vectorfields. We state a priori estimates in Besov spaces then apply our results to the study of a shallow water equation.

In the last chapter, we give an example of coupling between heat equation and transport equation, namely the compressible barotropic Navier-Stokes system. Well-posedness in Besov spaces with critical regularity is stated.

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Chapter 1

An introduction to Fourier analysis

1.1 Notations and definitions

- $\mathcal{S}$ stands for the *Schwartz space* of smooth functions over $\mathbb{R}^N$ whose derivatives of all order decay at infinity. The space $\mathcal{S}$ is endowed with the topology generated by the following family of semi-norms:

$$\|u\|_{M,\mathcal{S}} := \sup_{\substack{x \in \mathbb{R}^N \\ |\alpha| \leq M}} (1 + |x|)^M |\partial^\alpha u(x)| \quad \text{for all } u \in \mathcal{S} \text{ and } M \in \mathbb{N}.$$

- The set $\mathcal{S}'$ of *temperate distributions* is the dual set of $\mathcal{S}$ for the usual pairing.
- For any $u \in \mathcal{S}$, the *Fourier transform* of u denoted by $\hat{u}$ or $\mathcal{F}u$ is defined by

$$\forall \xi \in \mathbb{R}^N, \hat{u}(\xi) = \mathcal{F}u(\xi) := \int_{\mathbb{R}^N} e^{-i\xi \cdot x} u(x) dx.$$

The Fourier transform maps $\mathcal{S}$ into and onto itself, and the inverse Fourier transform is given by the formula $\mathcal{F}^{-1} = (2\pi)^{-N} \bar{\mathcal{F}}$.

- The Fourier transform is extended by duality to the whole $\mathcal{S}'$ by setting

$$\langle \hat{u}, \varphi \rangle := \langle u, \hat{\varphi} \rangle_{\mathcal{S}', \mathcal{S}}$$

whenever $u \in \mathcal{S}'$ and $\varphi \in \mathcal{S}$.

- Derivatives: for all multi-index $\alpha \in \mathbb{N}^N$, we have

$$\mathcal{F}(\partial_x^\alpha u) = (i\xi)^\alpha \mathcal{F}u \quad \text{and} \quad \mathcal{F}(x^\alpha u) = (-i)^{|\alpha|} \partial_\xi^\alpha \mathcal{F}u.$$

- Algebraic properties: for $(u, v) \in \mathcal{S} \times \mathcal{S}'$, we have $u * v \in \mathcal{S}'$ and

$$\mathcal{F}(u * v) = \mathcal{F}u \mathcal{F}v.$$

The above formula also holds true for couples of distributions with compact supports.

- Multipliers: if A is a smooth function with polynomial growth at infinity, and $u \in \mathcal{S}'(\mathbb{R}^N)$ then we set $A(D)u := \mathcal{F}^{-1}(A \mathcal{F}u)$.
- The open (resp. closed) *ball* with radius R centered at $x_0 \in \mathbb{R}^N$ is denoted by $\mathcal{B}(0, R)$ (resp. $\bar{\mathcal{B}}(0, R)$).
- The *shell* $\{\xi \in \mathbb{R}^N \mid R_1 \leq |\xi| \leq R_2\}$ is denoted by $\mathcal{C}(0, R_1, R_2)$.
- The notation $A \lesssim B$ means that $A \leq CB$ for some “irrelevant” constant C (which may change from line to line but whose meaning is clear from the context). Likewise, $A \approx B$ means that $A \lesssim B$ and $B \lesssim A$.

1.2 The Littlewood-Paley decomposition

1.2.1 Bernstein inequalities

As recalled in the previous section, in Fourier variables differentiating with respect to x_j amounts to multiplying by the function $\xi \mapsto i\xi_j$.

As far as one is concerned with estimates in Lebesgue spaces and whenever the distribution we consider is well localized in Fourier variables, Bernstein lemma states that differentiating amounts to multiplying by an appropriate constant

Lemma (Bernstein). *Let k be in $\mathbb{N}$. Let (R_1, R_2) satisfy $0 < R_1 < R_2$. There exists a constant C depending only on r_1, r_2, N , such that for all $1 \leq a \leq b \leq \infty$ and $u \in L^a$, we have*

$$(1.1) \quad \text{Supp } \widehat{u} \subset \mathcal{B}(0, R_1\lambda) \Rightarrow \sup_{|\alpha|=k} \|\partial^\alpha u\|_{L^b} \leq C^{k+1} \lambda^{k+N(\frac{1}{a}-\frac{1}{b})} \|u\|_{L^a},$$

$$(1.2) \quad \text{Supp } \widehat{u} \subset \mathcal{C}(0, R_1\lambda, R_2\lambda) \Rightarrow C^{-k-1} \lambda^k \|u\|_{L^a} \leq \sup_{|\alpha|=k} \|\partial^\alpha u\|_{L^a} \leq C^{k+1} \lambda^k \|u\|_{L^a}.$$

Proof: Arguing by rescaling, one can assume with no loss of generality that $\lambda = 1$.

Now, fix a smooth function ϕ compactly supported and such that $\phi \equiv 1$ in a neighborhood of the ball $\mathcal{B}(0, R_1)$. We notice that $\widehat{u} = \phi \widehat{u}$. Hence, denoting $g := \mathcal{F}^{-1}\phi$, we get for all multi-index α ,

$$\partial^\alpha u(x) = \int \partial^\alpha g(y) u(x-y) dy.$$

Taking advantage of Young inequality, we thus get

$$\|\partial^\alpha u\|_{L^b} \leq \|\partial^\alpha g\|_{L^c} \|u\|_{L^a} \quad \text{with} \quad \frac{1}{c} = 1 + \frac{1}{b} - \frac{1}{a}.$$

Because

$$\|\partial^\alpha g\|_{L^c} \leq \|\partial^\alpha g\|_{L^\infty} + \|\partial^\alpha g\|_{L^1} \leq C^{k+1},$$

the proof of the first inequality is complete.

For proving (1.2), we first notice that the inequality on the right is a particular case of (1.1). Next, we introduce a smooth function $\widetilde{\varphi}$ with compact support in $\mathbb{R}^N \setminus \{0\}$ and such that $\varphi \equiv 1$ in a neighborhood of the shell $\mathcal{C}(0, R_1, R_2)$.

As $\sum_{|\alpha|=k} (i\xi)^\alpha (-i\xi)^\alpha = |\xi|^{2k}$, we have

$$(1.3) \quad u = \sum_{|\alpha|=k} g_\alpha \star \partial^\alpha u \quad \text{with} \quad \widehat{g}_\alpha(\xi) := (i\xi)^\alpha |\xi|^{-2k} \widetilde{\varphi}(\xi).$$

Making use of Young inequality, one can now conclude to the left inequality in (1.2). $\blacksquare$

Remark. In other words, if $\widehat{u}$ is supported in a ball of radius λ then differentiating once is not worse than multiplying by λ . If $\widehat{u}$ is supported in the shell $\{\xi \in \mathbb{R}^N \mid R_1\lambda \leq |\xi| \leq R_2\lambda\}$ then, up to an irrelevant constant, differentiating once amounts to multiplying by λ .

In most applications, the functions we deal with are not spectrally supported in a shell or in a ball. Hence, if one wants to take advantage of the nice properties exhibited in Bernstein lemma, one first has to split the function into pieces which are spectrally supported in a shell or in a ball. This may be done by introducing a dyadic partition of unity in Fourier variables. There are two main ways to proceed. Either the decomposition is made indistinctly over the whole space $\mathbb{R}^N$ (and we say that the decomposition is *homogeneous*), or the low frequencies are treated separately (and the decomposition is said to be *nonhomogeneous*).

Both decompositions have advantages and drawbacks. The nonhomogeneous one is more suitable for characterizing the usual functional spaces whereas the properties of invariance by dilation of the homogeneous decomposition may be more adapted for studying certain PDE's or stating optimal functional inequalities having some scaling invariance.

1.2.2 The nonhomogeneous Littlewood-Paley decomposition

Let $\alpha > 1$ and (φ, χ) be a couple of smooth functions valued in $[0, 1]$, such that φ is supported in the shell $\{\xi \in \mathbb{R}^N \mid \alpha^{-1} \leq |\xi| \leq 2\alpha\}$, χ is supported in the ball $\{\xi \in \mathbb{R}^N \mid |\xi| \leq \alpha\}$ and

$$\forall \xi \in \mathbb{R}^N, \chi(\xi) + \sum_{q \in \mathbb{N}} \varphi(2^{-q}\xi) = 1.$$

For $u \in \mathcal{S}'$, one can define *nonhomogeneous dyadic blocks* as follows. Let

$$\begin{aligned} \Delta_q u &:= 0 \quad \text{if } q \leq -2, \\ \Delta_{-1} u &:= \chi(D)u = \tilde{h} \star u \quad \text{with } \tilde{h} := \mathcal{F}^{-1}\chi, \\ \Delta_q u &:= \varphi(2^{-q}D)u = 2^{qN} \int h(2^q y)u(x-y)dy \quad \text{with } h = \mathcal{F}^{-1}\varphi, \quad \text{if } q \geq 0. \end{aligned}$$

One can prove that we have

$$u = \sum_{q \in \mathbb{Z}} \Delta_q u \quad \text{in } \mathcal{S}'(\mathbb{R}^N)$$

for all temperate distribution u (see exercise 1.2). The right-hand side is called *nonhomogeneous Littlewood-Paley decomposition* of u .

It is also convenient to introduce the following low frequency cut-off:

$$S_q u := \sum_{p \leq q-1} \Delta_p u.$$

Of course, $S_0 u = \Delta_{-1} u$. Because $\varphi(\xi) = \chi(\xi/2) - \chi(\xi)$ for all $\xi \in \mathbb{R}^N$, one can prove that, more generally, we have

$$S_q u = \chi(2^{-q}D)u = \int \tilde{h}(2^q y)u(x-y)dy \quad \text{for all } q \in \mathbb{N}.$$

The Littlewood-Paley decomposition is “almost” orthogonal in L^2 . Assuming for instance that $\alpha = 4/3$, we have the following result¹:

Proposition 1.2.1. *For any $u \in \mathcal{S}'(\mathbb{R}^N)$ and $v \in \mathcal{S}'(\mathbb{R}^N)$, the following properties hold:*

$$\begin{aligned} \Delta_p \Delta_q u &\equiv 0 \quad \text{if } |p - q| \geq 2, \\ \Delta_q (S_{p-1} u \Delta_p v) &\equiv 0 \quad \text{if } |p - q| \geq 5. \end{aligned}$$

Remark. At this point, one can wonder why it is so important to choose *smooth* cut-off functions χ and φ for defining a Littlewood-Paley decomposition. Obviously, setting $\Delta'_{-1} u := 1_{|\xi| \leq 1}(D)u$ and $\Delta'_q u := 1_{2^q \leq |\xi| \leq 2^{q+1}}(D)u$ would define a dyadic spectral decomposition which, in addition, would be orthogonal in L^2 .

In most applications however, it is crucial that we have

$$\|\Delta_q u\|_{L^p} \leq C \|u\|_{L^p}$$

for some constant C independent of q .

Alas, unless $p = 2$, the above inequality fails to be true with $\Delta'_q u$ instead of $\Delta_q u$ (see exercise 1.5).

¹Of course, similar properties may be proved for any $\alpha > 1$.

1.2.3 About the periodic case

Throughout, $a_1, \dots, a_N$ denote N positive reals. We denote by $\mathbb{T}_a^N$ the periodic box with period $2\pi a_i$ in the i -th direction, and $Q_a^N := (0, 2\pi a_1) \times \dots \times (0, 2\pi a_N)$. We also introduce $\tilde{\mathbb{Z}}_a^N := \mathbb{Z}/a_1 \times \dots \times \mathbb{Z}/a_N$ the dual lattice associated to $\mathbb{T}_a^N$.

We claim that the analysis of the previous section for temperate distributions defined on the whole space $\mathbb{R}^N$ may be carried out to $\mathcal{S}'(\mathbb{T}_a^N)$ with very few changes.

Indeed, decompose $u \in \mathcal{S}'(\mathbb{T}_a^N)$ into Fourier series:

$$u(x) = \sum_{\beta \in \tilde{\mathbb{Z}}_a^N} \hat{u}_\beta e^{i\beta \cdot x} \quad \text{with} \quad \hat{u}_\beta := \frac{1}{|\mathbb{T}_a^N|} \int_{\mathbb{T}_a^N} e^{-i\beta \cdot y} u(y) dy.$$

Denoting

$$h_q(x) = \sum_{\beta \in \tilde{\mathbb{Z}}_a^N} \varphi(2^{-q}\beta) e^{i\beta \cdot x},$$

one can now define the *periodic dyadic blocks* as follows:

$$\Delta_q^{per} u(x) := \sum_{\beta \in \tilde{\mathbb{Z}}_a^N} \varphi(2^{-q}\beta) \hat{u}_\beta e^{i\beta \cdot x} = \frac{1}{|\mathbb{T}_a^N|} \int_{\mathbb{T}_a^N} h_q(y) u(x-y) dy \quad \text{for all } q \in \mathbb{Z}$$

and the low frequency cut-off:

$$S_q^{per} u(x) := \hat{u}_0 + \sum_{p \leq q-1} \Delta_p^{per} u(x) = \sum_{\beta \in \tilde{\mathbb{Z}}_a^N} \chi(2^{-q}\beta) \hat{u}_\beta e^{i\beta \cdot x}.$$

It is obvious that $\Delta_p^{per} u = 0$ for negative enough p (depending on a) and that

$$u = \hat{u}_0 + \sum_q \Delta_q^{per} u \quad \text{in } \mathcal{S}'(\mathbb{T}_a^N).$$

Now, to any temperate distribution u over $\mathbb{R}^N$, one can associate the periodic distribution u^{per} defined by

$$u^{per}(x) := \sum_{\alpha \in \mathbb{Z}_{2\pi a}^N} u(x + \alpha) \quad \text{where} \quad \mathbb{Z}_{2\pi a}^N := 2\pi a_1 \mathbb{Z} \times \dots \times 2\pi a_N \mathbb{Z}.$$

For u^{per} , both periodic and nonhomogeneous Littlewood-Paley decompositions are available. It turns out that periodic blocks and nonhomogeneous blocks coincide in the following sense:

Proposition 1.2.2. *For all temperate distribution u over $\mathbb{R}^N$, one has*

$$\forall q \in \mathbb{Z}, \quad (\varphi(2^{-q}D)u)^{per} = \Delta_q^{per} u^{per}.$$

Proof: This is actually an easy consequence of the following **Poisson formula** for $\theta \in \mathcal{S}'(\mathbb{R}^N)$:

$$\frac{1}{(2\pi)^N} \sum_{\beta \in \mathbb{Z}^N} \hat{\theta}(\beta) = \sum_{\alpha \in \mathbb{Z}^N} \theta(2\pi\alpha),$$

which, after a change of variables yields

$$(1.4) \quad \forall u \in \mathcal{S}'(\mathbb{R}^N), \forall x \in \mathbb{R}^N, \quad \frac{1}{|\mathbb{T}_a^N|} \sum_{\beta \in \tilde{\mathbb{Z}}_a^N} e^{i\beta \cdot x} \hat{u}(\beta) = \sum_{\alpha \in \mathbb{Z}_{2\pi a}^N} u(x + \alpha).$$

Now, on one hand, for all $x \in \mathbb{R}^N$, we have by definition of $\varphi(2^{-q}D)$,

$$\begin{aligned} (\varphi(2^{-q}D)u)^{per}(x) &= 2^{qN} \sum_{\alpha \in \mathbb{Z}_{2\pi a}^N} \int_{\mathbb{R}^N} h(2^q y) u(x + \alpha - y) dy, \\ &= 2^{qN} \int_{\mathbb{R}^N} h(2^q y) u^{per}(x - y) dz. \end{aligned}$$

On the other hand, by taking advantage of (1.4), we get

$$\frac{1}{|\mathbb{T}_a^N|} \sum_{\beta \in \tilde{\mathbb{Z}}_a^N} e^{i\beta \cdot (x-y)} \varphi(2^{-q}\beta) = 2^{qN} \sum_{\alpha \in \mathbb{Z}_{2\pi a}^N} h(2^q(x + \alpha - y)),$$

whence

$$\begin{aligned} \Delta_q^{per} u^{per}(x) &= \frac{1}{|\mathbb{T}_a^N|} \sum_{\beta \in \tilde{\mathbb{Z}}_a^N} \int_{\mathbb{T}_a^N} e^{i\beta(x-y)} \varphi(2^{-q}\beta) u^{per}(y) dy, \\ &= 2^{qN} \sum_{\alpha \in \mathbb{Z}_{2\pi a}^N} \int_{Q_a^N} h(2^q(x + \alpha - y)) u^{per}(y) dy, \\ &= 2^{qN} \int_{\mathbb{R}^N} h(2^q(x - z)) u^{per}(z) dz. \end{aligned}$$

The proof is complete. ■

In what follows, we shall focus on distributions defined on $\mathbb{R}^N$. We want to point out that all the properties described in the next sections remain true in the periodic setting provided the dyadic blocks have been defined as indicated above.

1.3 Littlewood-Paley decomposition and functional spaces

Many functional spaces over $\mathbb{R}^N$ (or $\mathbb{T}_a^N$) such as Hölder or Sobolev spaces may be characterized in terms of Littlewood-Paley decomposition.

1.3.1 Sobolev and Hölder spaces

Let us first recall how nonhomogeneous Sobolev spaces H^s are defined.

Definition. Let $s \in \mathbb{R}$. A temperate distribution u belongs to $H^s(\mathbb{R}^N)$ if $\widehat{u} \in L_{loc}^2(\mathbb{R}^N)$ and

$$\|u\|_{H^s} := \left(\int_{\mathbb{R}^N} (1 + |\xi|^2)^s |\widehat{u}(\xi)|^2 d\xi \right)^{\frac{1}{2}} < \infty.$$

It is classical that H^s endowed with the norm $\|\cdot\|_{H^s}$ is a Banach space². Now, from the definition of (χ, φ) , one easily infers that

$$(1.5) \quad \forall \xi \in \mathbb{R}^N, \quad \frac{1}{3} \leq \chi^2(\xi) + \sum_{q \in \mathbb{N}} \varphi^2(2^{-q}\xi) \leq 1,$$

whence the following result:

Proposition 1.3.1. *There exists a constant C such that for all $s \in \mathbb{R}$, we have*

$$\frac{1}{C^{|s|+1}} \|u\|_{H^s}^2 \leq \sum_q 2^{2qs} \|\Delta_q u\|_{L^2}^2 \leq C^{|s|+1} \|u\|_{H^s}^2.$$

Hence Littlewood-Paley decomposition supplies a simple characterization of Sobolev spaces: u belongs to H^s if and only if the sequence $(2^{qs} \|\Delta_q u\|_{L^2})_{q \in \mathbb{Z}}$ belongs to $\ell^2(\mathbb{Z})$.

²Actually it is even an Hilbert space.

Let us now focus on Hölder spaces.

Definition. Let $r \in (0, 1)$. We denote by C^r the set of bounded functions $u : \mathbb{R}^N \rightarrow \mathbb{R}$ such that there exists $C \geq 0$ with

$$(1.6) \quad \forall (x, y) \in \mathbb{R}^N \times \mathbb{R}^N, |u(x) - u(y)| \leq C|x - y|^r.$$

More generally, if $r > 0$ is not an integer, we denote by C^r the set of $[r]$ times³ differentiable functions u such that $\partial^\alpha u \in C^{r-[\alpha]}$ for all $|\alpha| \leq r$.

One can easily prove that the set C^r endowed with the norm

$$\|u\|_{C^r} := \sum_{|\alpha| \leq [r]} \left(\|\partial^\alpha u\|_{L^\infty} + \sup_{x \neq y} \frac{|\partial^\alpha u(x) - \partial^\alpha u(y)|}{|x - y|^{r-[\alpha]}} \right)$$

is a Banach space.

In the case $r \in \mathbb{R}^+ \setminus \mathbb{N}$, the Littlewood-Paley decomposition supplies a very simple characterization of C^r :

Proposition 1.3.2. *There exists a constant C such that for all $r \in \mathbb{R}^+ \setminus \mathbb{N}$ and $u \in C^r$ we have*

$$\sup_q 2^{qr} \|\Delta_q u\|_{L^\infty} \leq \frac{C^{r+1}}{[r]!} \|u\|_{C^r}.$$

Conversely, if the sequence $(2^{qr} \|\Delta_q u\|_{L^\infty})_{q \geq -1}$ is bounded then

$$\|u\|_{C^r} \leq C^{r+1} \left(\frac{1}{r - [r]} + \frac{1}{[r] + 1 - r} \right) \sup_q 2^{qr} \|\Delta_q u\|_{L^\infty}.$$

Proof: Let us just sketch the proof (for more details see [10]).

Let us first notice that, owing to $\int x^\alpha h(x) dx = 0$ for all multi-index α , one can write

$$(1.7) \quad \forall q \in \mathbb{N}, \Delta_q u(x) = 2^{qN} \int h(2^q(x - y)) \left(u(y) - \sum_{k=1}^{[r]} \frac{1}{k!} D^k u(x) (y - x)^{(k)} \right) dy.$$

Applying the $[r]$ -th order Taylor formula for bounding the right-hand side of (1.7) and using the fact that $\|\Delta_{-1} u\|_{L^\infty} \leq C \|u\|_{L^\infty}$ yields the first inequality.

For proving the converse inequality, we notice that since for all multi-index we have $\partial^\alpha u = \sum_q \partial^\alpha \Delta_q u$, Bernstein lemma insures that

$$\|\partial^\alpha u\|_{L^\infty} \leq \frac{C^{1+r}}{r - [r]} \sup_q 2^{qr} \|\Delta_q u\|_{L^\infty}$$

whenever $|\alpha| \leq [r]$.

Next, for all multi-index α of size $[r]$, all $(x, y) \in \mathbb{R}^N \times \mathbb{R}^N$ such that $|x - y| \leq 1$ and all $Q \in \mathbb{N}$, we have

$$|\partial^\alpha u(x) - \partial^\alpha u(y)| \leq \sum_{q=-1}^{Q-1} |\partial^\alpha \Delta_q u(x) - \partial^\alpha \Delta_q u(y)| + \sum_{q=Q}^{+\infty} |\partial^\alpha \Delta_q u(x) - \partial^\alpha \Delta_q u(y)|.$$

³From now on, the notation $[r]$ stands for the integer part of r .

Making use of Bernstein lemma, we end up with

$$|\partial^\alpha u(x) - \partial^\alpha u(y)| \leq C^{r+1} \left(\sup_{q \geq 0} 2^{qr} \|\Delta_q u\|_{L^\infty} \right) \left(\sum_{q=-1}^Q 2^{-q(r-[r]-1)} |x-y| + \sum_{q \geq Q+1} 2^{-q(r-[r])} \right).$$

Choosing $Q = \lceil -\log_2 |x-y| \rceil + 1$ yields the wanted inequality. $\blacksquare$

Remark. The above characterization of Hölder spaces is false if r is an integer (see exercise 1.9).

1.3.2 Besov spaces

From now on, we make the convention that for all non-negative sequence $(a_q)_{q \in \mathbb{Z}}$, the notation $\left(\sum_q a_q^r \right)^{\frac{1}{r}}$ stands for $\sup_q a_q$ in the case $r = \infty$.

The characterizations of Sobolev and Hölder spaces given in the previous part naturally lead to the following definition:

Definition. Let $1 \leq p, r \leq \infty$ and $s \in \mathbb{R}$. For $u \in \mathcal{S}'(\mathbb{R}^N)$, we set

$$\|u\|_{B_{p,r}^s} := \left(\sum_q (2^{qs} \|\Delta_q u\|_{L^p})^r \right)^{\frac{1}{r}}.$$

The Besov space $B_{p,r}^s$ is the set of temperate distributions u such that $\|u\|_{B_{p,r}^s} < \infty$.

Before going further into the study of Besov spaces, let us state two important lemmas. The first one reads:

Lemma 1.3.3. *Let $(u_q)_{q \in \mathbb{N}}$ be a sequence of bounded functions such that the Fourier transform of u_q is supported in dyadic shells. Let us assume that, for some $M \geq 0$, we have*

$$\|u_q\|_{L^\infty} \leq C 2^{qM}.$$

Then the series $\sum_q u_q$ is convergent in $\mathcal{S}'$.

Proof: After rescaling, relation (1.3) rewrites

$$(1.8) \quad u_q = 2^{-qk} \sum_{|\alpha|=k} 2^{qN} g_\alpha(2^q \cdot) \star \partial^\alpha u_q.$$

Therefore, for any test function ϕ in $\mathcal{S}$, we have

$$(1.9) \quad \langle u_q, \phi \rangle = (-1)^k 2^{-qk} \sum_{|\alpha|=k} \langle u_q, 2^{qN} \check{g}_\alpha(2^q \cdot) \star \partial^\alpha \phi \rangle \quad \text{with} \quad \check{g}(z) = g(-z).$$

Hence

$$|\langle u_q, \phi \rangle| \leq C 2^{-qk} \sum_{|\alpha|=k} 2^{qM} \|\partial^\alpha \phi\|_{L^1}.$$

Let us choose $k > M$. Then $\sum_q \langle u_q, \phi \rangle$ is a convergent series, the sum of which is less than $C \|\phi\|_{P,S}$ for some large enough integer P . Thus the formula

$$\langle u, \phi \rangle := \lim_{q \rightarrow \infty} \sum_{q' \leq q} \langle \Delta_{q'} u, \phi \rangle$$

defines a temperate distribution. $\blacksquare$

The second important lemma is the following one:

Lemma 1.3.4. *Let $s \in \mathbb{R}$ and $1 \leq p, r \leq \infty$. Let $(u_q)_{q \geq -1}$ be a sequence of functions such that*

$$\left(\sum_{q \geq -1} (2^{qs} \|u_q\|_{L^p})^r \right)^{\frac{1}{r}} < \infty.$$

(i) *If $\text{Supp } \hat{u}_{-1} \subset \mathcal{B}(0, R_2)$ and $\text{Supp } \hat{u}_q \subset \mathcal{C}(0, 2^q R_1, 2^q R_2)$ for some $0 < R_1 < R_2$ then $u := \sum_{q \geq -1} u_q$ belongs to $B_{p,r}^s$ and there exists a universal constant C such that*

$$\|u\|_{B_{p,r}^s} \leq C^{1+|s|} \left(\sum_{q \geq -1} (2^{qs} \|u_q\|_{L^p})^r \right)^{\frac{1}{r}}.$$

(ii) *If s is positive and $\text{Supp } \hat{u}_q \subset \mathcal{B}(0, 2^q R)$ for some $R > 0$ then $u := \sum_{q \geq -1} u_q$ belongs to $B_{p,r}^s$ and there exists a universal constant C such that*

$$\|u\|_{B_{p,r}^s} \leq \frac{C^{1+s}}{s} \left(\sum_{q \geq -1} (2^{qs} \|u_q\|_{L^p})^r \right)^{\frac{1}{r}}.$$

Proof: Under the hypothesis of the first assertion and according to Bernstein lemma, we have

$\|u_q\|_{L^\infty} \leq C 2^{q(\frac{N}{p}-s)}$. Lemma 1.3.3 thus implies that $\sum_q u_q$ is a convergent series in $\mathcal{S}'$. Next, we notice that there exists an integer N_0 so that

$$|q' - q| \geq N_0 \implies \Delta_{q'} u_q = 0.$$

Therefore, with the convention that $u_q = 0$ if $q \leq -2$, we can write that

$$\begin{aligned} \|\Delta_{q'} u\|_{L^p} &= \left\| \sum_{|q-q'| < N_0} \Delta_{q'} u_q \right\|_{L^p} \\ &\leq C \sum_{|q-q'| < N_0} \|u_q\|_{L^p}. \end{aligned}$$

So, we obtain that

$$\begin{aligned} 2^{q's} \|\Delta_{q'} u\|_{L^p} &\leq C \sum_{|q-q'| \leq N_0} 2^{q's} \|u_q\|_{L^p} \\ &\leq C^{1+|s|} \sum_{|q-q'| \leq N_0} 2^{qs} \|u_q\|_{L^p}, \end{aligned}$$

and we deduce from convolution inequalities that

$$\|u\|_{B_{p,r}^s} \leq C^{1+|s|} \left(\sum_{q \in \mathbb{N}} 2^{rqs} \|u_q\|_{L^p}^r \right)^{\frac{1}{r}},$$

which is exactly the first result.

For proving the second result, we first notice that for any q , we have

$$\|u_q\|_{L^p} \leq C 2^{-qs}.$$

As s is positive, this implies that $\sum_q u_q$ is a convergent series in L^p . Next, we notice that there exists some $N_1 \in \mathbb{N}$ such that

$$q' \geq q + N_1 \implies \Delta_{q'} u_q = 0.$$

Now, we write that

$$\begin{aligned} \|\Delta_{q'} u\|_{L^p} &= \left\| \sum_{q > q' - N_1} \Delta_{q'} u_q \right\|_{L^p}, \\ &\leq C \sum_{q > q' - N_1} \|u_q\|_{L^p}. \end{aligned}$$

So, we get that

$$2^{q's} \|\Delta_{q'} u\|_{L^p} \leq C \sum_{q \geq q' - N_1} 2^{(q' - q)s} 2^{qs} \|u_q\|_{L^p}.$$

In other words, we have

$$2^{q's} \|\Delta_{q'} u\|_{L^p} \leq C(c_k) \star (d_\ell) \quad \text{with} \quad c_k = \mathbf{1}_{[-N_1, +\infty[}(k) 2^{-ks} \quad \text{and} \quad d_\ell = 2^{\ell s} \|u_\ell\|_{L^p}.$$

Applying convolution inequalities for series completes the proof of the second property. ■

Corollary. *The definition of the space $B_{p,r}^s$ does not depend on the choice of the couple (χ, φ) defining the Littlewood-Paley decomposition.*

Remark. The Besov spaces have been obtained by taking first the L^p norm on each dyadic block, then taking a weighted ℓ^r norm. It turns out that taking first the weighted ℓ^r norm and next the L^p norm over $\mathbb{R}^N$ is also relevant. This yields new Banach spaces called *Triebel-Lizorkin* spaces and denoted by $F_{p,r}^s$. Using such spaces may be appropriate for studying certain problems. In particular, if $1 < p < \infty$, the space $F_{p,2}^0$ coincides with the Lebesgue space L^p and, more generally, $F_{p,2}^s$ coincide with the potential space H_p^s of temperate distributions u such that $(I - \Delta)^{\frac{s}{2}}$ belongs to L^p .

The reader is referred to [36] or [38] for a more complete study of Triebel-Lizorkin spaces.

1.3.3 A few properties of Besov spaces

In the following proposition, we give a first list of important properties of Besov spaces.

Proposition 1.3.5. *Let $s \in \mathbb{R}$ and $1 \leq p, r \leq \infty$.*

- (i) *Topological properties:* $B_{p,r}^s$ is a Banach space which is continuously embedded in $\mathcal{S}'$.
- (ii) *Density:* the space $\mathcal{C}_c^\infty$ of smooth functions with compact support is dense in $B_{p,r}^s$ if and only if p and r are finite.
- (iii) *Duality:* for all $s \in \mathbb{R}$ and $1 \leq p, r < \infty$, the space $B_{p',r'}^{-s}$ is the dual space of $B_{p,r}^s$.
If $1 \leq p < \infty$, the completion $\mathcal{B}_{p,\infty}^s$ of $\mathcal{C}_c^\infty$ for the norm $\|\cdot\|_{B_{p,\infty}^s}$ is the predual of $B_{p',1}^{-s}$.
- (iv) *Separability:* If $1 \leq p, r < \infty$ then the space $B_{p,r}^s$ is separable. The same holds for $\mathcal{B}_{p,\infty}^s$.
- (v) *Embeddings:* we have

- (a) $B_{p,r}^s \hookrightarrow B_{p,\tilde{r}}^{\tilde{s}}$ whenever $\tilde{s} < s$ or $\tilde{s} = s$ and $\tilde{r} \geq r$,
- (b) $B_{p,r}^s \hookrightarrow B_{p,\tilde{r}}^{s - N(\frac{1}{p} - \frac{1}{\tilde{p}})}$ whenever $\tilde{p} \geq p$,

(c) we have $B_{\infty,1}^0 \hookrightarrow \mathcal{C} \cap L^\infty$. If $p < \infty$ then the space $B_{p,1}^{\frac{N}{p}}$ is continuously embedded in the space $\mathcal{C}_0$ of continuous bounded functions which decay at infinity.

(vi) Fatou property: if $(u_n)_{n \in \mathbb{N}}$ is a bounded sequence of $B_{p,r}^s$ which tends to u in $\mathcal{S}'$ then $u \in B_{p,r}^s$ and

$$\|u\|_{B_{p,r}^s} \leq \liminf \|u_n\|_{B_{p,r}^s}.$$

(vii) Complex interpolation: if $u \in B_{p,r}^s \cap B_{p,r}^{\tilde{s}}$ and $\theta \in [0, 1]$ then $u \in B_{p,r}^{\theta s + (1-\theta)\tilde{s}}$ and

$$\|u\|_{B_{p,r}^{\theta s + (1-\theta)\tilde{s}}} \leq \|u\|_{B_{p,r}^s}^\theta \|u\|_{B_{p,r}^{\tilde{s}}}^{1-\theta}.$$

(viii) Real interpolation: if $u \in B_{p,\infty}^s \cap B_{p,\infty}^{\tilde{s}}$ and $s < \tilde{s}$ then u belongs to $B_{p,1}^{\theta s + (1-\theta)\tilde{s}}$ for all $\theta \in (0, 1)$ and there exists a universal constant C such that

$$\|u\|_{B_{p,1}^{\theta s + (1-\theta)\tilde{s}}} \leq \frac{C}{\theta(1-\theta)(\tilde{s}-s)} \|u\|_{B_{p,\infty}^s}^\theta \|u\|_{B_{p,\infty}^{\tilde{s}}}^{1-\theta}.$$

Proof: Let us first prove that $B_{p,r}^s$ is continuously embedded in $\mathcal{S}'$. By definition, $B_{p,r}^s$ is a subspace of $\mathcal{S}'$. Thus it suffices to prove that there exist a constant C and an integer M such that for any $\phi \in \mathcal{S}$ we have

$$(1.10) \quad |\langle u, \phi \rangle| \leq C \|u\|_{B_{p,r}^s} \|\phi\|_{M,\mathcal{S}}.$$

Taking advantage of (1.9) with $\Delta_q u$ instead of u_q , we see that for all $k \in \mathbb{N}$, there exists an integer M_k and a constant C_k such that

$$|\langle \Delta_q u, \phi \rangle| \leq C_k 2^{-q} (2^{q(1-k)} \|\Delta_q u\|_{L^\infty}) \|\phi\|_{M_k,\mathcal{S}}.$$

According to Bernstein lemma, we have $\|\Delta_q u\|_{L^\infty} \leq C 2^{q\frac{N}{p}} \|\Delta_q u\|_{L^p}$. Hence, if k has been chosen so large as to satisfy $s - N/p \geq 1 - k$, we have

$$|\langle \Delta_q u, \phi \rangle| \leq C_k 2^{-q} \|u\|_{B_{p,r}^s} \|\phi\|_{M_k,\mathcal{S}}$$

which, after summation on q , yields inequality (1.10).

Next, consider $(u^{(n)})_{n \in \mathbb{N}}$ a Cauchy sequence in $B_{p,r}^s$. Inequality (1.10) implies that for any test function ϕ in $\mathcal{S}$, sequence $(\langle u^{(n)}, \phi \rangle)_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$. Thus the formula

$$\langle u, \phi \rangle := \lim_{n \rightarrow \infty} \langle u^{(n)}, \phi \rangle$$

defines a temperate distribution. By definition of the norm of $B_{p,r}^s$, sequence $(\Delta_q u^{(n)})_{n \in \mathbb{N}}$ is a Cauchy sequence in L^p for any q . Thus an element u_q of L^p exists such that $(\Delta_q u^{(n)})_{n \in \mathbb{N}}$ converges to u_q in L^p . As the sequence $(u^{(n)})_{n \in \mathbb{N}}$ converges to u in $\mathcal{S}'$ we actually have $\Delta_q u = u_q$.

Fix a $Q \in \mathbb{N}$ and a positive ε . Since for all $q \geq -1$, $\Delta_q u^{(n)}$ tends to $\Delta_q u$ in L^p , we have for all $n \in \mathbb{N}$,

$$\left(\sum_{q \leq Q} (2^{qs} \|\Delta_q(u^{(n)} - u)\|_{L^p})^r \right)^{\frac{1}{r}} = \lim_{m \rightarrow \infty} \left(\sum_{q \leq Q} (2^{qs} \|\Delta_q(u^{(n)} - u^{(m)})\|_{L^p})^r \right)^{\frac{1}{r}}.$$

Because the argument of the limit in the right-hand side is bounded by $\|u^{(n)} - u^{(m)}\|_{B_{p,r}^s}$ and $(u^{(n)})_{n \in \mathbb{N}}$ is a Cauchy sequence in $B_{p,r}^s$, one can now conclude that there exists a n_0 (independent of Q) such that for all $n \geq n_0$, we have

$$\left(\sum_{q \leq Q} (2^{qs} \|\Delta_q(u^{(n)} - u)\|_{L^p})^r \right)^{\frac{1}{r}} \leq \varepsilon.$$

Letting Q go to infinity insures that $(u^{(n)})_{n \in \mathbb{N}}$ tends to u in $B_{p,r}^s$.

Let us tackle the proof of (ii). Consider first the case where r is finite. Let u be in $B_{p,r}^s$ and $\varepsilon > 0$. Since r is finite, there exists an integer q such that

$$(1.11) \quad \|u - S_q u\|_{B_{p,r}^s} < \frac{\varepsilon}{2}.$$

Now let ϕ be in $\mathcal{C}_c^\infty$. For any $q' \in \mathbb{N}$, Bernstein lemma insures that we have,

$$\begin{aligned} 2^{q's} \|\Delta_{q'}(\phi S_q u - S_q u)\|_{L^p} &\leq 2^{-q'} 2^{q'([s]+2)} \|\Delta_{q'}(\phi S_q u - S_q u)\|_{L^p} \\ &\leq C_s 2^{-q'} \sup_{|\alpha|=[s]+2} \|\partial^\alpha(\phi S_q u - S_q u)\|_{L^p}. \end{aligned}$$

From the above inequality, we get that

$$(1.12) \quad \|\phi S_q u - S_q u\|_{B_{p,r}^s} \leq C_s \left(\|(1-\phi)S_q u\|_{L^p} + \sup_{|\alpha|=[s]+2} \|\partial^\alpha((1-\phi)S_q u)\|_{L^p} \right).$$

Let us consider a sequence $(\phi_n)_{n \in \mathbb{N}}$ such that all the derivatives of ϕ_n of order less than or equal to $[s] + 2$ are uniformly bounded with respect to n and such that $\phi_n \equiv 1$ in a neighborhood of the ball $\mathcal{B}(0, n)$. If p is finite, combining Leibniz formula and Lebesgue theorem, we discover that

$$\lim_{n \rightarrow \infty} \|(1-\phi_n)S_q u\|_{L^p} + \sup_{|\alpha|=[s]+2} \|\partial^\alpha((1-\phi_n)S_q u)\|_{L^p} = 0.$$

Thus a function ϕ in $\mathcal{C}_c^\infty$ exists such that

$$C_s \left(\|(1-\phi)S_q u\|_{L^p} + \sup_{|\alpha|=[s]+2} \|\partial^\alpha((1-\phi)S_q u)\|_{L^p} \right) < \frac{\varepsilon}{2}.$$

Combining (1.11) and (1.12), we end up with

$$\|\phi S_q u - u\|_{B_{p,r}^s} < \varepsilon.$$

As $S_q u$ is a smooth function, this completes the proof in the case $p, r < \infty$.

Now, it is obvious that the set $\mathcal{C}_b^\infty$ of smooth functions with bounded derivatives at all orders is embedded in any space $B_{\infty,r}^s$. Therefore $\mathcal{C}_c^\infty$ cannot be a dense subset of $B_{\infty,r}^s$. Finally, the closure of $\mathcal{C}_c^\infty$ for the Besov norm $B_{p,\infty}^s$ is the space of temperate distributions u such that

$$\lim_{q \rightarrow \infty} 2^{qs} \|\Delta_q u\|_{L^p} = 0,$$

which is a strict subspace of $B_{p,\infty}^s$. This completes the proof of (ii).

In order to prove the properties of duality, we use the fact that the map

$$u \mapsto (\Delta_q u)_{q \geq -1}$$

is a (bi)continuous isomorphism between $B_{p,r}^s$ and⁴ $\ell_s^r(L^p)$.

Assume that $1 \leq p < \infty$. On one hand, if in addition $1 \leq r < \infty$, it is obvious that $\ell_{-s}^{r'}(L^{p'})$ is the dual of $\ell_s^r(L^p)$. On the other hand, $\ell_{-s}^1(L^{p'})$ is the dual space of the set of L^p valued sequences $(z_q)_{q \geq -1}$ such that

$$\lim_{q \rightarrow +\infty} 2^{qs} \|z_q\|_{L^p} = 0.$$

Now, one can prove that $\mathcal{B}_{p,\infty}^s$ is the space of temperate distributions u such that

$$\lim_{q \rightarrow +\infty} 2^{qs} \|\Delta_q u\|_{L^p} = 0,$$

whence the desired result.

The proof of (iv) also relies on the use of the map $u \mapsto (\Delta_q u)_{q \geq -1}$. The details are left to the reader.

Let us now prove (v). Considering that $\ell^r(\mathbb{Z}) \subset \ell^{\tilde{r}}(\mathbb{Z})$ for $r \leq \tilde{r}$, the first embedding is straightforward. In order to prove the second embedding, we apply Bernstein lemma and get

$$\|S_0 u\|_{L^{\tilde{p}}} \leq C \|S_0 u\|_{L^p} \quad \text{and} \quad \|\Delta_q u\|_{L^{\tilde{p}}} \leq C 2^{qN(\frac{1}{p} - \frac{1}{\tilde{p}})} \|\Delta_q u\|_{L^p} \quad \text{if } q \in \mathbb{N},$$

whence the result.

For proving that $B_{p,1}^{\frac{N}{p}} \hookrightarrow \mathcal{C}_b$ we use again Bernstein lemma and get that

$$\|\Delta_q u\|_{L^\infty} \leq 2^{q\frac{N}{p}} \|\Delta_q u\|_{L^p}.$$

This insures that the series $\sum \Delta_q u$ of continuous bounded functions converges uniformly on $\mathbb{R}^N$. Hence u is a bounded continuous function. Besides, it is obvious that the embedding is continuous. If p is finite, one can use in addition that $\mathcal{C}_c^\infty$ is dense in $B_{p,1}^{\frac{N}{p}}$ and conclude that u decays at infinity.

Let us now focus on the proof of (vi). Let $(u_n)_{n \in \mathbb{N}}$ be a bounded sequence of $B_{p,r}^s$ which tends to some u in $\mathcal{S}'$. This insures that for all $q \in \mathbb{Z}$, sequence $(\Delta_q u_n)_{n \in \mathbb{N}}$ tends to $\Delta_q u$ in $\mathcal{S}'$. Since $(\Delta_q u_n)_{n \in \mathbb{N}}$ is a bounded sequence in $L^p \cap \mathcal{C}_b^\infty$, one can conclude that $\Delta_q u$ belongs to $L^p \cap \mathcal{C}_b^\infty$ and that

$$\|\Delta_q u\|_{L^p} \leq \liminf \|\Delta_q u_n\|_{L^p}.$$

Now, for all $Q \in \mathbb{N}$, we have

$$\begin{aligned} \left(\sum_{q=-1}^Q (2^{qs} \|\Delta_q u\|_{L^p})^r \right)^{\frac{1}{r}} &\leq \left(\sum_{q=-1}^Q (2^{qs} \liminf \|\Delta_q u_n\|_{L^p})^r \right)^{\frac{1}{r}}, \\ &\leq \liminf \left(\sum_{q=-1}^Q (2^{qs} \|\Delta_q u_n\|_{L^p})^r \right)^{\frac{1}{r}}, \\ &\leq \liminf \|u_n\|_{B_{p,r}^s}. \end{aligned}$$

Letting Q go to infinity completes the proof of (vi).

⁴Here ℓ_s^r stands for the set of sequences $(z_q)_{q \geq -1}$ such that $\|(2^{qs} z_q)\|_{\ell^r} < \infty$.

Property (vii) is a straightforward consequence of Hölder inequality.

For proving (viii), we write

$$\|u\|_{B_{p,1}^{\theta s+(1-\theta)\tilde{s}}} = \sum_{q \leq Q} 2^{q(\theta s+(1-\theta)\tilde{s})} \|\Delta_q u\|_{L^p} + \sum_{q > Q} 2^{q(\theta s+(1-\theta)\tilde{s})} \|\Delta_q u\|_{L^p}$$

for some Q to be chosen hereafter.

Now, by definition of the Besov norms, we have

$$\begin{aligned} 2^{q(\theta s+(1-\theta)\tilde{s})} \|\Delta_q u\|_{L^p} &\leq 2^{q(1-\theta)(\tilde{s}-s)} \|u\|_{B_{p,\infty}^s}, \\ 2^{q(\theta s+(1-\theta)\tilde{s})} \|\Delta_q u\|_{L^p} &\leq 2^{-q\theta(\tilde{s}-s)} \|u\|_{B_{p,\infty}^{\tilde{s}}}. \end{aligned}$$

Thus we infer that

$$\begin{aligned} \|u\|_{B_{p,1}^{\theta s+(1-\theta)\tilde{s}}} &\leq \|u\|_{B_{p,\infty}^s} \sum_{q \leq Q} 2^{q(1-\theta)(\tilde{s}-s)} + \|u\|_{B_{p,\infty}^{\tilde{s}}} \sum_{q > Q} 2^{-q\theta(\tilde{s}-s)} \\ &\leq \|u\|_{B_{p,\infty}^s} \frac{2^{(Q+1)(1-\theta)(\tilde{s}-s)}}{2^{(1-\theta)(\tilde{s}-s)} - 1} + \|u\|_{B_{p,\infty}^{\tilde{s}}} \frac{2^{-Q\theta(\tilde{s}-s)}}{1 - 2^{-\theta(\tilde{s}-s)}}. \end{aligned}$$

In order to complete the proof of (viii), it suffices to choose Q such that

$$\frac{\|u\|_{B_{p,\infty}^{\tilde{s}}}}{\|u\|_{B_{p,\infty}^s}} \leq 2^{Q(\tilde{s}-s)} < 2 \frac{\|u\|_{B_{p,\infty}^{\tilde{s}}}}{\|u\|_{B_{p,\infty}^s}}.$$

■

One can wonder how far is $B_{p,\infty}^s$ from $B_{p,1}^s$, or in other words, what happens if one lets θ tends to 1 in proposition 1.3.5.(viii). Of course, one already knows that

$$B_{p,1}^s \hookrightarrow B_{p,r}^s \hookrightarrow B_{p,\infty}^s.$$

A more precise answer is given by the following logarithmic interpolation result:

Proposition 1.3.6. *There exists a constant C such that for all $s \in \mathbb{R}$, $\varepsilon > 0$ and $1 \leq p \leq \infty$, we have*

$$\|u\|_{B_{p,1}^s} \leq C \frac{1+\varepsilon}{\varepsilon} \|u\|_{B_{p,\infty}^s} \left(1 + \log \frac{\|u\|_{B_{p,\infty}^{s+\varepsilon}}}{\|u\|_{B_{p,\infty}^s}} \right).$$

Proof: Let Q be a positive integer to be fixed hereafter. We have

$$\|u\|_{B_{p,1}^s} = \sum_{q < Q} 2^{qs} \|\Delta_q u\|_{L^p} + \sum_{q \geq Q} 2^{qs} \|\Delta_q u\|_{L^p},$$

whence,

$$\|u\|_{B_{p,1}^s} \leq (Q+1) \|u\|_{B_{p,\infty}^s} + \frac{2^{-Q\varepsilon}}{1 - 2^{-\varepsilon}} \|u\|_{B_{p,\infty}^{s+\varepsilon}}.$$

Choosing for Q the closest positive integer to $\frac{1}{\varepsilon} \log_2 \frac{\|u\|_{B_{p,\infty}^{s+\varepsilon}}}{\|u\|_{B_{p,\infty}^s}}$ yields the result. ■

We now want to study how multipliers operate on Besov spaces. Before stating our result, we need to define the multipliers we are going to consider.

Definition. A smooth function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is said to be a S^m -multiplier if for all multi-index α , there exists a constant C_α such that

$$\forall \xi \in \mathbb{R}^N, |\partial^\alpha f(\xi)| \leq C_\alpha (1 + |\xi|)^{m-|\alpha|}.$$

Proposition 1.3.7. Let $m \in \mathbb{R}$ and f be a S^m -multiplier. Then for all $s \in \mathbb{R}$ and $1 \leq p, r \leq \infty$ the operator $f(D)$ is continuous from $B_{p,r}^s$ to $B_{p,r}^{s-m}$.

Proof: Let $\tilde{\varphi}$ be a smooth function supported in a shell and such that $\tilde{\varphi} \equiv 1$ on $\text{Supp } \varphi$. It is clear that we have

$$\Delta_q f(D)u = \tilde{\varphi}(2^{-q}D) f(D) \Delta_q u \quad \text{for all } q \in \mathbb{N}.$$

Hence, by virtue of convolution inequalities, we have

$$\|\Delta_q f(D)u\|_{L^p} \leq \|k_q\|_{L^1} \|\Delta_q u\|_{L^p}$$

with

$$k_q(x) := \frac{1}{(2\pi)^N} \int e^{ix \cdot \xi} f(\xi) \tilde{\varphi}(2^{-q}\xi) d\xi.$$

By performing an easy change of variables, we notice that $\|k_q\|_{L^1} = \|\ell_q\|_{L^1}$ with

$$\ell_q(y) := \frac{1}{(2\pi)^N} \int e^{iy \cdot \eta} \tilde{\varphi}(\eta) f(2^q \eta) d\eta.$$

Now, for all $M \in \mathbb{N}$, we have

$$\begin{aligned} (1 + |y|^2)^M \ell_q(y) &= \int (1 - \Delta_\eta)^M (e^{iy \cdot \eta} \tilde{\varphi}(\eta) f(2^q \eta)) d\eta, \\ &= \int e^{iy \cdot \eta} (1 - \Delta_\eta)^M (\tilde{\varphi}(\eta) f(2^q \eta)) d\eta, \\ &= \sum_{|\alpha|+|\beta| \leq 2M} c_{\alpha,\beta} 2^{q|\beta|} \int e^{iy \cdot \eta} \partial^\alpha \tilde{\varphi}(\eta) \partial^\beta f(2^q \eta) d\eta, \end{aligned}$$

for some integers $c_{\alpha,\beta}$ (whose exact value does not matter).

Hence, using the fact that integration may be restricted to $\text{Supp } \tilde{\varphi}$ and that $|\partial^\beta f(2^q \eta)| \leq C_\beta (1 + 2^q |\eta|)^{m-|\beta|}$, we get

$$(1 + |y|^2)^M |\ell_q(y)| \leq C_M 2^{qm}.$$

Choosing $M > N/2$, we thus conclude that

$$\|k_q\|_{L^1} = \|\ell_q\|_{L^1} \leq C 2^{qm}$$

whence

$$\forall q \in \mathbb{N}, 2^{q(s-m)} \|f(D) \Delta_q u\|_{L^p} \leq C 2^{qs} \|\Delta_q u\|_{L^p}.$$

Stating a similar inequality for $q = -1$ is left to the reader. This yields the proposition. ■

Proposition 1.3.8. A constant C exists which satisfies the following properties. Let $s < 0$, $(p, r) \in [1, \infty]^2$ and $u \in \mathcal{S}'$. Then u belongs to $B_{p,r}^s$ if and only if $(2^{qs} \|S_q u\|_{L^p})_{q \in \mathbb{N}} \in \ell^r(\mathbb{N})$. Moreover, we have

$$\frac{1}{2} \|u\|_{B_{p,r}^s} \leq \left(\sum_q (2^{qs} \|S_q u\|_{L^p})^r \right)^{\frac{1}{r}} \leq C \left(1 + \frac{1}{|s|} \right) \|u\|_{B_{p,r}^s}.$$

Proof: On one hand, we have

$$\begin{aligned} 2^{qs} \|\Delta_q u\|_{L^p} &\leq 2^{qs} (\|S_{q+1} u\|_{L^p} + \|S_q u\|_{L^p}) \\ &\leq 2^{-s} 2^{(q+1)s} \|S_{q+1} u\|_{L^p} + 2^{qs} \|S_q u\|_{L^p}, \end{aligned}$$

which proves the inequality on the left. On the other hand, we can write that

$$\begin{aligned} 2^{qs} \|S_q u\|_{L^p} &\leq 2^{qs} \sum_{q' \leq q-1} \|\Delta_{q'} u\|_{L^p} \\ &\leq \sum_{q' \leq q-1} 2^{(q-q')s} 2^{q's} \|\Delta_{q'} u\|_{L^p}. \end{aligned}$$

As s is negative, we get the result. ■

1.4 Paradifferential calculus

When dealing with nonlinear problems, one often has to study the functional properties of products of two temperate distributions u and v .

Characterizing distributions such that the product uv makes sense is an intricate question which is intimately related to the notion of *wavefront* (see e.g [1] for an elementary introduction).

In this section, we shall see that very simple arguments based on the use of Littlewood-Paley decomposition yield sufficient conditions for uv to be defined, and continuity results for the map $(u, v) \mapsto uv$.

1.4.1 Definitions

For u and v two temperate distributions, we have the following formal decomposition:

$$uv = \sum_{p,q} \Delta_p u \Delta_q v.$$

The fundamental idea of paradifferential calculus is to split uv into three parts, both of them being always defined. The first part, denoted by $T_u v$ and called *paraproduct* of v by u corresponds to terms $\Delta_p u \Delta_q v$ where p is small in comparison with q . The second term, $T_v u$ is the symmetric counterpart of $T_u v$ (i.e. we keep only the terms corresponding to large frequencies of u multiplied by small frequencies of v). The third and last term (the *remainder term*) corresponds to the dyadic blocks of u and v with comparable frequencies.

This very simple splitting device goes back to the pioneering work by J.-M. Bony in [4]. In what follows, we shall adopt the following definition for paraproduct and remainder:

Definition. Let u and v be two temperate distributions. We denote

$$T_u v = \sum_{p \leq q-2} \Delta_p u \Delta_q v = \sum_q S_{q-1} u \Delta_q v$$

and

$$R(u, v) = \sum_q \Delta_q u \widetilde{\Delta}_q v \quad \text{with} \quad \widetilde{\Delta}_q := \Delta_{q-1} + \Delta_q + \Delta_{q+1}.$$

At least formally, we have the following *Bony decomposition*:

$$(1.13) \quad uv = T_u v + T_v u + R(u, v).$$

Of course, it may happen that the product uv is not defined. However, the reader may retain the following principles:

- The paraproduct of two temperate distributions u and v is always defined. This is due to the fact that the general term of the paraproduct is spectrally localized in dyadic shells. Besides, the regularity of $T_u v$ is mainly determined by the regularity of v . In particular, $T_u v$ cannot be more regular than v .
- The remainder may not be defined. Roughly, it is defined as soon as u and v belong to functional spaces whose sum of regularity index is positive. In that case, the regularity exponent of $R(u, v)$ is the sum of the regularity exponents of u and v .

1.4.2 Results of continuity for the paraproduct and the remainder

The bilinear paraproduct and remainder operators benefit from continuity properties in most usual functional spaces. In the present chapter, we focus on Besov spaces. The reader is referred to [36] and [37] for a more complete study.

As regards the paraproduct, we have the following results:

Proposition 1.4.1. *Let $1 \leq p, r \leq \infty$ and $s \in \mathbb{R}$.*

- (i) *The paraproduct T is a bilinear continuous operator from $L^\infty \times B_{p,r}^s$ to $B_{p,r}^s$ and there exists a constant C such that*

$$\|T\|_{\mathcal{L}(L^\infty \times B_{p,r}^s; B_{p,r}^s)} \leq C^{|s|+1}.$$

- (ii) *If $\sigma > 0$ and $1 \leq r, r_1, r_2 \leq \infty$ are such that $1/r = 1/r_1 + 1/r_2$ then T is bilinear continuous from $B_{\infty, r_1}^{-\sigma} \times B_{p, r_2}^s$ to $B_{p, r}^{s-\sigma}$ and there exists a constant C such that*

$$\|T\|_{\mathcal{L}(B_{\infty, r_1}^{-\sigma} \times B_{p, r_2}^s; B_{p, r}^{s-\sigma})} \leq \frac{C^{|s-\sigma|+1}}{\sigma}.$$

Proof: According to proposition 1.2.1, the sequence $(\mathcal{F}(S_{q-1}u\Delta_q v))_{q \in \mathbb{Z}}$ is supported in dyadic shells. Hence, in view of proposition 1.3.5, it suffices to prove that

$$\left(\sum_q (2^{qs} \|S_{q-1}u\Delta_q v\|_{L^p})^r \right)^{\frac{1}{r}} \lesssim \|u\|_{L^\infty} \|v\|_{B_{p,r}^s}.$$

Since $\|S_{q-1}u\|_{L^\infty} \leq \|\tilde{h}\|_{L^1} \|u\|_{L^\infty}$, this is actually straightforward. This yields the first result.

For proving the second result, we use that, because $\sigma > 0$, we have

$$2^{q(s-\sigma)} \|S_{q-1}u\Delta_q v\|_{L^p} \leq (2^{qs} \|\Delta_q v\|_{L^p}) \sum_{q' \leq q-2} 2^{\sigma(q'-q)} 2^{-q'\sigma} \|\Delta_{q'} u\|_{L^\infty}.$$

Therefore, combining Hölder and convolution inequalities for series, we get

$$\left(\sum_{q \in \mathbb{Z}} (2^{q(s-\sigma)} \|S_{q-1}u\Delta_q v\|_{L^p})^r \right)^{\frac{1}{r}} \leq \left(\sum_{k \geq 2} 2^{-k\sigma} \right) \|u\|_{B_{\infty, r_1}^{-\sigma}} \|v\|_{B_{p, r_2}^s},$$

whence the desired inequality. ■

Remark. By combining the above results of continuity for the paraproduct with the embeddings stated in proposition 1.3.5, one can get a score of other results of continuity. For instance, by using that for all $\varepsilon > 0$, we have $B_{p_1, r_1}^{\frac{N}{p_1} - \varepsilon} \hookrightarrow B_{\infty, r_1}^{-\varepsilon}$, we discover that T is continuous from $B_{p_1, r_1}^{\frac{N}{p_1} - \varepsilon} \times B_{p, r_2}^s$ to $B_{p, r}^{s-\varepsilon}$ for all $1 \leq p, p_1, r, r_1, r_2 \leq \infty$ such that $1/r = 1/r_1 + 1/r_2$.

Proposition 1.4.2. *Let $(s_1, s_2) \in \mathbb{R}^2$ and $1 \leq p, p_1, p_2, r, r_1, r_2 \leq \infty$. Assume that*

$$(1.14) \quad \frac{1}{p} \leq \frac{1}{p_1} + \frac{1}{p_2} \leq 1, \quad \frac{1}{r} \leq \frac{1}{r_1} + \frac{1}{r_2} \quad \text{and} \quad s_1 + s_2 > 0.$$

Then the remainder R maps $B_{p_1, r_1}^{s_1} \times B_{p_2, r_2}^{s_2}$ in $B_{p, r}^{s_1 + s_2 + N(\frac{1}{p} - \frac{1}{p_1} - \frac{1}{p_2})}$ and there exists a constant C such that

$$\|R(u, v)\|_{B_{p, r}^{s_1 + s_2 + N(\frac{1}{p} - \frac{1}{p_1} - \frac{1}{p_2})}} \leq \frac{C^{|s_1 + s_2| + 1}}{s_1 + s_2} \|u\|_{B_{p_1, r_1}^{s_1}} \|v\|_{B_{p_2, r_2}^{s_2}}.$$

Proof: It suffices to treat the case where $1/r = 1/r_1 + 1/r_2$ and $1/p = 1/p_1 + 1/p_2$. The general case then follows from the embeddings of proposition 1.3.5.(v).

Now, by definition of the remainder operator, we have

$$R(u, v) = \sum_q \Delta_q u \widetilde{\Delta}_q v.$$

On one hand, by definition of Δ_q and $\widetilde{\Delta}_q$, the support of $\mathcal{F}(\Delta_q u \widetilde{\Delta}_q v)$ is included in $\mathcal{B}(0, 3 \cdot 2^{q+3})$. On the other hand, by virtue of Hölder inequality for functions, we have

$$2^{q(s_1 + s_2)} \|\Delta_q u \widetilde{\Delta}_q v\|_{L^p} \leq (2^{qs_1} \|\Delta_q u\|_{L^{p_1}}) (2^{qs_2} \|\widetilde{\Delta}_q v\|_{L^{p_2}}).$$

Applying Hölder inequality for series, we thus have

$$\|2^{q(s_1 + s_2)} \|\Delta_q u \widetilde{\Delta}_q v\|_{L^p}\|_{\ell^r} \leq C \|u\|_{B_{p_1, r_1}^{s_1}} \|v\|_{B_{p_2, r_2}^{s_2}}.$$

As $s_1 + s_2 > 0$ has been assumed, lemma 1.3.4 yields the desired result. $\blacksquare$

Remark. By combining proposition 1.3.5.(v) with the above proposition, one can get other results of continuity for the remainder. Moreover, condition (1.14) may be somewhat relaxed (see exercise 1.10).

1.4.3 Results of continuity for the product

We do not aim at giving an exhaustive list of the mapping properties of $(u, v) \mapsto uv$ in Besov spaces. As a matter of fact, memorizing such a list would be quite useless: it is actually far wiser to appeal to results of continuity for the paraproduct and remainder, and to Bony's decomposition.

For example, by combining propositions 1.4.1 and 1.4.2, and the continuous embeddings stated in proposition 1.3.5, one gets the following important results:

Proposition 1.4.3. *Let $s > 0$ and $1 \leq p, r \leq \infty$. Then $B_{p, r}^s \cap L^\infty$ is an algebra and we have*

$$\|uv\|_{B_{p, r}^s} \lesssim \|u\|_{L^\infty} \|v\|_{B_{p, r}^s} + \|v\|_{L^\infty} \|u\|_{B_{p, r}^s}.$$

Proof: According to propositions 1.4.2 and 1.4.1, we have

$$\begin{aligned} \|R(u, v)\|_{B_{p, r}^s} &\lesssim \|u\|_{B_{p, r}^s} \|v\|_{B_{\infty, \infty}^0}, \\ \|T_u v\|_{B_{p, r}^s} &\lesssim \|u\|_{L^\infty} \|v\|_{B_{p, r}^s}, \\ \|T_v u\|_{B_{p, r}^s} &\lesssim \|v\|_{L^\infty} \|u\|_{B_{p, r}^s}. \end{aligned}$$

One can easily check that $L^\infty \hookrightarrow B_{\infty, \infty}^0$, hence applying Bony's decomposition yields the proposition. $\blacksquare$

Proposition 1.4.4. *Let $1 \leq p_1, p_2, p_3, p_4, r \leq \infty$ and $(s_1, s_2, s_3, s_4) \in \mathbb{R}^4$ be such that*

$$s_1 + s_2 - \frac{N}{p_1} = s_3 + s_4 - \frac{N}{p_4}, \quad \frac{1}{p_1} + \frac{1}{p_2} \leq 1, \quad s_1 + s_2 > 0 \quad \text{and} \quad s_1, s_3 < \frac{N}{p_1}.$$

Then the product is continuous from $(B_{p_1, \infty}^{s_1} \cap B_{p_2, r}^{s_2})^2$ to $B_{p_2, r}^{s_1 + s_2 - \frac{N}{p_1}}$ and we have

$$\|uv\|_{B_{p_2, r}^{s_1 + s_2 - \frac{N}{p_1}}} \lesssim \|u\|_{B_{p_1, \infty}^{s_1}} \|v\|_{B_{p_2, r}^{s_2}} + \|v\|_{B_{p_3, \infty}^{s_3}} \|u\|_{B_{p_4, r}^{s_4}}.$$

Proof: By virtue of propositions 1.4.1 and 1.4.2 and embeddings, we have

$$\begin{aligned} \|T_u v\|_{B_{p_2, r}^{s_1 + s_2 - \frac{N}{p_1}}} &\lesssim \|u\|_{B_{\infty, \infty}^{s_1 - \frac{N}{p_1}}} \|v\|_{B_{p_2, r}^{s_2}} \lesssim \|u\|_{B_{p_1, \infty}^{s_1}} \|v\|_{B_{p_2, r}^{s_2}}, \\ \|R(u, v)\|_{B_{p_2, r}^{s_1 + s_2 - \frac{N}{p_1}}} &\lesssim \|u\|_{B_{p_1, \infty}^{s_1}} \|v\|_{B_{p_2, r}^{s_2}}, \\ \|T_v u\|_{B_{p_4, r}^{s_3 + s_4 - \frac{N}{p_3}}} &\lesssim \|u\|_{B_{\infty, \infty}^{s_3 - \frac{N}{p_3}}} \|v\|_{B_{p_4, r}^{s_4}} \lesssim \|u\|_{B_{p_3, \infty}^{s_3}} \|v\|_{B_{p_4, r}^{s_4}}. \end{aligned}$$

Taking advantage of Bony's decomposition completes the proof. $\blacksquare$

Proposition 1.4.5. *For all $\phi \in \mathcal{S}$, $1 \leq p, r \leq \infty$ and $s \in \mathbb{R}$, the map $M_\phi : u \rightarrow \phi u$ is continuous in $B_{p, r}^s$.*

Proof: The proof relies on Bony decomposition. Indeed, according to proposition 1.4.1 and Besov embeddings, we have

$$\begin{aligned} \|T_\phi u\|_{B_{p, r}^s} &\lesssim \|\phi\|_{L^\infty} \|u\|_{B_{p, r}^s}, \\ \|T_u \phi\|_{B_{p, r}^s} &\lesssim \begin{cases} \|\phi\|_{B_{p, r}^s} \|u\|_{L^\infty} & \lesssim \|\phi\|_{B_{p, r}^s} \|u\|_{B_{p, r}^s} & \text{if } s > \frac{N}{p}, \\ \|\phi\|_{B_{p, r}^{\frac{N}{p}+1}} \|u\|_{B_{\infty, \infty}^{s - \frac{N}{p} - 1}} & \lesssim \|\phi\|_{B_{p, r}^{\frac{N}{p}+1}} \|u\|_{B_{p, r}^s} & \text{if } s \leq \frac{N}{p}, \end{cases} \end{aligned}$$

and, by virtue of proposition 1.4.2,

$$\|R(u, \phi)\|_{B_{p, r}^s} \lesssim \begin{cases} \|\phi\|_{L^\infty} \|u\|_{B_{p, r}^s} & \text{if } s > 0, \\ \|\phi\|_{B_{\infty, \infty}^{1-s}} \|u\|_{B_{p, r}^s} & \text{if } s \leq 0, \end{cases}$$

which completes the proof. $\blacksquare$

1.4.4 A result of compactness in Besov spaces

One can now state a result of compactness for Besov spaces which will prove to be very useful for solving nonlinear PDE's.

Proposition 1.4.6. *Let $1 \leq p, r \leq \infty$, $s \in \mathbb{R}$ and $\varepsilon > 0$. For all $\phi \in \mathcal{C}_c^\infty$, the map $u \mapsto \phi u$ is compact from $B_{p, r}^{s+\varepsilon}$ to $B_{p, r}^s$.*

As a preliminary step, we need to state the following result :

Lemma 1.4.7. *Let $a_1, \dots, a_N$ be positive and $\delta \in (0, (\min_i a_i)/4)$. There exists a constant C such that for all $1 \leq p, r \leq \infty$ and $s \in \mathbb{R}$, and $u \in B_{p, r}^s(\mathbb{T}_a^N)$ supported in a cube of size $2\pi a_1 - 2\delta \times \dots \times 2\pi a_N - 2\delta$, we have*

$$C^{-1} \|u\|_{B_{p, r}^s(\mathbb{R}^N)} \leq \|u^{per}\|_{B_{p, r}^s(\mathbb{T}_a^N)} \leq C \|u\|_{B_{p, r}^s(\mathbb{R}^N)} \quad \text{with} \quad u^{per} := \sum_{\alpha \in \mathbb{Z}_{2\pi a}^N} u(\cdot + \alpha).$$

Proof: With no loss of generality, one can assume that

$$\text{Supp } u \subset Q_{a,\delta}^N := [\delta, 2\pi a_1 - \delta] \times \cdots \times [\delta, 2\pi a_N - \delta].$$

Let θ be a smooth function supported in $Q_{a,\delta/2}^N$ and equals to one on a neighborhood of $Q_{a,\delta}^N$. Denoting $h_q := 2^{qN} h(2^q \cdot)$, we have for all $q \in \mathbb{N}$, $M \in \mathbb{N}$ and $x \in \mathbb{R}^N$,

$$\begin{aligned} \Delta_q u(x) &= \langle u, \theta h_q(x - \cdot) \rangle, \\ &= \langle (\text{Id} - \Delta)^{-M} u, (\text{Id} - \Delta)^M (\theta h_q(x - \cdot)) \rangle. \end{aligned}$$

By virtue of Besov embeddings, one can choose M so large as to satisfy

$$B_{p,r}^s(\mathbb{R}^N) \hookrightarrow \{v \in \mathcal{S}'(\mathbb{R}^N) \mid (\text{Id} - \Delta)^{-M} v \in L^\infty(\mathbb{R}^N)\}.$$

By taking advantage of Leibniz formula, we thus get for some large enough C ,

$$|\Delta_q u(x)| \leq C \|u\|_{B_{p,r}^s(\mathbb{R}^N)} \sum_{|\alpha|+|\beta| \leq 2M} \int_{\mathbb{R}^N} |\partial_\alpha \theta(y)| |\partial_\beta h_q(x-y)| dy.$$

If one assumes that $x \notin Q_a^N$ then one has⁵ $|x-y| \geq |x-\pi a| - \pi|a| + \delta/2$ whenever y belongs to $\text{Supp } \theta$. Therefore one has for all $K \in \mathbb{N}$,

$$|\Delta_q u(x)| \leq \frac{C}{(|x-\pi a| - \pi|a| + \frac{\delta}{2})^K} \|u\|_{B_{p,r}^s(\mathbb{R}^N)} \sum_{|\alpha|+|\beta| \leq 2M} \int_{\mathbb{R}^N} |x-y|^K |\partial_\alpha \theta(y)| |\partial_\beta h_q(x-y)| dy$$

and it is easy to conclude that there exists a constant C_K such that

$$(1.15) \quad |\Delta_q u(x)| \leq C_K (|x-\pi a| - \pi|a| + \frac{\delta}{2})^{-K} 2^{(2M-K)q} \|u\|_{B_{p,r}^s(\mathbb{R}^N)} \quad \text{for all } x \in \mathbb{R}^N \setminus Q_a^N.$$

By choosing $K = 2M + 1$ with M large enough, one can now easily conclude that

$$(1.16) \quad \|\Delta_q u\|_{L^p(\mathbb{R}^N \setminus Q_a^N)} \leq C 2^{-q} \|u\|_{B_{p,r}^s(\mathbb{R}^N)}$$

for some constant C depending only on δ and on a .

Next, we notice that, by virtue of proposition 1.2.2, we have

$$\forall x \in \mathbb{R}^N, \Delta_q^{per} u^{per}(x) - \Delta_q u(x) = \sum_{\alpha \in \mathbb{Z}_{2\pi a}^N \setminus \{0\}} \Delta_q u(x + \alpha).$$

Therefore, taking M large enough and using (1.15) with $K = 2M + 1$, one gathers that for all $x \in Q_a^N$, we have

$$|\Delta_q^{per} u^{per}(x) - \Delta_q u(x)| \leq C 2^{-q} \|u\|_{B_{p,r}^s(\mathbb{R}^N)},$$

whence

$$(1.17) \quad \|\Delta_q^{per} u^{per} - \Delta_q u\|_{L^p(Q_a^N)} \leq C 2^{-q} \|u\|_{B_{p,r}^s(\mathbb{R}^N)}.$$

Note that if one replaces the function h_q by the function $2^{qN} \mathcal{F}^{-1} \chi(2^q \cdot)$ in the above computations then one gets

$$(1.18) \quad \max \left(\|S_q u\|_{L^p(\mathbb{R}^N \setminus Q_a^N)}, \|S_q^{per} u^{per} - S_q u\|_{L^p(Q_a^N)} \right) \leq C 2^{-q} \|u\|_{B_{p,r}^s(\mathbb{R}^N)}.$$

⁵We take the max norm in $\mathbb{R}^N$ to simplify the computations.

Let $q_0 \in \mathbb{N}$ be such that $8C2^{-q_0} \leq 1$. From (1.16), (1.17) and (1.18), we get

$$\left| \left(\|S_{q_0}^{per} u^{per}\|_{L^p(Q_a^N)}^r + \sum_{q \geq q_0} (2^{qs} \|\Delta_q^{per} u^{per}\|_{L^p(Q_a^N)})^r \right)^{\frac{1}{r}} - \left(\|S_{q_0} u\|_{L^p(\mathbb{R}^N)}^r + \sum_{q \geq q_0} (2^{qs} \|\Delta_q u\|_{L^p(\mathbb{R}^N)})^r \right)^{\frac{1}{r}} \right| \leq \frac{\|u\|_{B_{p,r}^s(\mathbb{R}^N)}}{2}.$$

One can easily show that

$$\left(\|S_{q_0}^{per} u^{per}\|_{L^p(Q_a^N)}^r + \sum_{q \geq q_0} (2^{qs} \|\Delta_q^{per} u^{per}\|_{L^p(Q_a^N)})^r \right)^{\frac{1}{r}} \approx \|u^{per}\|_{B_{p,r}^s(\mathbb{T}_a^N)}$$

and that

$$\left(\|S_{q_0} u\|_{L^p(\mathbb{R}^N)}^r + \sum_{q \geq q_0} (2^{qs} \|\Delta_q u\|_{L^p(\mathbb{R}^N)})^r \right)^{\frac{1}{r}} \approx \|u\|_{B_{p,r}^s(\mathbb{R}^N)}$$

whence the desired result. ■

One can now prove proposition 1.4.6. According to proposition 1.4.5, we already know that $T_\phi : u \mapsto \phi u$ maps $B_{p,r}^{s+\varepsilon}(\mathbb{R}^N)$ in $B_{p,r}^s(\mathbb{R}^N)$. We shall prove the compactness by decomposing T_ϕ into a product of continuous maps, one of them being compact.

With no loss of generality, one can assume that ϕ is supported in some cube $Q_{a,\delta}^N$ which satisfies the assumptions of proposition 1.4.7. Now, one can use the decomposition

$$T_\phi = \mathcal{J} \circ \mathcal{I} \circ \Pi \circ M_\phi$$

where

- $\mathcal{J}$ stands for the extension map by 0 outside Q_a^N from the subset of those distributions over $\mathbb{T}_a^N$ whose restriction to Q_a^N is supported in $Q_{a,\delta}^N$, to the set of temperate distributions over $\mathbb{R}^N$ supported in $Q_{a,\delta}^N$,
- $\mathcal{I}$ is the canonical embedding from $B_{p,r}^{s+\varepsilon}(\mathbb{T}_a^N)$ to $B_{p,r}^s(\mathbb{T}_a^N)$,
- Π is the map $u \mapsto u^{per}$ introduced in proposition 1.2.2,
- M_ϕ is the map $u \mapsto \phi u$ over $B_{p,r}^{s+\varepsilon}(\mathbb{R}^N)$ with values in $B_{p,r}^{s+\varepsilon}(\mathbb{R}^N)$.

According to proposition 1.4.5, the map M_ϕ is continuous. Besides, according to lemma 1.4.7, the map $\mathcal{J}$ (resp. Π) is continuous from the subspace of those distributions of $B_{p,r}^s(\mathbb{T}_a^N)$ whose restriction to Q_a^N is supported in $Q_{a,\delta}^N$, to $B_{p,r}^s(\mathbb{R}^N)$ (resp. from the subspace of distributions of $B_{p,r}^{s+\varepsilon}(\mathbb{R}^N)$ supported in $Q_{a,\delta}^N$, to $B_{p,r}^{s+\varepsilon}(\mathbb{T}_a^N)$).

We claim that $\mathcal{I}$ may be approximated by a sequence of operators with finite rank. This will yield compactness.

Indeed, introduce the finite rank operator $\mathcal{I}^n$ defined by

$$\mathcal{I}^n(v) := \sum_{q \leq n} \Delta_q^{per} v.$$

Using proposition 1.2.1, we discover that for all $q \in \mathbb{N}$, we have

$$\Delta_q^{per}(\mathcal{I} - \mathcal{I}^n)(v) = \sum_{\substack{j > n \\ |j-q| \leq 1}} \Delta_q^{per} \Delta_j^{per} v.$$

We thus have for some constant C independent of q ,

$$2^{qs} \|\Delta_q^{per}(\mathcal{I} - \mathcal{I}^n)(v)\|_{L^p(\mathbb{T}_a^N)} \leq C 2^{-n\varepsilon} (2^{q(s+\varepsilon)} \|\Delta_q^{per} v\|_{L^p(\mathbb{T}_a^N)}),$$

whence

$$\|\mathcal{I} - \mathcal{I}^n\|(v) \|_{B_{p,r}^s(\mathbb{T}_a^N)} \leq C 2^{-n\varepsilon} \|v\|_{B_{p,r}^{s+\varepsilon}(\mathbb{T}_a^N)}.$$

This insures that $\mathcal{I}_n$ tends to $\mathcal{I}$ in $\mathcal{L}(B_{p,r}^{s+\varepsilon}(\mathbb{T}_a^N); B_{p,r}^s(\mathbb{T}_a^N))$. The proof of proposition 1.4.6 is complete. $\blacksquare$

1.4.5 Results of continuity for the composition

Let us state the main result of this section:

Proposition 1.4.8. *Let I be an open interval of $\mathbb{R}$. Let $s > 0$ and σ be the smallest integer such that $\sigma \geq s$. Let $F : I \rightarrow \mathbb{R}$ satisfy $F(0) = 0$ and $F' \in W^{\sigma,\infty}(I; \mathbb{R})$. Assume that $v \in B_{p,r}^s$ has values in $J \subset I$. Then $F(v) \in B_{p,r}^s$ and there exists a constant C depending only on s , I , J , and N , and such that*

$$\|F(v)\|_{B_{p,r}^s} \leq C(1 + \|v\|_{L^\infty})^\sigma \|F'\|_{W^{\sigma,\infty}(I)} \|v\|_{B_{p,r}^s}.$$

Proof: The proof is based on Meyer's first linearization method (see e.g. [1], chapter 2).

Of course, one can change F for a function $\tilde{F} \in W^{\sigma+1,\infty}(\mathbb{R})$ compactly supported in I and such that $\tilde{F} \equiv F$ on a neighborhood of J . So let us assume that F belongs to $W^{\sigma+1,\infty}(\mathbb{R})$ and has compact support in I .

The starting point of the proof is the following formal decomposition⁶

$$(1.19) \quad F(v) = \sum_{q' \geq -1} F(S_{q'+1}v) - F(S_{q'}v).$$

According to first order Taylor's formula, we have for all $q' \geq -1$,

$$F(S_{q'+1}v) - F(S_{q'}v) = m_{q'} \Delta_{q'} v \quad \text{with} \quad m_{q'} := \int_0^1 F'(S_{q'}v + \tau \Delta_{q'}v) d\tau.$$

One can easily prove that the m_p 's are *Meyer multipliers*, namely

$$(1.20) \quad \forall k \in \{0, \dots, \sigma\}, \quad \|D^k m_{q'}\|_{L^\infty} \leq C_k 2^{q'k} (1 + \|v\|_{L^\infty})^k \|F'\|_{W^{k,\infty}}.$$

In particular, inequality (1.20) with $k = 0$ implies that

$$\|F(S_{q'+1}v) - F(S_{q'}v)\|_{L^p} \leq C 2^{-q's} \sup_q 2^{qs} \|\Delta_q v\|_{L^p}.$$

Since s is positive, we conclude that (1.19) holds true in L^p .

We now have to prove that $F(v)$ belongs to $B_{p,r}^s$. For all $q \geq -1$, we have

$$\Delta_q F(v) = \underbrace{\sum_{-1 \leq q' \leq q-1} \Delta_q(m_{q'} \Delta_{q'}v)}_{\Delta_q^1} + \underbrace{\sum_{q' \geq q} \Delta_q(m_{q'} \Delta_{q'}v)}_{\Delta_q^2}.$$

⁶Remind that $S_{-1}v \equiv 0$.

Taking advantage of Bernstein lemma, we get for all (q', q) such that $-1 \leq q' \leq q - 1$,

$$\|\Delta_q(m_{q'} \Delta_{q'} v)\|_{L^p} \lesssim 2^{-q\sigma} \|D^\sigma \Delta_q(m_{q'} \Delta_{q'} v)\|_{L^p},$$

whence, combining Leibniz formula and (1.20),

$$2^{qs} \|\Delta_q(m_{q'} \Delta_{q'} v)\|_{L^p} \lesssim \|F'\|_{W^{\sigma,\infty}(I)} (1 + \|v\|_{L^\infty})^\sigma 2^{(q'-q)(\sigma-s)} (2^{q'\sigma} \|\Delta_{q'} v\|_{L^p}).$$

Since $\sigma > s$, convolution inequalities enable us to conclude that

$$\left(\sum_q (2^{qs} \|\Delta_q^1\|_{L^p})^r \right)^{\frac{1}{r}} \lesssim \|F'\|_{W^{\sigma,\infty}(I)} (1 + \|v\|_{L^\infty})^\sigma \|v\|_{B_{p,r}^s}$$

and proposition 1.3.4 insures that $\sum \Delta_q^1$ belongs to $B_{p,r}^s$.

Bounding the term pertaining to Δ_q^2 is easy. Indeed, we have according to (1.20),

$$2^{qs} \|\Delta_q(m_{q'} \Delta_{q'} v)\|_{L^p} \lesssim \|F'\|_{L^\infty(I)} 2^{(q-q')s} (2^{q's} \|\Delta_{q'} v\|_{L^p}),$$

so that, since $s > 0$,

$$\left(\sum_q (2^{qs} \|\Delta_q^2\|_{L^p})^r \right)^{\frac{1}{r}} \lesssim \|F'\|_{L^\infty(I)} \|v\|_{B_{p,r}^s}.$$

Applying once again proposition 1.3.4 completes the proof. ■

Finally, combining propositions 1.4.3 and 1.4.8 with the following equality:

$$F(v) - F(u) = (v - u) \int_0^1 F'(u + \tau(v - u)) d\tau,$$

we readily get the following

Corollary 1.4.9. *Let I be an open interval of $\mathbb{R}$ and $F : I \rightarrow \mathbb{R}$. Let $s > 0$ and σ be the smallest integer such that $\sigma \geq s$. Assume that $F'(0) = 0$ and that F'' belongs to $W^{\sigma,\infty}(I; \mathbb{R})$. Let $u, v \in B_{p,r}^s$ have values in $J \subset\subset I$. There exists a constant $C = C_{s,I,J,N}$ such that*

$$\begin{aligned} \|F(v) - F(u)\|_{B_{p,r}^s} &\leq C(1 + \|v\|_{L^\infty})^\sigma \|F''\|_{W^{\sigma,\infty}(I)} \\ &\quad \times \left(\|v - u\|_{B_{p,r}^s} \sup_{\tau \in [0,1]} \|u + \tau(v - u)\|_{L^\infty} + \|v - u\|_{L^\infty} \sup_{\tau \in [0,1]} \|u + \tau(v - u)\|_{B_{p,r}^s} \right). \end{aligned}$$

1.5 Calculus in homogeneous functional spaces

Nonhomogeneous functional spaces are not the most natural spaces for studying mathematical problems which have a property of invariance by dilation. For instance, it is well known that the Sobolev embedding $H^1(\mathbb{R}^3) \hookrightarrow L^6(\mathbb{R}^3)$ involves only the L^2 norm of the gradient and not the whole H^1 norm. We shall also see in the next chapters that many interesting PDE's have some properties of scaling invariance.

This is a good motivation for introducing a *homogeneous Littlewood-Paley decomposition* where the low frequencies are treated exactly as the high frequencies, and to define *homogeneous functional spaces*.

1.5.1 Homogeneous Littlewood-Paley decomposition

Let (χ, φ) be as in section 1.2.2. The *homogeneous dyadic blocks* are defined by

$$\dot{\Delta}_q u := \varphi(2^{-q}D)u \quad \text{for all } q \in \mathbb{Z}.$$

We also introduce the following low frequency cut-off:

$$\dot{S}_q u := \chi(2^{-q}D) \quad \text{for all } q \in \mathbb{Z}.$$

The above definition deserves two important preliminary remarks:

- For $u \in \mathcal{S}'$, we have $\sum_{q \in \mathbb{Z}} \dot{\Delta}_q u = u$ modulo a polynomial only.
- In contrast with the nonhomogeneous case, we *do not* have $\dot{S}_q u = \sum_{p \leq q-1} \dot{\Delta}_p u$.

In light of the first remark, working with distributions modulo polynomials seems to be the most appropriate choice. As a matter of fact, this is the viewpoint of most authors of textbooks on abstract functional analysis (see e.g. [38] or [36]).

Since in the next chapters, we aim at applying homogeneous Littlewood-Paley decomposition for solving nonlinear PDE's however, it is not suitable to work with distributions defined modulo polynomials. This motivates the following definition (after J.-Y. Chemin in [12]):

Definition. We denote by $\mathcal{S}'_h$ the space of temperate distributions u such that

$$\lim_{j \rightarrow -\infty} \dot{S}_j u = 0 \quad \text{in } \mathcal{S}'.$$

Remarks. (i) A polynomial u does not belong to $\mathcal{S}'_h$ unless it is identically 0. Indeed, if u is a polynomial then we have $\dot{S}_j u = u$ for all j in $\mathbb{Z}$.

(ii) It is obvious that $\mathcal{S}'_h$ is the space of temperate distributions u which satisfy

$$u = \sum_j \dot{\Delta}_j u \quad \text{in } \mathcal{S}'.$$

(iii) The space $\mathcal{S}'_h$ is not a closed subspace of $\mathcal{S}'$ for the topology of weak convergence. Indeed, consider a sequence $(f_n)_{n \in \mathbb{N}}$ with $f_n(x) = f(x/n)$ and $f \in \mathcal{S}$ such that $f(0) = 1$. Then $(f_n)_{n \in \mathbb{N}}$ tends weakly to the constant function 1 which does not belong to $\mathcal{S}'_h$.

Examples. (i) If a temperate distribution u is such that its Fourier transform $\hat{u}$ is locally integrable near 0, then u belongs to $\mathcal{S}'_h$.

(ii) If u is a temperate distribution such that for some function θ in $\mathcal{C}_c^\infty(\mathbb{R}^N)$ with value 1 near the origin, we have $\theta(D)u$ in L^p for some $p \in [1, +\infty[$, then u belongs to $\mathcal{S}'_h$. In particular, when p is finite, any nonhomogeneous Besov space $B_{p,r}^s$ is included in $\mathcal{S}'_h$.

1.5.2 Homogeneous Besov spaces

Definition 1.5.1. Let u be a temperate distribution, s a real number, and $1 \leq p, r \leq \infty$. Then we set

$$\|u\|_{\dot{B}_{p,r}^s} := \left(\sum_{q \in \mathbb{Z}} 2^{rqs} \|\dot{\Delta}_q u\|_{L^p}^r \right)^{\frac{1}{r}}$$

with the usual change if $r = \infty$.

For the semi-norms we have defined, we can prove the following inequalities

Theorem 1.5.2. *A constant C exists such that for all $s \in \mathbb{R}$,*

$$\begin{aligned} r_1 \leq r_2 &\Rightarrow \|u\|_{\dot{B}_{p,r_2}^s} \leq \|u\|_{\dot{B}_{p,r_1}^s}, \\ p_1 \leq p_2 &\Rightarrow \|u\|_{\dot{B}_{p_2,r}^{s-N(\frac{1}{p_1}-\frac{1}{p_2})}} \leq C \|u\|_{\dot{B}_{p_1,r}^s}. \end{aligned}$$

Moreover we have the following interpolation inequalities for any θ in $]0,1[$ and $s < \tilde{s}$:

$$\begin{aligned} \|u\|_{\dot{B}_{p,r}^{\theta s+(1-\theta)\tilde{s}}} &\leq \|u\|_{\dot{B}_{p,r}^s}^\theta \|u\|_{\dot{B}_{p,r}^{\tilde{s}}}^{1-\theta}, \\ \|u\|_{\dot{B}_{p,1}^{\theta s+(1-\theta)\tilde{s}}} &\leq \frac{C}{\theta(1-\theta)(\tilde{s}-s)} \|u\|_{\dot{B}_{p,\infty}^s}^\theta \|u\|_{\dot{B}_{p,\infty}^{\tilde{s}}}^{1-\theta}. \end{aligned}$$

The proof is the same as in the nonhomogeneous framework, and thus omitted. Finally, the following logarithmic interpolation inequalities are available:

Proposition 1.5.3. *There exists a constant C such that for all $s \in \mathbb{R}$, $\varepsilon > 0$ and $1 \leq p \leq \infty$, we have*

$$(1.21) \quad \|u\|_{\dot{B}_{p,1}^s} \leq C \frac{1+\varepsilon}{\varepsilon} \|u\|_{\dot{B}_{p,\infty}^s} \left(1 + \log \left(\frac{\|u\|_{\dot{B}_{p,\infty}^{s-\varepsilon}} + \|u\|_{\dot{B}_{p,\infty}^{s+\varepsilon}}}{\|u\|_{\dot{B}_{p,\infty}^s}} \right) \right),$$

$$(1.22) \quad \|u\|_{\dot{B}_{p,1}^s} \leq C \frac{1+\varepsilon}{\varepsilon} \left(1 + \|u\|_{\dot{B}_{p,\infty}^s} \log(e + \|u\|_{\dot{B}_{p,\infty}^{s-\varepsilon}} + \|u\|_{\dot{B}_{p,\infty}^{s+\varepsilon}}) \right).$$

Proof: The proof is almost the same as in the nonhomogeneous framework. We write for some positive integer Q to be fixed hereafter:

$$\|u\|_{\dot{B}_{p,1}^s} = \sum_{q \leq -Q} 2^{qs} \|\Delta_q u\|_{L^p} + \sum_{|q| < Q} 2^{qs} \|\Delta_q u\|_{L^p} + \sum_{q \geq Q} 2^{qs} \|\Delta_q u\|_{L^p},$$

whence, from elementary computations,

$$\|u\|_{\dot{B}_{p,1}^s} \leq (2Q-1) \|u\|_{\dot{B}_{p,\infty}^s} + \frac{2^{-Q\varepsilon}}{1-2^{-\varepsilon}} (\|u\|_{\dot{B}_{p,\infty}^{s-\varepsilon}} + \|u\|_{\dot{B}_{p,\infty}^{s+\varepsilon}}).$$

Choosing for Q the closest positive integer to $\frac{1}{\varepsilon} \log_2 \left(\frac{\|u\|_{\dot{B}_{p,\infty}^{s-\varepsilon}} + \|u\|_{\dot{B}_{p,\infty}^{s+\varepsilon}}}{\|u\|_{\dot{B}_{p,\infty}^s}} \right)$ yields (1.21).

The second inequality may be easily deduced from (1.21). ■

Homogeneous Besov spaces have some invariance properties by dilation. More precisely, if for any temperate distribution u and $\lambda > 0$, we introduce the temperate distribution u_λ defined for all $x \in \mathbb{R}^N$ by $u_\lambda(x) := u(\lambda x)$, then we have

Proposition 1.5.4. *If $\|u\|_{\dot{B}_{p,r}^s}$ is finite, so is $\|u_\lambda\|_{\dot{B}_{p,r}^s}$ and we have*

$$\|u_\lambda\|_{\dot{B}_{p,r}^s} \approx \lambda^{s-\frac{N}{p}} \|u\|_{\dot{B}_{p,r}^s}.$$

Besides, equality is true if $\lambda = 2^m$ for some $m \in \mathbb{Z}$.

Proof: Let α be a positive real. We have

$$\varphi(\alpha^{-1}D)u_\lambda(x) = \alpha^N \int h(\alpha(x-y))u(\lambda y) dy.$$

By the change of variables $z = \lambda y$, we get that

$$\begin{aligned}\varphi(\alpha^{-1}D)u_\lambda(x) &= \alpha^N \lambda^{-N} \int h(\alpha x - \alpha \lambda^{-1}z) u(z) dz, \\ &= [\varphi(\lambda \alpha^{-1}D)u](\alpha x).\end{aligned}$$

For $q \in \mathbb{Z}$, let us denote $v_q := \varphi(2^{\lfloor \log_2 \lambda \rfloor - \log_2 \lambda - q} D)u_\lambda$. Taking $\alpha = 2^{q - \lfloor \log_2 \lambda \rfloor} \lambda$ in the above equality, we get

$$2^{qs} \|v_q\|_{L^p} \approx \lambda^{s - \frac{N}{p}} 2^{(q - \lfloor \log_2 \lambda \rfloor)s} \|\dot{\Delta}_{q - \lfloor \log_2 \lambda \rfloor} u\|_{L^p}$$

with an equality if $\log_2 \lambda$ is an integer.

Thus we deduce that

$$\left(\sum_q (2^{qs} \|v_q\|_{L^p}) \right)^{\frac{1}{r}} \approx \lambda^{s - \frac{N}{p}} \|u\|_{\dot{B}_{p,r}^s}.$$

Since $\text{Supp } \hat{v}_q \subset \mathcal{C}(0, \frac{3}{4}2^q, \frac{16}{3}2^q)$, we have

$$\dot{\Delta}_q u_\lambda = \sum_{|p-q| \leq 2} \dot{\Delta}_p u_q.$$

Now, straightforward computations yield the desired result. ■

Next, we notice that $\|\cdot\|_{\dot{B}_{p,r}^s}$ is actually only a semi-norm in the sense that, if u is a polynomial then, for any integer q we have $\dot{\Delta}_q u = 0$, (because the support of $\hat{u}$ is the origin), and so $\|u\|_{\dot{B}_{p,r}^s} = 0$. Therefore, if we define the homogeneous Besov space $\dot{B}_{p,r}^s$ as the set of temperate distributions u such that $\|u\|_{\dot{B}_{p,r}^s}$ is finite, we may run into troubles later when studying non linear problems because we will not be able to tell a polynomial from a null function !

In the present lecture notes, we adopt the following definition:

Definition. Let s be a real number and (p, r) be in $[1, \infty]^2$. The space $\dot{B}_{p,r}^s$ is the set of distributions u in $\mathcal{S}'_h$ such that $\|u\|_{\dot{B}_{p,r}^s}$ is finite.

In the homogeneous framework, the results corresponding to proposition 1.3.4 read:

Proposition 1.5.5. *Let $s \in \mathbb{R}$ and $1 \leq p, r \leq \infty$ satisfy*

$$(1.23) \quad s < \frac{N}{p}, \quad \text{or} \quad s = \frac{N}{p} \quad \text{and} \quad r = 1.$$

(i) *Let $(u_q)_{q \in \mathbb{Z}}$ be a sequence of functions such that*

$$\left(\sum_q (2^{qs} \|u_q\|_{L^p})^r \right)^{\frac{1}{r}} < \infty.$$

If $\text{Supp } \hat{u}_q \subset \mathcal{C}(0, 2^q R_1, 2^q R_2)$ for some $0 < R_1 < R_2$ then $u := \sum_{q \in \mathbb{Z}} u_q$ belongs to $\dot{B}_{p,r}^s$ and there exists a constant C such that

$$(1.24) \quad \|u\|_{\dot{B}_{p,r}^s} \leq C^{1+|s|} \left(\sum_q (2^{qs} \|u_q\|_{L^p})^r \right)^{\frac{1}{r}}.$$

Let $(u_q)_{q \in \mathbb{Z}}$ be a sequence of functions such that

$$\left(\sum_q (2^{qs} \|u_q\|_{L^p})^r \right)^{\frac{1}{r}} < \infty.$$

If $\text{Supp } \hat{u}_q \subset \mathcal{B}(0, 2^q R)$ for some positive R and if in addition s is positive then $u := \sum_{q \in \mathbb{Z}} u_q$ belongs to $\dot{B}_{p,r}^s$ and there exists a constant C such that

$$(1.25) \quad \|u\|_{\dot{B}_{p,r}^s} \leq \frac{C^{1+s}}{s} \left(\sum_q (2^{qs} \|u_q\|_{L^p})^r \right)^{\frac{1}{r}}.$$

Proof: Let us prove (i).

On one hand, using Bernstein lemma, it is easy to see that the series $\sum_{q \leq 0} u_q$ is convergent in L^∞ and that $u^- := \sum_{q \leq 0} u_q$ belongs to $\mathcal{S}'_h$. On the other hand, since $\|u_q\|_{L^\infty} \leq C 2^{q(\frac{N}{p}-s)}$, lemma 1.3.3 insures that the series $\sum_{q > 0} u_q$ is convergent in $\mathcal{S}'$. Besides, the Fourier transform of $u^+ := \sum_{q > 0} u_q$ is supported away from the origin, hence u^+ belongs to $\mathcal{S}'_h$. Since $u = u^- + u^+$, one can conclude that u belongs to $\mathcal{S}'_h$. Now, the proof of (1.24) may be done exactly as in lemma 1.3.4.

Proving (ii) is similar. Indeed, we have for any $q \in \mathbb{Z}$,

$$\|u_q\|_{L^p} \leq C 2^{-qs} \quad \text{and} \quad \|u_q\|_{L^\infty} \leq C 2^{q(\frac{N}{p}-s)}.$$

Hence $\sum_{q \leq 0} u_q$ converges in L^∞ and belongs to $\mathcal{S}'_h$ whereas $\sum_{q > 0} u_q$ converges in L^p (and also belongs to $\mathcal{S}'_h$ since its Fourier transform is bounded away from 0). Next, the proof of (1.25) goes along the lines of lemma 1.3.4. $\blacksquare$

As an important corollary of proposition 1.5.5, we get that, under condition (1.23), the definition of $\dot{B}_{p,r}^s$ does not depend on φ .

The following proposition, the proof of which is left to the reader, describes the relations between homogeneous and nonhomogeneous spaces.

Proposition 1.5.6. *Let s be a negative number (or $s = 0$ and $r = 1$). Then $\dot{B}_{p,r}^s \hookrightarrow B_{p,r}^s$. Besides, if $s < 0$, a constant C (independent of s) exists so that, for any u belonging to $\dot{B}_{p,r}^s$, we have*

$$\|u\|_{B_{p,r}^s} \leq \frac{C}{-s} \|u\|_{\dot{B}_{p,r}^s}.$$

Let s be a positive number (or $s = 0$ and $r = \infty$). Then $B_{p,r}^s \hookrightarrow \dot{B}_{p,r}^s$ when p is finite, and $B_{\infty,r}^s \cap \mathcal{S}'_h$ is a subset of $\dot{B}_{\infty,r}^s$. If $s > 0$, a constant C exists (independent of s) so that, for any u belonging to $B_{p,r}^s$, we have

$$\|u\|_{\dot{B}_{p,r}^s} \leq \frac{C}{s} \|u\|_{B_{p,r}^s}.$$

Let us notice that there is no monotonicity property with respect to s for homogeneous spaces. The reason why is that homogeneous Besov spaces carry on informations about both low and high frequencies.

In homogeneous spaces, the counterpart of proposition 1.3.8 reads:

Proposition 1.5.7. *There exists a constant C which satisfies the following properties. Let $s < 0$, $(p, r) \in [1, \infty]^2$ and u a distribution in $\mathcal{S}'_h$. Then u belongs to $\dot{B}_{p,r}^s$ if and only if*

$$(2^{qs} \|\dot{S}_q u\|_{L^p})_{q \in \mathbb{Z}} \in \ell^r(\mathbb{Z}).$$

Moreover, we have

$$\frac{1}{2} \|u\|_{\dot{B}_{p,r}^s} \leq \left(\sum_q (\|2^{qs} \|\dot{S}_q u\|_{L^p}\|_{L^p})^r \right)^{\frac{1}{r}} \leq C \left(1 + \frac{1}{|s|} \right) \|u\|_{\dot{B}_{p,r}^s}.$$

Proof: The proof goes along the lines of proposition 1.3.8. ■

Proposition 1.5.8. *Let f be a smooth function on $\mathbb{R}^N \setminus \{0\}$ which is homogeneous of degree m . Let $1 \leq p, r \leq \infty$. Assume that*

$$s - m < \frac{N}{p}, \quad \text{or} \quad r = 1 \quad \text{and} \quad s - m \leq \frac{N}{p}.$$

Then $f(D)$ is continuous from $\dot{B}_{p,r}^s$ to $\dot{B}_{p,r}^{s-m}$.

Proof: Assume that $r > 1$. Introduce a smooth function $\tilde{\varphi}$ supported in a shell and such that $\tilde{\varphi} \equiv 1$ on $\text{Supp } \varphi$. As f is homogeneous of degree m , we have

$$\dot{\Delta}_q f(D)u = 2^{qm}[\tilde{\varphi}f](2^{-q}D)\dot{\Delta}_q u.$$

Since $\mathcal{F}^{-1}(\tilde{\varphi}f)$ belongs to $\mathcal{S}$, we readily get

$$\left\| \dot{\Delta}_q f(D)u \right\|_{L^p} \leq C 2^{qm} \left\| \dot{\Delta}_q u \right\|_{L^p},$$

whence

$$\left(\sum_q (2^{q(s-m)} \left\| \dot{\Delta}_q f(D)u \right\|_{L^p})^r \right)^{\frac{1}{r}} \lesssim \|u\|_{\dot{B}_{p,r}^s}.$$

Now, we have

$$\dot{S}_q f(D)u = \sum_{q' \leq q} \dot{S}_q f(D)\dot{\Delta}_{q'} u,$$

whence according to Bernstein lemma,

$$\left\| \dot{S}_q f(D)u \right\|_{L^\infty} \lesssim \sum_{q' \leq q} 2^{q'(\frac{N}{p} + m - s)} 2^{q's} \left\| \dot{\Delta}_{q'} u \right\|_{L^p} \lesssim 2^{q(\frac{N}{p} + m - s)} \|u\|_{\dot{B}_{p,\infty}^s}.$$

Since $N/p + m - s > 0$, it is now clear that $\dot{S}_q f(D)u$ tends to 0 when q goes to $-\infty$. Hence $f(D)u$ belongs to $\dot{B}_{p,r}^{s-m}$. The proof in the case $r = 1$ is left to the reader. ■

Let us now focus on the topological properties of the spaces $\dot{B}_{p,r}^s$.

Proposition 1.5.9. *For all $s \in \mathbb{R}$ and $1 \leq p, r \leq \infty$, the couple $(\dot{B}_{p,r}^s, \|\cdot\|_{\dot{B}_{p,r}^s})$ is a normed space. If besides r is finite then $\mathcal{C}_c^\infty \cap \dot{B}_{p,r}^s$ is densely embedded in $\dot{B}_{p,r}^s$.*

Proof: It is obvious that $\|\cdot\|_{\dot{B}_{p,r}^s}$ is a semi-norm. Let us assume that $\|u\|_{\dot{B}_{p,r}^s} = 0$ for some u in $\mathcal{S}'_h$. This implies that $\text{Supp } \hat{u} \subset \{0\}$ and thus that for any $j \in \mathbb{Z}$ we have $\dot{S}_j u = u$. As u belongs to $\mathcal{S}'_h$, we must have $\lim_{j \rightarrow -\infty} \dot{S}_j u = 0$ so that we can conclude that $u = 0$.

Now, if r is finite and $u \in \dot{B}_{p,r}^s$, it is obvious that the sequence of general term

$$\sum_{|q| \leq n} \dot{\Delta}_q u$$

belongs to $\mathcal{C}^\infty \cap \dot{B}_{p,r}^s$ and tends to u in $\dot{B}_{p,r}^s$. Arguing like in 1.3.5.(ii), it is then easy to exhibit a sequence of functions of $\mathcal{C}_c^\infty \cap \dot{B}_{p,r}^s$ which tends to u in $\dot{B}_{p,r}^s$. ■

Theorem 1.5.10. *If $s < \frac{N}{p}$ or $s = \frac{N}{p}$ and $r = 1$ then $(\dot{B}_{p,r}^s, \|\cdot\|_{\dot{B}_{p,r}^s})$ is a Banach space which is continuously embedded in $\mathcal{S}'$.*

Proof: Let us first prove that $\dot{B}_{p,r}^s$ is continuously embedded in $\mathcal{S}'$. The case $r = 1$ and $s = N/p$ is easy because the series $\sum \dot{\Delta}_j u$ is convergent in L^∞ . As $u = \sum_j \dot{\Delta}_j u$, this implies that u belongs to L^∞ . Besides, we have

$$(1.26) \quad \dot{B}_{p,1}^{\frac{N}{p}} \hookrightarrow \dot{B}_{\infty,1}^0 \hookrightarrow L^\infty \hookrightarrow \mathcal{S}'.$$

Let us now assume that $s < N/p$. Using that $\dot{B}_{p,r}^s \hookrightarrow \dot{B}_{\infty,\infty}^{s-\frac{N}{p}}$ and arguing like for proving (1.10), one can find a large integer M such that for all nonnegative j , we have

$$|\langle \dot{\Delta}_j u, \phi \rangle| \leq 2^{-j} \|u\|_{\dot{B}_{\infty,\infty}^{s-\frac{N}{p}}} \|\phi\|_{M,\mathcal{S}}.$$

For negative j , one can write that for large enough M , we have

$$\begin{aligned} |\langle \dot{\Delta}_j u, \phi \rangle| &\lesssim 2^{j(\frac{N}{p}-s)} \|u\|_{\dot{B}_{\infty,\infty}^{s-\frac{N}{p}}} \|\phi\|_{L^1} \\ &\lesssim 2^{j(\frac{N}{p}-s)} \|u\|_{\dot{B}_{p,r}^s} \|\phi\|_{M,\mathcal{S}}. \end{aligned}$$

Because u belongs to $\mathcal{S}'_h$, we have $\langle u, \phi \rangle = \sum_j \langle \dot{\Delta}_j u, \phi \rangle$. Therefore, for large enough M ,

$$(1.27) \quad |\langle u, \phi \rangle| \leq C_s \|u\|_{\dot{B}_{p,r}^s} \|\phi\|_{M,\mathcal{S}}$$

and we can conclude that $\dot{B}_{p,r}^s \hookrightarrow \mathcal{S}'$.

We still have to prove that for all triplet (s, p, r) satisfying the hypothesis of the theorem, the set $\dot{B}_{p,r}^s$ is a Banach space. So let us consider a Cauchy sequence $(u_n)_{n \in \mathbb{N}}$ in $\dot{B}_{p,r}^s$. Using (1.26) or (1.27), this implies that a temperate distribution u exists such that the sequence $(u_n)_{n \in \mathbb{N}}$ converges to u in $\mathcal{S}'$. We now have to state that u belongs to $\mathcal{S}'_h$. Let us first assume that $s < N/p$. Since u_n belongs to $\mathcal{S}'_h$, we have, thanks to (1.27),

$$\forall j \in \mathbb{Z}, \forall n \in \mathbb{N}, |\langle \dot{S}_j u_n, \phi \rangle| \lesssim 2^{j(\frac{N}{p}-s)} \|u_n\|_{\dot{B}_{p,r}^s} \|\phi\|_{M,\mathcal{S}}.$$

As the sequence $(u_n)_{n \in \mathbb{N}}$ tends to u in $\mathcal{S}'$, we have

$$\forall j \in \mathbb{Z}, |\langle \dot{S}_j u, \phi \rangle| \leq C_s 2^{j(\frac{N}{p}-s)} \|\phi\|_{M,\mathcal{S}} \sup_n \|u_n\|_{\dot{B}_{p,r}^s}.$$

Thus u belongs to $\mathcal{S}'_h$.

The case when u belongs to $\dot{B}_{p,1}^{\frac{N}{p}}$ is a little bit different. Let $\varepsilon > 0$. As $(u_n)_{n \in \mathbb{N}}$ is a Cauchy sequence in $\dot{B}_{p,1}^{\frac{N}{p}} \hookrightarrow \dot{B}_{\infty,1}^0$, there exists an integer n_0 such that

$$\forall j \in \mathbb{Z}, \forall n \geq n_0, \sum_{k \leq j} \|\dot{\Delta}_k u_n\|_{L^\infty} \leq \frac{\varepsilon}{2} + \sum_{k \leq j} \|\dot{\Delta}_k u_{n_0}\|_{L^\infty}.$$

Let us choose j_0 small enough so that

$$\sum_{k \leq j_0} \|\dot{\Delta}_k u_{n_0}\|_{L^\infty} \leq \frac{\varepsilon}{2}.$$

As u_n belongs to $\mathcal{S}'_h$, we have

$$\forall j \leq j_0, \forall n \geq n_0, \|\dot{S}_j u_n\|_{L^\infty} \leq \varepsilon.$$

As sequence $(u_n)_{n \in \mathbb{N}}$ tends to u in L^∞ , this implies that

$$\forall j \leq j_0, \|\dot{S}_j u\|_{L^\infty} \leq \varepsilon.$$

This proves that u belongs to $\mathcal{S}'_h$. Next, arguing like in the nonhomogeneous case completes the proof. $\blacksquare$

Remark. It turns out that when $s > N/p$ (or $s = N/p$ and $r > 1$) the space $\dot{B}_{p,r}^s$ is no longer a Banach space. In the one dimensional case for instance, it can be easily seen that the sequence $(f_n)_{n \in \mathbb{N}}$ defined by

$$\widehat{f_n}(\xi) = \frac{\chi(\xi)}{\xi \log |\xi|} \quad \text{if } |\xi| \geq 2^{-n}, \quad \text{and } 0 \quad \text{elsewhere,}$$

is a Cauchy sequence in $\dot{B}_{2,\infty}^{\frac{1}{2}}$ but cannot have a limit in $\dot{B}_{2,\infty}^{\frac{1}{2}}$ since the function $\xi \mapsto \frac{\chi(\xi)}{\xi \log |\xi|}$ does not belong to $\mathcal{S}'$!

This defect of convergence due to low frequencies is typical to homogeneous functional spaces with high regularity index. It is sometimes called *infrared divergence*.

There is a way to modify the definition of homogeneous Besov spaces so as to get a Banach space regardless of the regularity index. This is called *realizing* homogeneous Besov spaces. It turns out that realizations coincide with definition 1.5.2 when $s < N/p$ or $s = N/p$ and $r = 1$. In the other cases however, realizations *are not* functional spaces but spaces defined up to a polynomial whose degree depends on $s - N/p$ and on r (see e.g [5] or [32]). It goes without saying that solving PDE's in such spaces may be quite unpleasant.

1.5.3 Paradifferential calculus in homogeneous spaces

We designate *homogeneous paraproduct* of v by u and denote by $\dot{T}_u v$ the bilinear operator:

$$\dot{T}_u v := \sum_q \dot{S}_{q-1} u \dot{\Delta}_q v.$$

We designate *homogeneous remainder* of u and v and denote by $\dot{R}(u, v)$ the bilinear operator:

$$\dot{R}(u, v) = \sum_{|p-q| \leq 1} \dot{\Delta}_p u \dot{\Delta}_q v.$$

It is clear that, formally, we have the following *homogeneous* Bony decomposition:

$$(1.28) \quad uv = \dot{T}_u v + \dot{T}_v u + \dot{R}(u, v).$$

The properties of continuity of homogeneous paraproduct and remainder on homogeneous Besov spaces are described in the following propositions.

Proposition 1.5.11. *There exists a constant C such that for any couple of real numbers (s, σ) with σ positive and for any (p, r, r_1, r_2) in $[1, +\infty]^4$ with $1/r = 1/r_1 + 1/r_2$, we have*

$$\|\dot{T}\|_{\mathcal{L}(L^\infty \times \dot{B}_{p,r}^s; \dot{B}_{p,r}^s)} \leq C^{|s|+1}$$

if condition (1.23) is satisfied, and

$$\|\dot{T}\|_{\mathcal{L}(\dot{B}_{\infty,r_1}^{-\sigma} \times \dot{B}_{p,r_2}^s; \dot{B}_{p,r}^{s-\sigma})} \leq \frac{C^{|s-\sigma|+1}}{\sigma}.$$

if $s - \sigma < N/p$, or $s - \sigma = N/p$ and $r = 1$.

Proof: Let $w_q := \dot{S}_{q-1}u \dot{\Delta}_q v$. Arguing like in proposition 1.4.1, one can easily prove that

$$\left(\sum_q (2^{qs} \|w_q\|_{L^p})^r \right)^{\frac{1}{r}} \lesssim \|u\|_{L^\infty} \|v\|_{\dot{B}_{p,r}^s} \quad \text{and} \quad \left(\sum_q (2^{q(s-\sigma)} \|w_q\|_{L^p})^r \right)^{\frac{1}{r}} \lesssim \|u\|_{\dot{B}_{\infty,r_1}^{-\sigma}} \|v\|_{\dot{B}_{p,r_2}^s}.$$

Since the sequence $(\mathcal{F}w_q)_{q \in \mathbb{Z}}$ is supported in dyadic shells, applying proposition 1.5.5 completes the proof. $\blacksquare$

Proposition 1.5.12. *There exists a constant C which satisfies the following inequalities. For any (s_1, s_2) , any $1 \leq p_1, p_2, p \leq \infty$ and any $1 \leq r, r_1, r_2 \leq \infty$ such that*

$$s_1 + s_2 > 0, \quad \frac{1}{p} \leq \frac{1}{p_1} + \frac{1}{p_2} \leq 1 \quad \text{and} \quad \frac{1}{r} \leq \frac{1}{r_1} + \frac{1}{r_2} \leq 1,$$

we have

$$\|\dot{R}\|_{\mathcal{L}(\dot{B}_{p_1,r_1}^{s_1} \times \dot{B}_{p_2,r_2}^{s_2}; \dot{B}_{p,r}^{\sigma_{12}})} \leq \frac{C^{|s_1+s_2|+1}}{s_1 + s_2} \quad \text{with} \quad \sigma_{12} := s_1 + s_2 - N \left(\frac{1}{p_1} + \frac{1}{p_2} - \frac{1}{p} \right),$$

provided that $\sigma_{12} < N/p$, or $\sigma_{12} = N/p$ and $r = 1$.

Proof: Denoting $w_q := \Delta_q u \widetilde{\Delta}_q v$ and arguing like in proposition 1.4.2, we easily get

$$\left(\sum_q (2^{q\sigma_{12}} \|w_q\|_{L^p})^r \right)^{\frac{1}{r}} \lesssim \|u\|_{\dot{B}_{p_1,r_1}^{s_1}} \|v\|_{\dot{B}_{p_2,r_2}^{s_2}}.$$

Since the sequence the sequence $(\mathcal{F}w_q)_{q \in \mathbb{Z}}$ is supported in dyadic balls, applying proposition 1.5.5 completes the proof. $\blacksquare$

Proposition 1.5.13. *Let I be an open interval of $\mathbb{R}$. Let $s > 0$ and σ be the smallest integer such that $\sigma \geq s$. Let $F : I \rightarrow \mathbb{R}$ satisfy $F(0) = 0$ and $F' \in W^{\sigma,\infty}(I; \mathbb{R})$. Assume that $v \in \dot{B}_{p,r}^s$ has values in $J \subset I$. Assume that condition (1.23) is satisfied. Then $F(v) \in \dot{B}_{p,r}^s$ and there exists a constant $C = C_{s,I,J,N}$ such that*

$$\|F(v)\|_{\dot{B}_{p,r}^s} \leq C(1 + \|v\|_{L^\infty})^\sigma \|F'\|_{W^{\sigma,\infty}(I)} \|v\|_{\dot{B}_{p,r}^s}.$$

Proof: Arguing like in the proof of proposition 1.4.8 and using Bernstein lemma, we get for all $j \in \mathbb{Z}$,

$$\begin{aligned} \|F(\dot{S}_{j+1}v) - F(\dot{S}_jv)\|_{L^p} &\leq C 2^{-js} (2^{js} \|\Delta_j v\|_{L^p}), \\ \|F(\dot{S}_{j+1}v) - F(\dot{S}_jv)\|_{L^\infty} &\leq C 2^{j(\frac{N}{p}-s)} (2^{js} \|\Delta_j v\|_{L^p}). \end{aligned}$$

Since s is positive, $F(0) = 0$ and condition (1.23) is satisfied, this insures that the series

$$\sum_{j \in \mathbb{Z}} F(\dot{S}_{j+1}v) - F(\dot{S}_jv)$$

converges to $F(v)$ in $L^p + L^\infty$, thus in $\mathcal{S}'$.

Next, for all $q \in \mathbb{Z}$, we split $\dot{\Delta}_q F(v)$ into

$$\dot{\Delta}_q F(v) = \sum_{j < q} \Delta_q(m_j \dot{\Delta}_j v) + \sum_{j \geq q} \dot{\Delta}_q(m_j \dot{\Delta}_j v).$$

Following the lines of the proof of proposition 1.4.8 and applying proposition 1.5.5 leads to the desired result. $\blacksquare$

1.6 Exercises

Exercise 1.1. Prove that for any $\alpha > 1$, there exists two smooth functions φ and χ such that φ is supported in the shell $\{\xi \in \mathbb{R}^N \mid \alpha^{-1} \leq |\xi| \leq 2\alpha\}$ and χ is supported in the ball $\{\xi \in \mathbb{R}^N \mid |\xi| \leq \alpha\}$, and

$$\forall \xi \in \mathbb{R}^N, \chi(\xi) + \sum_{q \in \mathbb{N}} \varphi(2^{-q}\xi) = 1.$$

Exercise 1.2. Prove that for all temperate distribution u , the equality $u = \sum_{q \in \mathbb{Z}} \Delta_q u$ holds true in $\mathcal{S}'(\mathbb{R}^N)$.

Exercise 1.3. Prove proposition 1.2.1.

Exercise 1.4. Prove inequality (1.5).

Exercise 1.5. For $q \in \mathbb{Z}$, denote $\Delta'_q u := 1_{2^q \leq |\xi| \leq 2^{q+1}}(D)u$. Prove that the inequality

$$\|\Delta'_q u\|_{L^p} \leq C \|u\|_{L^p}$$

for some constant C independent of q is false in the case $p \neq 2$.

Hint: Try with the function $u = \chi$.

Exercise 1.6. Compare the Besov space $B_{p,r}^0$ with the Lebesgue space L^p .

Exercise 1.7. Let $\mathcal{B}_{p,\infty}^s$ be the completion of $\mathcal{C}_c^\infty$ for the $\|\cdot\|_{B_{p,\infty}^s}$ norm. Prove that $u \in \mathcal{B}_{p,\infty}^s$ if and only if $\lim_{q \rightarrow +\infty} 2^{qs} \|\Delta_q u\|_{L^p} = 0$.

Exercise 1.8. Let $r \in (0, 1)$. Prove that $B_{\infty,\infty}^{r-1}(\mathbb{R}^N)$ is the set of temperate distributions u such that there exist $N+1$ functions $u^0, \dots, u^N$ of C^r verifying

$$u = u^0 + \sum_{j=1}^N \partial_j u^j.$$

Exercise 1.9. Let $C_\star^1$ be the Zygmund space of bounded functions u such that

$$\forall (x, y) \in \mathbb{R}^N \times \mathbb{R}^N, |u(x+y) + u(x-y) - 2u(x)| \leq C|y|$$

for some constant C .

Prove that $C_\star^1 = B_{\infty,\infty}^1$.

Exercise 1.10. Let $1 \leq p, p_1, p_2, r \leq \infty$ and $s \in \mathbb{R}$. Let $r' := r/(r-1)$. Assume that $1/p \leq 1/p_1 + 1/p_2$.

Prove that the remainder R maps $B_{p_1,r}^s \times B_{p_2,r'}^{-s}$ in $B_{p,\infty}^{-N(\frac{1}{p} - \frac{1}{p_1} - \frac{1}{p_2})}$.

Exercise 1.11. Prove (1.20).

Exercise 1.12. Let I be an open interval of $\mathbb{R}$ and $F : I \rightarrow \mathbb{R}$. Assume that $F(0) = 0$ and that F' is bounded.

Prove that for all $v \in B_{p,1}^0$ with values in $J \subset\subset I$, the function $F(v)$ belongs to $B_{p,1}^0$ and that

$$\|F(v)\|_{B_{p,1}^0} \lesssim \|F'\|_{L^\infty(I)} \|v\|_{B_{p,1}^0}.$$

Exercise 1.13. Prove that u belongs to $\mathcal{S}'_h$ if and only if for any θ in $\mathcal{C}_c^\infty(\mathbb{R}^N)$ with value 1 near the origin, we have $\lim_{\lambda \rightarrow \infty} \theta(\lambda D)u = 0$ in $\mathcal{S}'$.

Exercise 1.14. Prove proposition 1.5.6.

Exercise 1.15. We say that a temperate distribution u tends weakly to 0 at infinity if $u(\lambda \cdot)$ tends to 0 in $\mathcal{S}'$ when λ goes to infinity.

Prove that $u \in \mathcal{S}'$ belongs to $\mathcal{S}'_h$ if and only if u tends weakly to 0 at infinity.

Exercise 1.16. Let s be in $]0, N[$. Prove that for any p in $[1, \infty]$, we have

$$\frac{1}{|\cdot|^s} \in \dot{B}_{p,\infty}^{\frac{N}{p}-s}.$$

Exercise 1.17. Find divergent Cauchy sequences for the space $\dot{B}_{p,r}^s$ when $s > N/p$ or $s = N/p$ and $r > 1$.

Exercise 1.18. Let $1 \leq p_1, r_1, p_2, r_2 \leq \infty$ and $(s_1, s_2) \in \mathbb{R}^2$. Assume that

$$s_1 < \frac{N}{p_1} \quad \text{or} \quad s_1 = \frac{N}{p_1} \quad \text{and} \quad r_1 = 1.$$

Prove that $L^{p_1} \cap \dot{B}_{p_2,r_2}^{s_2}$ and $\dot{B}_{p_1,r_1}^{s_1} \cap \dot{B}_{p_2,r_2}^{s_2}$ are Banach spaces.

Exercise 1.19. Let $1 \leq p, r \leq +\infty$, $s \in \mathbb{R}$ and $\varepsilon > 0$. Let $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^N)$.

Prove that the map $u \mapsto \phi u$ is compact from $\dot{B}_{p,r}^s$ to $\dot{B}_{p,r}^{s-\varepsilon} + \dot{B}_{p,r}^s$.

Exercise 1.20. Let s be a positive real number and $(p, r) \in [1, \infty]^2$. Prove the existence of a constant C such that for all u in $\mathcal{S}'_h$, we have

$$C^{-1} \|u\|_{\dot{B}_{p,r}^{-2s}} \leq \left\| \|t^s e^{t\Delta} u\|_{L^p} \right\|_{L^r(\mathbb{R}^+, \frac{dt}{t})} \leq C \|u\|_{\dot{B}_{p,r}^{-2s}}.$$

Hint: use that for any positive s and c , we have

$$\sup_{t>0} \sum_{j \in \mathbb{Z}} t^s 2^{2js} e^{-ct2^{2j}} < \infty.$$

Exercise 1.21. Let s be in $]0, 1[$ and $(p, r) \in [1, \infty]^2$. Prove that there exists a constant C such that for any u in $\mathcal{S}'_h$, we have

$$C^{-1} \|u\|_{\dot{B}_{p,r}^s} \leq \left\| \frac{\|\tau_{-z} u - u\|_{L^p}}{|z|^s} \right\|_{L^r(\mathbb{R}^N; \frac{dz}{|z|^N})} \leq C \|u\|_{\dot{B}_{p,r}^s}$$

Exercise 1.22. 1) Let (s_1, s_2) , (p_1, p_2, p) and (r_1, r_2) be such that

$$s_1 + s_2 \geq 0, \quad \frac{1}{p} \leq \frac{1}{p_1} + \frac{1}{p_2} \leq 1 \quad \text{and} \quad \frac{1}{r_1} + \frac{1}{r_2} = 1.$$

Assume that $\sigma_{12} := s_1 + s_2 - N \left(\frac{1}{p_1} + \frac{1}{p_2} - \frac{1}{p} \right)$ satisfies $\sigma_{12} < N/p$, or $\sigma_{12} = N/p$ and $r = 1$.

Prove that the remainder is continuous from $\dot{B}_{p_1,r_1}^{s_1} \times \dot{B}_{p_2,r_2}^{s_2}$ to $\dot{B}_{p,\infty}^{\sigma_{12}}$ and that

$$\|\dot{R}\|_{\mathcal{L}(\dot{B}_{p_1,r_1}^{s_1} \times \dot{B}_{p_2,r_2}^{s_2}; \dot{B}_{p,\infty}^{\sigma_{12}})} \leq C^{|s_1+s_2|+1}.$$

2) Adapt exercise 1.12 to the homogeneous framework.

Chapter 2

The heat equation

In this chapter, we state estimates in Besov spaces for the heat equation. Such estimates are fundamental for solving certain nonlinear PDE's of parabolic type. As an example, we show that incompressible Navier-Stokes equations are locally well-posed in Besov spaces with critical index of regularity.

2.1 Generalities

The basic *heat equation* reads

$$(H) \quad \begin{cases} \partial_t u - \mu \Delta u = f, \\ u|_{t=0} = u_0. \end{cases}$$

Above, the external source term $f = f(t, x)$ and the initial data $u_0 = u_0(x)$ are given. The diffusion μ is a positive constant. We restrict ourselves to the evolution for *positive* times t , and, for the sake of simplicity, we always assume that x belongs to the whole space $\mathbb{R}^N$. Similar result would hold in the torus $\mathbb{T}_a^N$, though.

Giving an exhaustive list of properties of the heat equation is not our goal here. We however have to recall a few important facts for (H) that will be needed for stating estimates in Besov spaces.

Let $u_0 \in \mathcal{S}(\mathbb{R}^N)$ and $f \in \mathcal{C}(\mathbb{R}^+; \mathcal{S}(\mathbb{R}^N))$. Applying the partial Fourier transform with respect to the space variable (still denoted by $\widehat{\cdot}$), we get the following linear ordinary differential equation for all $\xi \in \mathbb{R}^N$:

$$(H) \quad \begin{cases} \partial_t \widehat{u}(t, \xi) + \mu |\xi|^2 \widehat{u}(t, \xi) = \widehat{f}(t, \xi), \\ \widehat{u}|_{t=0}(\xi) = \widehat{u}_0(\xi), \end{cases}$$

whence

$$(2.1) \quad \widehat{u}(t, \xi) = e^{-\mu |\xi|^2 t} \widehat{u}_0(\xi) + \int_0^t e^{-\mu |\xi|^2 (t-\tau)} \widehat{f}(\tau, \xi) d\tau.$$

Performing the inverse Fourier transform, we end up with the following well known representation formula for all $t \in \mathbb{R}^+$ and $x \in \mathbb{R}^N$:

$$(2.2) \quad u(t, x) = \frac{1}{(4\pi\mu t)^{\frac{N}{2}}} \int_{\mathbb{R}^N} e^{-\frac{|x-y|^2}{4\mu t}} u_0(y) dy + \int_0^t \frac{1}{(4\pi\mu(t-\tau))^{\frac{N}{2}}} \left(\int_{\mathbb{R}^N} e^{-\frac{|x-y|^2}{4\mu(t-\tau)}} f(\tau, y) dy \right) d\tau.$$

By arguing by duality, formulae (2.1) and (2.2) may be extended to temperate distributions.

Introducing the heat semi-group $(e^{s\Delta})_{s \geq 0}$ for the Laplacian operator, equality (2.2) may be rewritten in a more concise way:

$$(2.3) \quad u(t) = e^{\mu t \Delta} u_0 + \int_0^t e^{\mu(t-\tau)\Delta} f(\tau) d\tau.$$

Although formulae (2.1) and (2.2) are explicit, they are not so convenient for getting a priori estimates in functional spaces. In fact, even for stating the basic energy equality, it is far more efficient to multiply (H) by u , integrate over $\mathbb{R}^N$ and perform an integration by parts in the term with the Laplacian. At least formally, we end up with

$$\frac{1}{2} \frac{d}{dt} \|u\|_{L^2}^2 + \mu \|Du\|_{L^2}^2 = \int f u dx,$$

thus, integrating over the time interval $[0, T]$, we get

$$\|u(T)\|_{L^2}^2 + 2\mu \int_0^T \|Du(t)\|_{L^2}^2 dt = \|u_0\|_{L^2}^2 + 2 \int_0^T \int_{\mathbb{R}^N} f(t, x) u(t, x) dt dx.$$

We notice that starting from $u_0 \in L^2$ and $f \in L^2(0, T; \dot{H}^{-1})$, the above equality provides an estimate for Du in $L^2(0, T \times \mathbb{R}^N)$. In other words, it gives a gain of one derivative for u with respect to u_0 and of two derivatives with respect to f . One can wonder if it is possible to gain more than one derivative with respect to u_0 by considering L^ρ norms in time ($\rho < 2$) with values in Sobolev spaces.

In the next section, we shall see that very simple arguments based on Littlewood-Paley decomposition enable us to gain *two* derivatives when taking a L^1 norm in time. Besides, the method we are going to present apply indistinctly to the L^p framework for all $p \in [1, +\infty]$.

2.2 A priori estimates in Besov spaces for the heat equation

The fundamental idea is to localize the heat equation through a Littlewood-Paley decomposition. It is then easy to prove $L^\rho(0, T; L^p)$ estimates for each dyadic block in term of norms of $\dot{\Delta}_q u_0$ and $\dot{\Delta}_q f$. If one assumes that u_0 and f belong to some Besov spaces, performing a (weighted) ℓ^r summation is the most natural next step. In doing so however, one does not obtain an estimate in a space of type $L^\rho(0, T; \dot{B}_{p,r}^s)$ since the time integration has been performed *before* the summation.

This leads to the definition of the following spaces first introduced by J.-Y. Chemin and N. Lerner in [13] then extended in [9].

Definition. For $T > 0$, $s \in \mathbb{R}$, $1 \leq r, \rho \leq \infty$, we set (with the usual convention if $r = \infty$):

$$\|u\|_{\tilde{L}_T^\rho(\dot{B}_{p,r}^s)} := \left(\sum_q (2^{qs} \|\dot{\Delta}_q u\|_{L_T^\rho(L^p)})^r \right)^{\frac{1}{r}}.$$

We then define the space $\tilde{L}_T^\rho(\dot{B}_{p,r}^s)$ as the set of temperate distributions u over $(0, T) \times \mathbb{R}^N$ such that $\lim_{q \rightarrow -\infty} \dot{\Delta}_q u = 0$ in $\mathcal{S}'(0, T \times \mathbb{R}^N)$ and $\|u\|_{\tilde{L}_T^\rho(\dot{B}_{p,r}^s)} < \infty$.

In a similar way, we set

$$\|u\|_{\tilde{L}^\rho(\dot{B}_{p,r}^s)} := \left(\sum_q (2^{qs} \|\dot{\Delta}_q u\|_{L^\rho(\mathbb{R}^+; L^p)})^r \right)^{\frac{1}{r}},$$

and define the space $\tilde{L}^\rho(\dot{B}_{p,r}^s)$ as the set of temperate distributions u over $\mathbb{R}^+ \times \mathbb{R}^N$ such that $\lim_{q \rightarrow -\infty} \dot{\Delta}_q u = 0$ in $\mathcal{S}'(\mathbb{R}^+ \times \mathbb{R}^N)$ and $\|u\|_{\tilde{L}^\rho(\dot{B}_{p,r}^s)} < \infty$.

Remark 2.2.1. (i) According to Minkowski inequality, we have

$$\|u\|_{\tilde{L}_T^\rho(\dot{B}_{p,r}^s)} \leq \|u\|_{L_T^\rho(\dot{B}_{p,r}^s)} \quad \text{if } r \geq \rho, \quad \|u\|_{\tilde{L}_T^\rho(\dot{B}_{p,r}^s)} \geq \|u\|_{L_T^\rho(\dot{B}_{p,r}^s)} \quad \text{if } r \leq \rho.$$

(ii) All the properties of continuity for the product, composition, remainder and paraproduct stated in chapter 1 may be easily generalized to the spaces $\tilde{L}_T^\rho(\dot{B}_{p,r}^s)$. The general principle is that the time exponent ρ behaves according to Hölder inequality.

For instance, we have the following tame estimate :

$$\|uv\|_{\tilde{L}_T^\rho(\dot{B}_{p,r}^s)} \lesssim \|u\|_{L_T^{\rho_1}(L^\infty)} \|v\|_{\tilde{L}_T^{\rho_2}(\dot{B}_{p,r}^s)} + \|v\|_{L_T^{\rho_3}(L^\infty)} \|u\|_{\tilde{L}_T^{\rho_4}(\dot{B}_{p,r}^s)}$$

whenever $s > 0$, $1 \leq p \leq \infty$, $1 \leq \rho, \rho_1, \rho_2, \rho_3, \rho_4 \leq \infty$ and

$$\frac{1}{\rho} = \frac{1}{\rho_1} + \frac{1}{\rho_2} = \frac{1}{\rho_3} + \frac{1}{\rho_4}.$$

Remark. Of course similar definitions may be given in the nonhomogeneous framework, leading to some functional spaces denoted by $\tilde{L}_T^\rho(B_{p,r}^s)$. The details are left to the reader.

2.2.1 Spectral localization

In this section, we prove estimates for the semi-group of the heat equation restricted to functions with compact supports away from the origin in Fourier variables. These estimates are based on the following result.

Lemma 2.2.2. Let ϕ be a smooth function supported in the shell $\mathcal{C}(0, R_1, R_2)$ with $0 < R_1 < R_2$. There exist two positive constants κ and C depending only on ϕ and such that for all $1 \leq p \leq \infty$, $\tau \geq 0$ and $\lambda > 0$, we have

$$\|\phi(\lambda^{-1}D)e^{\tau\Delta}u\|_{L^p} \leq Ce^{-\kappa\tau\lambda^2} \|\phi(\lambda^{-1}D)u\|_{L^p}.$$

Proof: Performing a change of variable, one can assume with no loss of generality that $\lambda = 1$.

Now, let $\tilde{\phi}$ be a smooth function supported in $\mathcal{C}(0, R'_1, R'_2)$ for some $R'_1 < R_1$ and $R'_2 > R_2$ and such that $\tilde{\phi} \equiv 1$ in a neighborhood of $\mathcal{C}(0, R_1, R_2)$. We have

$$\mathcal{F}(\phi(D)e^{\tau\Delta}u)(\xi) = \left(\tilde{\phi}(\xi)e^{-\tau|\xi|^2}\right) \mathcal{F}(\phi(D)u)(\xi).$$

Thus, $\phi(D)e^{\tau\Delta}u = k_\tau \star \phi(D)u$ with

$$k_\tau(z) := (2\pi)^{-N} \int_{\mathbb{R}^N} e^{-\tau|\xi|^2} e^{iz \cdot \xi} \tilde{\phi}(\xi) d\xi.$$

According to convolution inequalities, we have

$$\|\phi(D)e^{\tau\Delta}u\|_{L^p} \leq \|k_\tau\|_{L^1} \|\phi(D)u\|_{L^p}.$$

Therefore it only remains to prove that there exist two positive constants κ and C such that

$$(2.4) \quad \forall \tau \in \mathbb{R}^+, \quad \|k_\tau\|_{L^1} \leq Ce^{-\kappa\tau}.$$

For that, we use the fact that for all $m \in \mathbb{N}$, we have

$$\begin{aligned} (1 + |z|^2)^m k_\tau(z) &= (2\pi)^{-N} \int e^{-\tau|\xi|^2} \tilde{\phi}(\xi) (\text{Id} - \Delta_\xi)^m (e^{iz \cdot \xi}) d\xi, \\ &= (2\pi)^{-N} \int e^{iz \cdot \xi} (\text{Id} - \Delta_\xi)^m (e^{-\tau|\xi|^2} \tilde{\phi}(\xi)) d\xi. \end{aligned}$$

From the last equality and the fact that the integration may be restricted to the shell $\mathcal{C}(0, R'_1, R'_2)$, we easily conclude that there exists a constant C_m such that

$$\forall z \in \mathbb{R}^N, (1 + |z|^2)^m |k_\tau(z)| \leq C_m e^{-\kappa\tau},$$

whence inequality (2.4). ■

2.2.2 Estimates for the heat equation

Let us now state our main result for the heat equation.

Theorem 2.2.3. *Let $T > 0$, $s \in \mathbb{R}$ and $1 \leq \rho, p, r \leq \infty$. Assume that $u_0 \in \dot{B}_{p,r}^s$ and $f \in \tilde{L}_T^\rho(\dot{B}_{p,r}^{s-2+\frac{2}{\rho}})$. Then (H) has a unique solution u in $\tilde{L}_T^\rho(\dot{B}_{p,r}^{s+\frac{2}{\rho}}) \cap \tilde{L}_T^\infty(\dot{B}_{p,r}^s)$ and there exists a constant C depending only on N and such that for all $\rho_1 \in [\rho, +\infty]$, we have*

$$\mu^{\frac{1}{\rho_1}} \|u\|_{\tilde{L}_T^{\rho_1}(\dot{B}_{p,r}^{s+\frac{2}{\rho_1}})} \leq C \left(\|u_0\|_{\dot{B}_{p,r}^s} + \mu^{\frac{1}{\rho}-1} \|f\|_{\tilde{L}_T^\rho(\dot{B}_{p,r}^{s-2+\frac{2}{\rho}})} \right).$$

If in addition r is finite then u belongs to $\mathcal{C}([0, T]; \dot{B}_{p,r}^s)$.

Proof: Since u_0 and f are temperate distributions, equation (H) has a unique solution u in $\mathcal{S}'(0, T \times \mathbb{R}^N)$, which satisfies

$$\hat{u}(t, \xi) = e^{-\mu t |\xi|^2} \hat{u}_0(\xi) + \int_0^t e^{-\mu(t-\tau)} \hat{f}(\tau, \xi) d\tau.$$

As $\mathcal{F}(\dot{S}_q u_0)$ (resp. $\mathcal{F}(\dot{S}_q f)$) tends to 0 in $\mathcal{S}'(\mathbb{R}^N)$ (resp. $\mathcal{S}'(0, T \times \mathbb{R}^N)$) when q goes to $-\infty$, we easily gather that also $\dot{S}_q u$ goes to zero in $\mathcal{S}'(0, T \times \mathbb{R}^N)$ when q goes to $-\infty$.

Next, we notice that, applying $\dot{\Delta}_q$ to (H) and using formula (2.3) yields

$$\dot{\Delta}_q u(t) = e^{\mu t \Delta} \dot{\Delta}_q u_0 + \int_0^t e^{\mu(t-\tau) \Delta} \dot{\Delta}_q f(\tau) d\tau.$$

Therefore,

$$\left\| \dot{\Delta}_q u(t) \right\|_{L^p} \leq \left\| e^{\mu t \Delta} \dot{\Delta}_q u_0 \right\|_{L^p} + \int_0^t \left\| e^{\mu(t-\tau) \Delta} \dot{\Delta}_q f(\tau) \right\|_{L^p} d\tau.$$

By virtue of lemma 2.2.2, we thus have for some $\kappa > 0$,

$$\left\| \dot{\Delta}_q u(t) \right\|_{L^p} \lesssim e^{-\kappa \mu 2^{2q} t} \left\| \dot{\Delta}_q u_0 \right\|_{L^p} + \int_0^t e^{-\kappa \mu 2^{2q}(t-\tau)} \left\| \dot{\Delta}_q f(\tau) \right\|_{L^p} d\tau.$$

Applying convolution inequalities, we get

$$\left\| \dot{\Delta}_q u \right\|_{L_T^{\rho_1}(L^p)} \lesssim \left(\frac{1 - e^{-\kappa \mu T \rho_1 2^{2q}}}{\kappa \mu \rho_1 2^{2q}} \right)^{\frac{1}{\rho_1}} \left\| \dot{\Delta}_q u_0 \right\|_{L^p} + \left(\frac{1 - e^{-\kappa \mu T \rho_2 2^{2q}}}{\kappa \mu \rho_2 2^{2q}} \right)^{\frac{1}{\rho_2}} \left\| \dot{\Delta}_q f \right\|_{L_T^\rho(L^p)}$$

with $1/\rho_2 = 1 + 1/\rho_1 - 1/\rho$.

Finally, taking the $\ell^r(\mathbb{Z})$ norm, we conclude that (with the usual convention if $r = +\infty$)

$$(2.5) \quad \|u\|_{\tilde{L}_T^{\rho_1}(\dot{B}_{p,r}^{s+\frac{2}{\rho_1}})} \lesssim \left[\sum_q \left(\frac{1 - e^{-\kappa \mu T \rho_1 2^{2q}}}{\kappa \mu \rho_1 2^{2q}} \right)^{\frac{r}{\rho_1}} (2^{qs} \left\| \dot{\Delta}_q u_0 \right\|_{L^p})^r \right]^{\frac{1}{r}} \\ + \left[\sum_q \left(\frac{1 - e^{-\kappa \mu T \rho_2 2^{2q}}}{\kappa \mu \rho_2 2^{2q}} \right)^{\frac{1}{\rho_2}} (2^{q(s-2+\frac{2}{\rho})} \left\| \dot{\Delta}_q f \right\|_{L_T^\rho(L^p)})^r \right]^{\frac{1}{r}},$$

which insures that $u \in \tilde{L}_T^\rho(B_{p,r}^{s+\frac{2}{p}}) \cap \tilde{L}_T^\infty(\dot{B}_{p,r}^s)$ and yields the desired inequality.

That u belongs to $\mathcal{C}([0, T]; \dot{B}_{p,r}^s)$ in the case where r is finite may be easily deduced from the density of $\mathcal{S} \cap \dot{B}_{p,r}^s$ in $\dot{B}_{p,r}^s$ (see proposition 1.5.9). $\blacksquare$

Remark. In the case $u_0 \equiv 0$ (resp. $f \equiv 0$) the use of $\tilde{L}_T^\rho(\dot{B}_{p,r}^s)$ spaces enables us to gain *two* derivatives for u compare to the regularity of f (resp. u_0). Remind that the classical result of maximal regularity for the heat equation which states that $D^2u \in L^p(0, T \times \mathbb{R}^N)$ whenever $u_0 \in B_{p,p}^{2-\frac{2}{p}}$ and $f \in L^p(0, T \times \mathbb{R}^N)$ breaks down for $p = 1$.

2.2.3 A counterexample

In this section, we aim at convincing the reader that the results stated in the previous section are optimal and that the appearance of $\tilde{L}_T^\rho(B_{p,r}^s)$ spaces is not a technical artifact.

Let us consider the simple case of the free heat equation $\partial_t u - \Delta u = 0$ with initial data u_0 in L^2 . From (2.2), it is easy to see that the function $(t, x) \mapsto tD^2u(t, x)$ belongs to $L^1(\mathbb{R}^+; L^2)$. Can we expect u to be in $L_{loc}^1(\mathbb{R}^+; H^2)$? The answer is negative. One can even state a more accurate result:

Proposition 2.2.4. *Let u_0 be in $\mathcal{S}'_h$. The solution to (H) with $f \equiv 0$ belongs to $L^1(\mathbb{R}^+; \dot{H}^2)$ if and only if u_0 is in $\dot{B}_{2,1}^0$.*

Proof: If u_0 is in $\dot{B}_{2,1}^0$ then theorem 2.2.3 states that u belongs to $\tilde{L}^1(\dot{B}_{2,1}^2)$. As $\tilde{L}^1(\dot{B}_{2,1}^2) = L^1(\mathbb{R}^+; \dot{B}_{2,1}^2)$ and $\dot{B}_{2,1}^2 \hookrightarrow \dot{H}^2$, we thus have $u \in L^1(\mathbb{R}^+; \dot{H}^2)$ as expected.

Conversely, let us assume that $u \in L^1(\mathbb{R}^+; \dot{H}^2)$. According to Parseval formula and to (2.1), we thus have

$$I := \int_0^{+\infty} \left(\int_{\mathbb{R}^N} |\xi|^4 e^{-2t|\xi|^2} |\widehat{u_0}(\xi)|^2 d\xi \right)^{\frac{1}{2}} dt < \infty.$$

Now, denoting $c_q := \left(\int_{2^q \leq |\xi| \leq 2^{q+1}} |\widehat{u_0}(\xi)|^2 d\xi \right)^{\frac{1}{2}}$, easy computations yield

$$I \geq \sum_{p \in \mathbb{Z}} \int_{4^{-p}}^{4^{1-p}} \left(\sum_{q \in \mathbb{Z}} 2^{4q} e^{-2^{2(q-p)}} c_q^2 \right)^{\frac{1}{2}} dt.$$

Keeping only the term $q = p$ in the second sum, we end up with

$$I \geq \alpha \sum_{p \in \mathbb{Z}} c_p$$

for some positive constant α .

Hence $\sum c_q$ has to be finite. In other words, u_0 has to belong to $B_{2,1}^0$. $\blacksquare$

2.2.4 Estimates in nonhomogeneous Besov spaces, and the periodic case

One can wonder if the estimates stated in theorem 2.2.3 remain true in *nonhomogeneous* Besov spaces. On one hand, the block $\Delta_{-1}u$ corresponding to the low frequencies of u cannot be handled by mean of lemma 2.2.2. On the other hand, by using the representation formula (2.2), we easily get

$$\|\Delta_{-1}u(T)\|_{L^p} \leq \|\Delta_{-1}u_0\|_{L^p} + \int_0^T \|\Delta_{-1}f\|_{L^p} dt,$$

whence, if $1 \leq \rho \leq \rho_1 \leq \infty$,

$$\|\Delta_{-1}u\|_{L_T^{\rho_1}(L^p)} \leq T^{\frac{1}{\rho_1}} \|\Delta_{-1}u_0\|_{L^p} + T^{1+\frac{1}{\rho_1}-\frac{1}{\rho}} \|\Delta_{-1}f\|_{L_T^{\rho}(L^p)}.$$

Of course the other dyadic blocks may be treated as in the homogeneous case. We end up with the following statement:

Theorem 2.2.5. *Let $s \in \mathbb{R}$ and $1 \leq \rho, p, r \leq \infty$. Let $T > 0$, $u_0 \in B_{p,r}^s$ and $f \in \tilde{L}_T^{\rho}(B_{p,r}^{s-2+\frac{2}{\rho}})$. Then (H) has a unique solution u in $\tilde{L}_T^{\rho}(B_{p,r}^{s+\frac{2}{\rho}}) \cap \tilde{L}_T^{\infty}(B_{p,r}^s)$ and there exists a constant C depending only on N and such that for all $\rho_1 \in [\rho, +\infty]$, we have*

$$\mu^{\frac{1}{\rho_1}} \|u\|_{\tilde{L}_T^{\rho_1}(B_{p,r}^{s+\frac{2}{\rho_1}})} \leq C \left((1 + T^{\frac{1}{\rho_1}}) \|u_0\|_{B_{p,r}^s} + (1 + T^{1+\frac{1}{\rho_1}-\frac{1}{\rho}}) \mu^{\frac{1}{\rho}-1} \|f\|_{\tilde{L}_T^{\rho}(B_{p,r}^{s-2+\frac{2}{\rho}})} \right).$$

If in addition r is finite then u belongs to $\mathcal{C}([0, T]; B_{p,r}^s)$.

Remark. Compare to the homogeneous case, the constant appearing in the above inequality now depends on T . When dealing with nonlinear parabolic PDE's, this may preclude from proving global in time existence results (even for small data). Avoiding the time dependence is thus a good motivation for using homogeneous spaces when dealing with such PDE's.

Let us finally make a short remark concerning the periodic case. In the case where u_0 and f are periodic and have zero average, it is clear that the unique solution u to (H) has also zero average. The above analysis may be carried out and leads again to the estimates stated in theorem 2.2.3.

In the general case where u_0 and f need not have zero average, the above estimates hold for $u - |\mathbb{T}_a^N|^{-1} \int_{\mathbb{T}_a^N} u \, dx$ and the study of $\int_{\mathbb{T}} u \, dx$ has to be done separately.

2.3 Optimal well-posedness results for Navier-Stokes equations

In this section, we give an example of application of theorem 2.2.3 for solving a nonlinear PDE's related to the heat equation. We focus on the incompressible Navier-Stokes equations which have been extensively studied recently. It goes without saying that the same method works for a great deal of semi-linear heat equations.

2.3.1 The model

The incompressible Navier-Stokes equations write

$$(NS) \quad \begin{cases} \partial_t u + u \cdot \nabla u - \mu \Delta u + \nabla \Pi = f, \\ \operatorname{div} u = 0, \\ u|_{t=0} = u_0. \end{cases}$$

Above $u = u(t, x)$ stands for the unknown velocity field which is a time dependent vectorfield and the scalar function $\Pi = \Pi(t, x)$ stands for the pressure. From a mathematical viewpoint, $\nabla \Pi$ is the Lagrange multiplier associated to the divergence free constraint. The initial velocity u_0 and the external force f are given functions. It is understood that the convective term $u \cdot \nabla u$ stands for the vector field whose i -th entry is $\sum_{j=1}^N u^j \partial_j u^i$. Finally, the parameter μ (the viscosity) has to be positive¹.

¹By performing the change of unknown function $u(t, x) = v(t, x/\sqrt{\mu})$, one can restrict the study to the case $\mu = 1$, an assumption which is made in most of papers devoted to Navier-Stokes equations. Since we aim at keeping track of the dependence with respect to μ , we will *not* perform this change of function.

Note that, strictly speaking, (NS) does not enter in the class of semi-linear parabolic equations. Since we will consider only the case where the space variable x belongs to the whole space however, we shall see in a moment that (NS) may be reduced to a (system of) semilinear heat equations with nonlocal nonlinearity.

We notice that for suitably smooth divergence-free vectorfields, the convective term rewrites $\operatorname{div} u \otimes u$. This leads to the following definition of weak solution:

Definition. A distribution $u \in \mathcal{S}'(0, T \times \mathbb{R}^N; \mathbb{R}^N)$ is called a *weak solution* of (NS) if $\operatorname{div} u = 0$ in $\mathcal{S}'(0, T \times \mathbb{R}^N; \mathbb{R})$ and if ²

$$\int_0^T \int_{\mathbb{R}^N} u \cdot \partial_t \varphi \, dx dt + \int_0^T \int_{\mathbb{R}^N} u^i u^j \partial_j \varphi_i \, dx dt - \mu \int_0^T \int_{\mathbb{R}^N} \nabla u \cdot \nabla \varphi \, dx dt + \int_0^T \int_{\mathbb{R}^N} f \cdot \varphi \, dx dt = \int_{\mathbb{R}^N} u_0 \varphi(0) \, dx$$

for all divergence free φ in $\mathcal{C}_c^\infty([0, T]; \mathcal{S}(\mathbb{R}^N; \mathbb{R}^N))$.

If u is a smooth enough weak solution of (NS), taking $\varphi = u$ in the above relation yields the following energy equality:

$$(2.6) \quad \|u(t)\|_{L^2}^2 + 2\mu \int_0^t \|Du(\tau)\|_{L^2}^2 \, d\tau = \|u_0\|_{L^2}^2 + 2 \int_0^t \int f \cdot u \, dx \, dt.$$

Taking advantage of (2.6) and of compactness arguments, J. Leray in 1934 proved the existence of global weak solutions with bounded energy in the case where $f \equiv 0$. Let us state his main result (see [33] for more details):

Theorem (Leray). *Let $N \geq 2$. For all divergence free vectorfield u_0 with coefficients in L^2 , system (NS) with $f \equiv 0$ has a global weak solution u in $L^\infty(\mathbb{R}^+; L^2)$ with $Du \in L^2(\mathbb{R}^+; L^2)$, which satisfies*

$$(2.7) \quad \|u(t)\|_{L^2}^2 + 2\mu \int_0^t \|Du(\tau)\|_{L^2}^2 \, d\tau \leq \|u_0\|_{L^2}^2.$$

If $N = 2$, then (2.6) is true. Besides the solution u belongs to $\mathcal{C}(\mathbb{R}^+; L^2)$ and is unique in the set of divergence-free vector fields with coefficients in $L^\infty(\mathbb{R}^+; L^2)$ and gradient in $L^2(\mathbb{R}^+; L^2)$.

Since the work by J. Leray in 1934, the problem of uniqueness in the energy space when $N \geq 3$ has remained unsolved. Considering smoother data and restricting the set of admissible solutions is the usual way to get existence and uniqueness results. Again, this has been first noticed by J. Leray in [33]:

Theorem (Leray). *Let $N = 3$. There exists a positive constant c such that for all divergence-free vector field u_0 with coefficients in H^1 which satisfies*

$$\|u_0\|_{L^2} \|\nabla u_0\|_{L^2} \leq c\mu^2 \quad \text{or} \quad \|u_0\|_{L^2}^2 \|u_0\|_{L^\infty} \leq c\mu^3,$$

system (NS) with no external force has a unique global solution $u \in \mathcal{C}(\mathbb{R}^+; H^1)$ which also satisfies $D^2 u \in L^2(\mathbb{R}^+; L^2)$.

One can alternately consider large data in H^1 . One still obtain existence and uniqueness of a solution but for *small time* only.

Whether global existence and uniqueness holds true is an outstanding open problem.

²From now on, we adopt the summation convention over repeated indices.

2.3.2 About scaling and critical spaces

In this section, we aim at finding functional spaces E as large as possible for which any vectorfield u_0 in E generates a unique solution on a small time interval. If one restricts to the Sobolev spaces framework, the optimal result is due to H. Fujita and T. Kato in [28]. Their original statement pertains to the three-dimensional case in bounded domains. The statement below in dimension $N = 2, 3$ has been proved by J.-Y. Chemin in [9].

Theorem (Fujita-Kato). *Let u_0 be a solenoidal vector-field with coefficients in $\dot{H}^{\frac{N}{2}-1}$. Let f have coefficients in $L^2(0, T; \dot{H}^{\frac{N}{2}-2})$. There exists a positive time T such that (NS) has a unique solution in $\mathcal{C}([0, T]; \dot{H}^{\frac{N}{2}-1})$ with gradient in $L^2(0, T; \dot{H}^{\frac{N}{2}-1})$.*

Moreover, there exists a constant c depending only on N and such that (NS) has a unique global solution in $\mathcal{C}_b(\mathbb{R}^+; \dot{H}^{\frac{N}{2}-1})$ with gradient in $L^2(\mathbb{R}^+; \dot{H}^{\frac{N}{2}-1})$ whenever

$$(2.8) \quad \|u_0\|_{\dot{H}^{\frac{N}{2}-1}} + \mu^{\frac{1}{2}} \|f\|_{L^2(\dot{H}^{\frac{N}{2}-2})} \leq c\mu.$$

Compare to Leray's results, there are two important advances in Fujita and Kato's approach. The first one is that suitably smooth solutions to (NS) may be interpreted as a fixed point of some functional over Banach spaces.

Indeed, let $\mathcal{P}$ denote the orthogonal projector of $L^2(\mathbb{R}^N; \mathbb{R}^N)$ over solenoidal vectorfields. One can easily show that $\mathcal{P}$ is the 0 order multiplier defined by $\mathcal{P} := \text{Id} - \nabla(-\Delta)^{-1} \text{div}$. In other words, in Fourier variables, we have for any u with coefficients in L^2 ,

$$\widehat{\mathcal{P}u}(\xi) = \widehat{u}(\xi) - \frac{\xi \cdot \widehat{u}(\xi)}{|\xi|^2} \xi.$$

Note that the above definition may be extended to distributions of $\mathcal{S}'_h$.

Now, we remark that Navier-Stokes equations rewrite

$$(\widetilde{NS}) \quad \partial_t u - \mathcal{P} \text{div}(u \otimes u) - \mu \Delta u = \mathcal{P}f, \quad u|_{t=0} = u_0.$$

Hence, by virtue of (2.3), we have at least formally,

$$(2.9) \quad u(t) = e^{\mu t \Delta} u_0 + \int_0^t e^{\mu(t-\tau)\Delta} \mathcal{P}f(\tau) d\tau - \int_0^t e^{\mu(t-\tau)\Delta} \mathcal{P} \text{div}(u(\tau) \otimes u(\tau)) d\tau.$$

Any solution of (NS) which satisfies (2.9) is called (after the theory of analytic semi-groups) a *mild solution* of Navier-Stokes equations. One can prove that any suitably smooth mild solution is also a weak solution to (NS) (see [32] for a detailed study).

Thus solving (NS) amounts to finding a fixed point for the functional $v \mapsto \Phi(v)$ defined by

$$\Phi(v)(t) := e^{\mu t \Delta} u_0 + \int_0^t e^{\mu(t-\tau)\Delta} \mathcal{P}((f - \text{div}(v \otimes v))(\tau)) d\tau.$$

The second advance in Fujita-Kato's approach has to do with the choice of the functional framework for the solution u and for the data.

The idea is that when appealing to contracting mapping arguments for solving $(\widetilde{NS})$, it is suitable that u belongs to a functional space X such that the linear term $\partial_t u - \mu \Delta u$ and the nonlinear term $\mathcal{P} \text{div}(u \otimes u)$ have the same regularity.

This may be interpreted in terms of *scaling invariance* for $(\widetilde{NS})$. Indeed, we notice that if u solves $(\widetilde{NS})$ with data u_0 and f , so does for any $\lambda > 0$ the vectorfield

$$u_\lambda : (t, x) \mapsto \lambda u(\lambda^2 t, \lambda x)$$

with data

$$u_{0,\lambda} : x \mapsto \lambda u(\lambda x) \quad \text{and} \quad f_\lambda : (t, x) \mapsto \lambda^3 f(\lambda^2 t, \lambda x).$$

It is also clear that a space where the heat and convective terms have the same regularity must have a norm invariant for all λ by the change $u \mapsto u_\lambda$. This leads to the following definition:

Definition. A *critical space* for initial data is any Banach space $E \subset \mathcal{S}'(\mathbb{R}^N)$ whose norm is invariant for all λ by $u_0 \mapsto u_{0,\lambda}$.

A critical space for external forces is any Banach space $F \subset \mathcal{S}'(\mathbb{R}^+ \times \mathbb{R}^N)$ whose norm is invariant for all λ by $f \mapsto f_\lambda$.

A critical space for solutions to (NS) is any Banach space $X \subset \mathcal{S}'(\mathbb{R}^+ \times \mathbb{R}^N)$ whose norm is invariant for all λ by $u \mapsto u_\lambda$.

One can easily check that Fujita-Kato's theorem enters in this framework. Indeed, $\dot{H}^{\frac{N}{2}-1}$ is invariant by $u_0 \mapsto u_{0,\lambda}$, the space $L^2(\mathbb{R}^+; \dot{H}^{\frac{N}{2}-2})$ is invariant by $f \mapsto f_\lambda$ and the space of divergence-free vector-fields with coefficients in $\mathcal{C}(\mathbb{R}^+; \dot{H}^{\frac{N}{2}-1})$ and gradient in $L^2(\mathbb{R}^+; \dot{H}^{\frac{N}{2}-1})$ is invariant by $u \mapsto u_\lambda$.

2.3.3 Global well-posedness for small data

In the present section, we investigate the problem of existence of global (mild) solutions to Navier-Stokes equations for small data which belong to functional spaces which are invariant by the scaling exhibited in the previous section.

Our main result is the following:

Theorem 2.3.1. *Let $1 \leq r \leq \infty$ and $1 \leq p < \infty$. There exists a constant $c > 0$ independent of μ such that for all divergence free vector-field u_0 with coefficients in $\dot{B}_{p,r}^{\frac{N}{p}-1}$ and external force f with coefficients in $\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}-1})$ such that*

$$(2.10) \quad \|u_0\|_{\dot{B}_{p,r}^{\frac{N}{p}-1}} + \|\mathcal{P}f\|_{\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}-1})} < c\mu,$$

system (NS) has a unique solution u in $\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}+1}) \cap \tilde{L}^\infty(\dot{B}_{p,r}^{\frac{N}{p}-1})$ which satisfies

$$(2.11) \quad \|u\|_{\tilde{L}^\infty(\dot{B}_{p,r}^{\frac{N}{p}-1})} + \mu \|u\|_{\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}+1})} < 2c\mu.$$

Besides, if r is finite then u belongs to $\mathcal{C}(\mathbb{R}^+; \dot{B}_{p,r}^{\frac{N}{p}-1})$ and uniqueness holds true in $\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}+1}) \cap \tilde{L}^\infty(\dot{B}_{p,r}^{\frac{N}{p}-1})$ with no smallness condition.

According to proposition 1.5.4, all the spaces $\dot{B}_{p,r}^{\frac{N}{p}-1}$ are scaling invariant for the Navier-Stokes equations. Besides, we have $\dot{H}^{\frac{N}{2}-1} = \dot{B}_{2,2}^{\frac{N}{2}-1}$. Hence theorem 2.3.1 is a natural generalization of Fujita and Kato's theorem.

Well-posedness in critical Besov spaces has been first proved by M. Cannone in [7] for $3 < p \leq 6$ and $r = \infty$, then extended in [8] and [35]. The idea of using $\tilde{L}_T^\rho(\dot{B}_{p,r}^s)$ spaces for solving Navier-Stokes equations is due to J.-Y. Chemin in [11]. Related results in more general functional spaces have been proved in [31]. Critical spaces in which global well-posedness for small data may be proved with the method below have been characterized in [2].

The proof of theorem 2.3.1 is based on the following lemma.

Lemma 2.3.2. *Let $(X, \|\cdot\|_X)$ be a Banach space and $B : X \times X \rightarrow X$ be a bilinear continuous operator with norm K . Then for all $y \in X$ such that $4K\|y\|_X < 1$, equation $x = y + B(x, x)$ has a unique solution x in the ball $\mathcal{B}(0, \frac{1}{2K})$. Besides x satisfies $\|x\|_X \leq 2\|y\|_X$.*

Proof: We rule out the case $K = 0$ which is obvious. Now, the result is a mere consequence of the contracting mapping theorem.

Indeed, let $R := 4K\|y\|_X$ and $F : x \mapsto y + B(x, x)$. On one hand, F maps $\overline{\mathcal{B}}(0, \frac{R}{2K})$ into itself provided that $R \leq 1$. On the other hand, for all $(x, x') \in \overline{\mathcal{B}}(0, \frac{R}{2K})^2$, we have

$$\|F(x') - F(x)\|_X \leq R\|x' - x\|_X.$$

Hence F is a contracting mapping on $\overline{\mathcal{B}}(0, \frac{R}{2K})$ whenever $R < 1$, which insures the existence of x .

Routine computations then lead to $\|x\|_X \leq 2\|y\|_X$ and to uniqueness in $\mathcal{B}(0, \frac{1}{2K})$. ■

Let us now prove theorem 2.3.1:

First step: Existence

The existence of a solution for (NS) with data u_0 and f will be obtained by applying lemma 2.3.2 for convenient y , B and $(X, \|\cdot\|_X)$.

For X , we shall take the space of divergence free distributions over $\mathbb{R}^+ \times \mathbb{R}^N$ with coefficients in $\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}+1}) \cap \tilde{L}^\infty(\dot{B}_{p,r}^{\frac{N}{p}-1})$ endowed with the norm

$$\|v\|_X := \|v\|_{\tilde{L}^\infty(\dot{B}_{p,r}^{\frac{N}{p}-1})} + \mu\|v\|_{\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}+1})}.$$

We then set $y : t \mapsto e^{\mu t \Delta} u_0 + \int_0^t e^{\mu(t-\tau)\Delta} \mathcal{P} f d\tau$ and define the bilinear functional B by the formula

$$B(v, w)(t) = - \int_0^t e^{\mu(t-\tau)\Delta} \mathcal{P} \operatorname{div}(v(\tau) \otimes w(\tau)) d\tau.$$

We claim that y belongs to X , that B maps $X \times X$ in X and that there exists some constant C such that

$$(2.12) \quad \|y\|_X \leq C(\|u_0\|_{\dot{B}_{p,r}^{\frac{N}{p}-1}} + \|\mathcal{P} f\|_{\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}-1})}),$$

$$(2.13) \quad \forall (v, w) \in X^2, \|B(v, w)\|_X \leq C\mu^{-1}\|v\|_X\|w\|_X.$$

Indeed, as u_0 is divergence free and belongs to $\dot{B}_{p,r}^{\frac{N}{p}-1}$, and as f is in $\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}-1})$, theorem 2.2.3 insures that y belongs to X and satisfies (2.12).

Next, using Bony's decomposition and $\operatorname{div} v = \operatorname{div} w = 0$, one can write

$$\operatorname{div}(v \otimes w) = \dot{T}_{\partial_j v} w^j + \dot{T}_{w^j} \partial_j v + \partial_j \dot{R}(v, w^j)$$

with the summation convention over repeated indices.

Hence, combining propositions 1.4.1 and 1.4.2, remark 2.2.1 and the embedding $\tilde{L}^\rho(\dot{B}_{p,r}^{\frac{N}{p}-\frac{1}{2}}) \hookrightarrow \tilde{L}^\rho(\dot{B}_{\infty,\infty}^{-\frac{1}{2}})$ for $\rho = 4/3$ or $\rho = 4$, we get

$$\|\operatorname{div}(v \otimes w)\|_{\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}-1})} \leq C\|v\|_{\tilde{L}^{\frac{4}{3}}(\dot{B}_{p,r}^{\frac{N}{p}+\frac{1}{2}})}\|w\|_{\tilde{L}^4(\dot{B}_{p,r}^{\frac{N}{p}-\frac{1}{2}})}.$$

Putting forward complex interpolation, we have

$$\begin{aligned} \|v\|_{\tilde{L}^{\frac{4}{3}}(\dot{B}_{p,r}^{\frac{N}{p}+\frac{1}{2}})} &\leq \|v\|_{\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}+1})}^{\frac{3}{4}} \|v\|_{\tilde{L}^\infty(\dot{B}_{p,r}^{\frac{N}{p}-1})}^{\frac{1}{4}} \leq \mu^{-\frac{3}{4}} \|v\|, \\ \|w\|_{\tilde{L}^4(\dot{B}_{p,r}^{\frac{N}{p}-\frac{1}{2}})} &\leq \|w\|_{\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}+1})}^{\frac{1}{4}} \|w\|_{\tilde{L}^\infty(\dot{B}_{p,r}^{\frac{N}{p}-1})}^{\frac{3}{4}} \leq \mu^{-\frac{1}{4}} \|w\|, \end{aligned}$$

whence

$$\|\operatorname{div}(v \otimes w)\|_{\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}-1})} \leq C\mu^{-1} \|v\|_X \|w\|_X.$$

Finally, by using the fact that $\mathcal{P}$ is an homogeneous multiplier of degree 0, and by applying theorem 2.2.3, we conclude that $B(v, w)$ belongs to X and that (2.13) is satisfied.

Now, lemma 2.3.2 may be applied provided that $4C\|y\|_X < \mu$. According to (2.12) this condition will be satisfied if

$$\|u_0\|_{\dot{B}_{p,r}^{\frac{N}{p}-1}} + \|\mathcal{P}f\|_{\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}-1})} < c\mu$$

for some small enough constant c .

This achieves the proof of existence of a global solution in X for (NS), and of uniqueness under condition (2.11).

Second step: Uniqueness in the case where r is finite

Let u_1 and u_2 be two solutions of (NS) in X . Denoting $\delta u := u_2 - u_1$, we have for all $t \in \mathbb{R}^+$,

$$\delta u(t) = - \int_0^t e^{\mu(t-\tau)\Delta} \mathcal{P} \operatorname{div}(\delta u \otimes u^2 + u^1 \otimes \delta u) d\tau.$$

By going along the lines of the proof of (2.13), one can easily check that for all positive T and divergence free (v, w) in $\tilde{L}_T^{\frac{4}{3}}(\dot{B}_{p,r}^{\frac{N}{p}+\frac{1}{2}}) \times \tilde{L}_T^4(\dot{B}_{p,r}^{\frac{N}{p}-\frac{1}{2}})$, we have

$$\|\mathcal{P} \operatorname{div}(v \otimes w)\|_{\tilde{L}_T^1(\dot{B}_{p,r}^{\frac{N}{p}-1})} \leq C \|v\|_{\tilde{L}_T^{\frac{4}{3}}(\dot{B}_{p,r}^{\frac{N}{p}+\frac{1}{2}})} \|w\|_{\tilde{L}_T^4(\dot{B}_{p,r}^{\frac{N}{p}-\frac{1}{2}})}.$$

Hence, denoting X_T the space of functions of X restricted to $[0, T] \times \mathbb{R}^N$, and applying theorem 2.2.3, we get with obvious notation,

$$\|\delta u\|_{X_T} \leq C \left(\|\delta u\|_{\tilde{L}_T^{\frac{4}{3}}(\dot{B}_{p,r}^{\frac{N}{p}+\frac{1}{2}})} \|u_2\|_{\tilde{L}_T^4(\dot{B}_{p,r}^{\frac{N}{p}-\frac{1}{2}})} + \|u_1\|_{\tilde{L}_T^{\frac{4}{3}}(\dot{B}_{p,r}^{\frac{N}{p}+\frac{1}{2}})} \|\delta u\|_{\tilde{L}_T^4(\dot{B}_{p,r}^{\frac{N}{p}-\frac{1}{2}})} \right).$$

Using complex interpolation, we conclude that

$$(2.14) \quad \|\delta u\|_{X_T} \leq Z(T) \|\delta u\|_{X_T}$$

with

$$Z(T) := C \left(\mu^{-\frac{3}{4}} \|u_2\|_{\tilde{L}_T^4(\dot{B}_{p,r}^{\frac{N}{p}-\frac{1}{2}})} + \mu^{-\frac{1}{4}} \|u_1\|_{\tilde{L}_T^{\frac{4}{3}}(\dot{B}_{p,r}^{\frac{N}{p}+\frac{1}{2}})} \right).$$

Now, Lebesgue dominated convergence theorem insures that Z is a continuous nondecreasing function which vanishes at zero. Hence $\delta u \equiv 0$ in X_T for small enough T .

Finally, because the function $t \mapsto \|\delta u\|_{\tilde{L}_t^1(\dot{B}_{p,r}^{\frac{N}{p}-1})}$ is also continuous, a standard connectivity argument enable us to conclude that $\delta u \equiv 0$ on $\mathbb{R}^+ \times \mathbb{R}^N$.

Last step: Continuity

We still assume that r is finite. Let $u \in X$ be a global solution to (NS). We have

$$\partial_t u - \mu \Delta u = \mathcal{P}(f - \operatorname{div}(u \otimes u)) \quad \text{and} \quad u|_{t=0} = u_0.$$

Arguing like in the first step, one easily state that the right-hand side of the first equation belongs to $\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}-1})$. Since r is finite, theorem 2.2.3 insures that $u \in \mathcal{C}(\mathbb{R}^+; \dot{B}_{p,r}^{\frac{N}{p}-1})$. ■

Remark. By going along the lines of the proof of uniqueness, it is easy to prove that, under condition (2.10), the map $(u_0, f) \mapsto u$ with u_0 a divergence-free vectorfield with coefficients in $\dot{B}_{p,r}^{\frac{N}{p}-1}$ and $f \in (\tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}+1}))^N$, is continuous with values in X . Hence, in the case where r is finite, Navier-Stokes equations with small data are globally well-posed in $\dot{B}_{p,r}^{\frac{N}{p}-1}$ in the sense of Hadamard.

2.3.4 Further results

- In the case $r = \infty$, one can prove that if u_0 belongs to the closure $\dot{B}_{p,\infty}^{\frac{N}{p}-1}$ of $\mathcal{S} \cap \dot{B}_{p,\infty}^{\frac{N}{p}-1}$ in $\dot{B}_{p,\infty}^{\frac{N}{p}-1}$ and satisfies (2.10) then system (NS) has a unique solution in $\tilde{L}^1(\dot{B}_{p,\infty}^{\frac{N}{p}+1}) \cap \tilde{L}^\infty(\dot{B}_{p,\infty}^{\frac{N}{p}-1})$ which is also continuous with values in $\dot{B}_{p,\infty}^{\frac{N}{p}-1}$ (see exercise 2.4).
- For any (possibly large) $u_0 \in \dot{B}_{p,r}^{\frac{N}{p}-1}$ with $\operatorname{div} u_0 = 0$, and $f \in \tilde{L}^1(\dot{B}_{p,r}^{\frac{N}{p}-1})$, it is actually possible to prove the existence and uniqueness of a solution on a small time interval.

The reader is referred to exercise 2.7 for more details.

- The space $\dot{B}_{\infty,\infty}^{-1}$ is the largest homogeneous Besov space invariant for the scaling of Navier-Stokes equation. Hence one can wonder whether well-posedness holds true for data in $\dot{B}_{\infty,\infty}^{-1}$. This is unlikely to be true if no additional condition as the solution u is expected to be in $L^\infty(0, T; \dot{B}_{\infty,\infty}^{-1}) \cap \tilde{L}_T^1(\dot{B}_{\infty,\infty}^1)$ only. With such a bad regularity, the meaning of $\operatorname{div}(u \otimes u)$ becomes unclear.

Let us mention however that well-posedness has been proved in [30] for data in the so-called BMO^{-1} space. This space satisfies $\dot{B}_{\infty,1}^{-1} \hookrightarrow BMO^{-1} \hookrightarrow \dot{B}_{\infty,\infty}^{-1}$.

- From the previous section, one has to retain that our approach is not very sensitive to the exact structure of the term $\operatorname{div}(u \otimes u)$. As a matter of fact, the same method would lead to global results for small data (or local results for large data) in $\dot{B}_{p,r}^{\frac{N}{p}-1}$ for any (possibly not physical) Partial or Pseudo Differential Equation having the heat equation as a linear part and a first order term which resembles $\operatorname{div}(u \otimes u)$.

It has been noticed in [34] that the approach presented here works for the scalar equation

$$\partial_t u - \Delta u + |D|(u^2) = 0$$

but that one can find a class of (large) data for which finite time blow-up does occur.

Therefore, one cannot expect our method to give better results for (NS) as far as the very structure of $\operatorname{div}(u \otimes u)$ has not been used.

2.4 Exercises

Exercise 2.1. Let $1 \leq p, r, r_1, r_2, \rho, \rho_1, \rho_2 \leq \infty$ with $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2}$ and $\frac{1}{\rho} = \frac{1}{\rho_1} + \frac{1}{\rho_2}$. Let $\sigma > 0$ and $s \in \mathbb{R}$ be such that

$$s - \sigma < \frac{N}{p} \quad \text{or} \quad s - \sigma = \frac{N}{p} \quad \text{and} \quad r = 1.$$

Prove that for all $T \in (0, +\infty]$, the homogeneous paraproduct $\dot{T}$ maps $\tilde{L}_T^{\rho_1}(\dot{B}_{\infty, r_1}^{-\sigma}) \times \tilde{L}_T^{\rho_2}(\dot{B}_{p_2, r_2}^s)$ in $\tilde{L}_T^\rho(\dot{B}_{p, r}^{s-\sigma})$ continuously.

Exercise 2.2. Let $1 \leq p, p_1, p_2, r, r_1, r_2, \rho, \rho_1, \rho_2 \leq \infty$ with

$$\frac{1}{r} \leq \frac{1}{r_1} + \frac{1}{r_2} \leq 1, \quad \frac{1}{\rho} = \frac{1}{\rho_1} + \frac{1}{\rho_2} \quad \text{and} \quad \frac{1}{p} \leq \frac{1}{p_1} + \frac{1}{p_2} \leq 1.$$

Let $s \in \mathbb{R}$ and $\sigma_{12} := s_1 + s_2 - N \left(\frac{1}{p_1} + \frac{1}{p_2} - \frac{1}{p} \right)$. Assume that

$$\sigma_{12} < \frac{N}{p} \quad \text{or} \quad \sigma_{12} = \frac{N}{p} \quad \text{and} \quad r = 1.$$

Prove that for all $T \in (0, +\infty]$, the homogeneous remainder $\dot{R}$ maps $\tilde{L}_T^{\rho_1}(\dot{B}_{p_1, r_1}^{s_1}) \times \tilde{L}_T^{\rho_2}(\dot{B}_{p_2, r_2}^{s_2})$ in $\tilde{L}_T^\rho(\dot{B}_{p, r}^{\sigma_{12}})$ continuously.

Exercise 2.3. Let $m \in \mathbb{R}$, $0 < R_1 < R_2$ and $1 \leq p \leq \infty$. Let ϕ be a smooth function supported in the shell $\mathcal{C}(0, R_1, R_2)$ and let A be a positive homogeneous function of degree m , smooth on $\mathbb{R}^N \setminus \{0\}$.

- 1) Prove that there exist two positive constants κ and C depending only on ϕ and A , and such that for all $\tau \geq 0$, $\lambda > 0$, we have

$$\left\| \phi(\lambda^{-1}D)e^{-\tau A(D)}u \right\|_{L^p} \leq Ce^{-\kappa\tau\lambda^m} \left\| \phi(\lambda^{-1}D)u \right\|_{L^p}.$$

- 2) Generalize theorem 2.2.3 to the solutions of

$$\partial_t u + A(D)u = f, \quad u|_{t=0} = u_0.$$

Exercise 2.4. Let $\dot{\mathcal{B}}_{p, \infty}^s$ be the closure of $\mathcal{S} \cap \dot{B}_{p, r}^s$ for the norm of $\dot{B}_{p, r}^s$.

Show that for all $u_0 \in \dot{\mathcal{B}}_{p, \infty}^s$ and $f \in \tilde{L}_T^\rho(\dot{\mathcal{B}}_{p, \infty}^{s-2+\frac{2}{\rho}})$, equation (H) has a unique solution u in $\tilde{L}_T^\rho(\dot{\mathcal{B}}_{p, \infty}^{s+\frac{2}{\rho}}) \cap \tilde{L}_T^\infty(\dot{\mathcal{B}}_{p, \infty}^s)$ and that $u \in \mathcal{C}([0, T]; \dot{\mathcal{B}}_{p, \infty}^s)$.

Exercise 2.5. Find counterexamples in $L^\rho(0, T; \dot{B}_{p, r}^s)$ for the gain of two derivatives in ordinary Besov spaces for the nonhomogeneous heat equation.

Hint: use exercise 1.20.

Exercise 2.6. Let $(X, \|\cdot\|)$ be a Banach space, $B : X \times X \rightarrow X$ a bilinear continuous operator with norm K and $L : X \rightarrow X$ a continuous linear operator with norm $M < 1$. Let $y \in X$ satisfy $4K\|y\| < (1 - M)^2$.

Prove that equation $x = y + L(x) + B(x, x)$ has a unique solution in the ball $\mathcal{B}(0, \frac{1-M}{2K})$.

Exercise 2.7. Let $u_0 \in \dot{B}_{p, r}^{\frac{N}{p}-1}$ with $\operatorname{div} u_0 = 0$, and $f \in \tilde{L}^1(\dot{B}_{p, r}^{\frac{N}{p}-1})$. We assume that p and r are finite.

- 1) Let $u_L \in \mathcal{S}'(\mathbb{R}^+ \times \mathbb{R}^N)$ satisfy

$$\partial_t u_L - \mu \Delta u_L = \mathcal{P}f \quad \text{and} \quad u_L(0) = u_0.$$

Prove that $\lim_{T \rightarrow 0} \|u_L\|_{\tilde{L}_T^\rho(\dot{B}_{p,r}^{\frac{N}{p}-1+\frac{2}{\rho}})} = 0$ whenever ρ is finite.

- 2) Check that u is a mild solution of (NS) on $[0, T] \times \mathbb{R}^N$ with data u_0 and f if and only if $u = u_L + \bar{u}$ with

$$\forall t \in [0, T], \quad \bar{u}(t) = - \int_0^t e^{\mu(t-\tau)\Delta} \mathcal{P}(u_L \cdot \nabla u_L + \bar{u} \cdot \nabla u_L + u_L \cdot \nabla \bar{u} + \bar{u} \cdot \nabla \bar{u}).$$

- 3) Prove that there exists a positive T such that (NS) has a unique solution u on $[0, T]$ in $\tilde{L}_T^\infty(\dot{B}_{p,r}^{\frac{N}{p}-1}) \times \tilde{L}_T^1(\dot{B}_{p,r}^{\frac{N}{p}+1})$ and that $u \in \mathcal{C}([0, T]; \dot{B}_{p,r}^{\frac{N}{p}-1})$.

Hint: use the previous exercise.

Chapter 3

The transport equation

This chapter is devoted to the study of the following transport equation:

$$(T) \quad \begin{cases} \partial_t f + v \cdot \nabla f = g, \\ f|_{t=0} = f_0 \end{cases}$$

where $v : \mathbb{R} \times \mathbb{R}^N \rightarrow \mathbb{R}^N$ stands for a given time dependent vectorfield, $f_0 : \mathbb{R}^N \rightarrow \mathbb{R}^M$ and $g : \mathbb{R} \times \mathbb{R}^N \rightarrow \mathbb{R}^M$ are known data.

Transport equations arise in many mathematical problems and, in particular, in most PDE's related to fluid mechanics. Although, most of the time, the velocity field v and the source term g depend (nonlinearly) on f , having a good theory for the linear transport equation (T) is an important first step for studying such PDE's.

In this chapter, we aim at stating estimates in Besov spaces for (T). For the sake of conciseness, we shall always assume that x belongs to $\mathbb{R}^N$ and restrict ourselves to the nonhomogeneous setting. Our results, however, may be easily carried out to the periodic case and to homogeneous spaces. We focus on the case where v has enough smoothness so that the initial regularity is conserved during the evolution. Roughly, it means that v must be at least lipschitz with respect to the space variable (see [25] and references therein for the case where Dv is only quasi-lipschitz).

In the last section of the chapter, we use these estimates for proving local well-posedness for a nonlinear shallow water equation commonly called Camassa-Holm equation.

3.1 Framework and basic properties

In this section, we briefly list some properties of smooth solutions to (T).

Let us first recall that for all $x \in \mathbb{R}^N$, the *flow* $\psi_{t,s}(x)$ of v starting from x at time s is, by definition, the solution of the following ordinary differential equation¹:

$$\begin{cases} \partial_t \psi_{t,s}(x) = v(t, \psi_{t,s}(x)), & x \in \mathbb{R}^N, \\ (\psi_{t,s}(x))|_{t=s} = x. \end{cases}$$

Then we have

$$(3.1) \quad \forall x \in \mathbb{R}^N, \quad \frac{d}{dt} f(t, \psi_{t,s}(x)) = (\partial_t + v \cdot \nabla) f(t, \psi_{t,s}(x)).$$

In other words, the operator $\partial_t + v \cdot \nabla$ may be interpreted as the derivative along the flow. For that reason, it is often called *material derivative*.

¹Of course, depending on the assumptions on v , the flow $\psi_{t,s}$ may not be defined for all s, t .

As a consequence of (3.1), we gather that, at least formally, the solution of (T) is given by the formula

$$(3.2) \quad f(t, x) = f_0(\psi_{0,t}(x)) + \int_0^t g(\tau, \psi_{\tau,t}(x)) d\tau.$$

To simplify the presentation, we restrict ourselves from now on to evolution for *positive* times ² and assume that v and f are defined on $[0, T] \times \mathbb{R}^N$.

Then the previous calculations may be made rigorous for all $(s, t) \in [0, T]^2$ if v is smooth enough, say, v is continuous and $Dv \in L^\infty([0, T] \times \mathbb{R}^N)$. Under this latter assumption, the flow is uniquely defined for all $(t, s, x) \in [0, T] \times [0, T] \times \mathbb{R}^N$ and the map $(t, s, x) \mapsto \psi_{t,s}(x)$ is $\mathcal{C}^1$ with respect to s , t , and x . Besides, as a corollary of uniqueness, we have

$$\forall (s, t, t') \in [0, T]^3, \quad \psi_{t,s} \circ \psi_{s,t'} = \psi_{t,t'}.$$

Hence denoting $\psi_t := \psi_{t,0}$, we see that ψ_t is a diffeomorphism with inverse $\psi_{0,t}$ and that formula (3.2) rewrites

$$(3.3) \quad f(t, x) = f_0(\psi_t^{-1}(x)) + \int_0^t g(\tau, \psi_\tau(\psi_t^{-1}(x))) d\tau.$$

It is then easy to prove by direct calculation that if $f_0 \in L^p$ and $g \in L^1_{loc}(0, T; L^p)$ then $f \in L^\infty(0, T; L^p)$ and satisfies

$$(3.4) \quad \|f(t)\|_{L^p} \leq e^{\frac{1}{p} \int_0^t \|\operatorname{div} v(\tau)\|_{L^\infty} d\tau} \left(\|f_0\|_{L^p} + \int_0^t e^{-\frac{1}{p} \int_0^\tau \|\operatorname{div} v(\tau')\|_{L^\infty} d\tau'} \|g(\tau)\|_{L^p} d\tau \right)$$

where it is understood that $1/p = 0$ if $p = \infty$.

In the next section, we want to investigate estimates in the same spirit with L^p replaced by some Besov space $B^s_{p,r}$.

3.2 A priori estimates in Besov spaces

In this section, we want to prove the following result³:

Proposition 3.2.1. *Let $1 \leq p \leq p_1 \leq \infty$, $1 \leq r \leq \infty$ and $p' := (1 - 1/p)^{-1}$. Assume that*

$$(3.5) \quad \sigma > -N \min\left(\frac{1}{p_1}, \frac{1}{p'}\right) \quad \text{or} \quad \sigma > -1 - N \min\left(\frac{1}{p_1}, \frac{1}{p'}\right) \quad \text{if} \quad \operatorname{div} v = 0.$$

There exists a constant C depending only on N , p , p_1 , r and σ , such that the following estimates hold true:

$$(3.6) \quad \|f\|_{\tilde{L}^\infty_t(B^{\sigma}_{p,r})} \leq \left(\|f_0\|_{B^{\sigma}_{p,r}} + \int_0^t e^{-C \int_0^\tau Z(\tau') d\tau'} \|g(\tau)\|_{B^{\sigma}_{p,r}} d\tau \right) e^{C \int_0^t Z(\tau) d\tau},$$

$$\text{with} \quad \begin{cases} Z(t) = \|\nabla v(t)\|_{B^{\frac{N}{p_1}, \infty} \cap L^\infty} & \text{if } \sigma < 1 + \frac{N}{p_1}, \\ Z(t) = \|\nabla v(t)\|_{B^{\sigma-1}_{p_1, r}} & \text{if } \sigma > 1 + \frac{N}{p_1} \quad \text{or} \quad \left\{ \sigma = 1 + \frac{N}{p_1} \quad \text{and} \quad r = 1 \right\}. \end{cases}$$

If $f = v$ then for all $\sigma > 0$ ($\sigma > -1$ if $\operatorname{div} v = 0$) estimates (3.6) and (3.12) hold with

$$Z(t) = \|\nabla v(t)\|_{L^\infty}.$$

²It goes without saying that similar results may be proved for negative times.

³From now on, we agree that, if $\operatorname{div} v = 0$ then $v \cdot \nabla f$ stands for $\operatorname{div}(fv)$

Proof: Applying the operator Δ_q to (T) yields

$$\begin{cases} (\partial_t + v \cdot \nabla) \Delta_q f = \Delta_q g + R_q, \\ \Delta_q f|_{t=0} = \Delta_q f_0, \end{cases} \quad (\mathcal{T}_q)$$

with $R_q := v \cdot \nabla \Delta_q f - \Delta_q(v \cdot \nabla f)$.

Since $Dv \in L^1(0, T; L^\infty)$, one readily gets

$$(3.7) \quad \begin{aligned} \|\Delta_q f(t)\|_{L^p} &\leq \|\Delta_q f_0\|_{L^p} + \int_0^t \|\Delta_q g(\tau)\|_{L^p} d\tau \\ &\quad + \int_0^t \left(\|R_q(\tau)\|_{L^p} + \frac{1}{p} \|\operatorname{div} v(\tau)\|_{L^\infty} \|\Delta_q f(\tau)\|_{L^p} \right) d\tau. \end{aligned}$$

Note that $\|\Delta_q f(t)\|_{L^p}$ may be replaced by $\sup_{\tau \in [0, t]} \|\Delta_q f(\tau)\|_{L^p}$ in the left-hand side.

Let us admit for a while the following lemma:

Lemma 3.2.2. *Let $\sigma \in \mathbb{R}$, $1 \leq r \leq \infty$ and $1 \leq p \leq p_1 \leq \infty$. Assume that (3.5) is satisfied. There exists a constant C depending only on p , p_1 , σ , r and N , and such that the following inequalities are true:*

$$(3.8) \quad \|2^{q\sigma} \|R_q\|_{L^p}\|_{\ell^r} \leq C \|\nabla v\|_{B_{p_1, \infty}^{\frac{N}{p}} \cap L^\infty} \|f\|_{B_{p, r}^\sigma} \quad \text{if } \sigma < 1 + \frac{N}{p_1},$$

$$(3.9) \quad \|2^{q\sigma} \|R_q\|_{L^p}\|_{\ell^r} \leq C \|\nabla v\|_{B_{p_1, r}^{\sigma-1}} \|f\|_{B_{p, r}^\sigma} \quad \text{if } \sigma > 1 + \frac{N}{p_1}, \quad \text{or } \sigma = 1 + \frac{N}{p_1} \quad \text{and } r = 1.$$

If besides $f = v$ then we also have

$$(3.10) \quad \|2^{q\sigma} \|R_q\|_{L^p}\|_{\ell^r} \leq C \|\nabla v\|_{L^\infty} \|f\|_{B_{p, r}^\sigma} \quad \text{if } \sigma > 0, \quad (\text{or } \sigma > -1 \quad \text{if } \operatorname{div} v = 0).$$

Now, inserting inequalities (3.8), (3.9) or (3.10) into (3.7) yields

$$(3.11) \quad \|f\|_{L_t^\infty(B_{p, r}^\sigma)} \leq \|f_0\|_{B_{p, r}^\sigma} + C \int_0^t Z(\tau) \|f\|_{L_\tau^\infty(B_{p, r}^\sigma)} d\tau + \int_0^t \|g(\tau)\|_{B_{p, r}^\sigma} d\tau$$

with Z defined as in the statement of the proposition.

Applying Gronwall lemma completes the proof. ■

Remark. Note that, starting from (3.7) and using lemma 3.2.2, one can also get

$$\|f\|_{\tilde{L}_t^\infty(B_{p, r}^\sigma)} \leq \|f_0\|_{B_{p, r}^\sigma} + C \int_0^t Z(\tau) \|f\|_{\tilde{L}_\tau^\infty(B_{p, r}^\sigma)} d\tau + \|g\|_{\tilde{L}_t^1(B_{p, r}^\sigma)},$$

and we thus have:

$$(3.12) \quad \|f\|_{\tilde{L}_t^\infty(B_{p, r}^\sigma)} \leq \left(\|f_0\|_{B_{p, r}^\sigma} + \|g\|_{\tilde{L}_t^1(B_{p, r}^\sigma)} \right) e^{C \int_0^t Z(\tau) d\tau}.$$

For completeness, we now give the proof of lemma 3.2.2. In order to show that only the gradient part of v is involved in the estimates, we have to split v into low and high frequencies : $v = \Delta_{-1}v + \tilde{v}$. Obviously, there exists a constant C such that

$$(3.13) \quad \forall a \in [1, \infty], \quad \|\Delta_{-1} \nabla v\|_{L^a} \leq C \|\nabla v\|_{L^a} \quad \text{and} \quad \|\nabla \tilde{v}\|_{L^a} \leq C \|\nabla v\|_{L^a}.$$

Since there exists a $R > 0$ so that $\text{Supp } \mathcal{F}\tilde{v} \cap B(0, R) = \emptyset$, Bernstein lemma yields

$$(3.14) \quad \forall a \in [1, \infty], \forall q \geq -1, \|\Delta_q \nabla \tilde{v}\|_{L^a} \approx 2^q \|\Delta_q \tilde{v}\|_{L^a}.$$

Now, we have (with the summation convention over repeated indices):

$$\begin{aligned} R_q &= v \cdot \nabla \Delta_q f - \Delta_q(v \cdot \nabla f), \\ &= [\tilde{v}^j, \Delta_q] \partial_j f + [\Delta_{-1} v^j, \Delta_q] \partial_j f. \end{aligned}$$

Hence, taking advantage of Bony's decomposition, we end up with $R_q = \sum_{i=1}^6 R_q^i$ where

$$\begin{aligned} R_q^1 &= [T_{\tilde{v}^j}, \Delta_q] \partial_j f, \\ R_q^2 &= T_{\partial_j \Delta_q f} \tilde{v}^j, \\ R_q^3 &= -\Delta_q T_{\partial_j f} \tilde{v}^j, \\ R_q^4 &= \partial_j R(\tilde{v}^j, \Delta_q f) - \partial_j \Delta_q R(\tilde{v}^j, f), \\ R_q^5 &= \Delta_q R(\text{div } \tilde{v}, f) - R(\text{div } \tilde{v}, \Delta_q f), \\ R_q^6 &= [\Delta_{-1} v^j, \Delta_q] \partial_j f. \end{aligned}$$

In the following computations, the constant C depends only on σ , p , p_1 , r and N and we denote by (c_q) a sequence such that $\|(c_q)\|_{\ell^r} \leq 1$.

Bounds for $2^{q\sigma} \|R_q^1\|_{L^p}$:

By virtue of proposition 1.2.1, we have

$$R_q^1 = \sum_{|q-q'| \leq 4} [S_{q'-1} \tilde{v}^j, \Delta_q] \partial_j \Delta_{q'} f.$$

Using the definition of the operator Δ_q leads to

$$[S_{q'-1} \tilde{v}^j, \Delta_q] \partial_j \Delta_{q'} f(x) = \int h(y) \left[S_{q'-1} \tilde{v}^j(x) - S_{q'-1} \tilde{v}^j(x - 2^{-q}y) \right] \partial_j \Delta_{q'} f(x - 2^{-q}y) dy$$

so that applying first order Taylor's formula, convolution inequalities and (3.13) yields

$$\begin{aligned} 2^{q\sigma} \|R_q^1\|_{L^p} &\leq C \|\nabla v\|_{L^\infty} \sum_{|q'-q| \leq 4} 2^{q'\sigma} \|\Delta_{q'} f\|_{L^p}, \\ (3.15) \quad &\leq C c_q \|\nabla v\|_{L^\infty} \|f\|_{B_{p,r}^\sigma}. \end{aligned}$$

Bounds for $2^{q\sigma} \|R_q^2\|_{L^p}$:

By virtue of proposition 1.2.1, we have

$$R_q^2 = \sum_{q' \geq q-3} S_{q'-1} \partial_j \Delta_q f \Delta_{q'} \tilde{v}^j.$$

Hence, using inequalities (3.13) and (3.14) yields

$$(3.16) \quad 2^{q\sigma} \|R_q^2\|_{L^p} \leq C c_q \|\nabla v\|_{L^\infty} \|f\|_{B_{p,r}^\sigma}.$$

Bounds for $2^{q\sigma} \|R_q^3\|_{L^p}$:

One proceeds as follows :

$$(3.17) \quad R_q^3 = - \sum_{|q'-q|\leq 4} \Delta_q \left(S_{q'-1} \partial_j f \Delta_{q'} \tilde{v}^j \right),$$

$$(3.18) \quad = - \sum_{\substack{|q'-q|\leq 4 \\ q''\leq q'-2}} \Delta_q \left(\Delta_{q''} \partial_j f \Delta_{q'} \tilde{v}^j \right).$$

Therefore, denoting $1/p_2 = 1/p - 1/p_1$ and taking advantage of (3.13) and (3.14),

$$\begin{aligned} 2^{q\sigma} \|R_q^3\|_{L^p} &\leq C \sum_{\substack{|q'-q|\leq 4 \\ q''\leq q'-2}} 2^{q\sigma} \|\Delta_{q''} \partial_j f\|_{L^{p_2}} \|\Delta_{q'} \tilde{v}^j\|_{L^{p_1}}, \\ &\leq C \sum_{\substack{|q'-q|\leq 4 \\ q''\leq q'-2}} 2^{(q-q'')(\sigma-1-\frac{N}{p_1})} 2^{q''\sigma} \|\Delta_{q''} f\|_{L^p} 2^{q'\frac{N}{p_1}} \|\Delta_{q'} \nabla v\|_{L^{p_1}}. \end{aligned}$$

Hence, if $\sigma < 1 + N/p_1$,

$$(3.19) \quad 2^{q\sigma} \|R_q^3\|_{L^p} \leq C c_q \|\nabla v\|_{B_{p_1,\infty}^{\frac{N}{p_1}}} \|f\|_{B_{p,r}^\sigma}.$$

Note that, starting from (3.17), one can alternately get

$$2^{q\sigma} \|R_q^3\|_{L^p} \leq C \sum_{|q'-q|\leq 4} \|\nabla S_{q'-1} f\|_{L^{p_2}} 2^{q'(\sigma-1)} \|\Delta_{q'} \nabla v\|_{L^{p_1}}.$$

As it may be proved that $B_{p,r}^{\sigma-1} \hookrightarrow L^{p_2}$ in the cases $\sigma > 1 + N/p_1$, or $\sigma = 1 + N/p_1$ and $r = 1$, we eventually get

$$(3.20) \quad 2^{q\sigma} \|R_q^3\|_{L^p} \leq C c_q \|\nabla v\|_{B_{p_1,r}^{\sigma-1}} \|f\|_{B_{p,r}^\sigma}.$$

Bounds for $2^{q\sigma} \|R_q^4\|_{L^p}$:

$$R_q^4 = \underbrace{\sum_{|q'-q|\leq 2} \partial_j (\Delta_{q'} \tilde{v}^j \Delta_q \tilde{\Delta}_{q'} f)}_{R_q^{4,1}} - \underbrace{\sum_{q'\geq q-3} \partial_j \Delta_q (\Delta_{q'} \tilde{v}^j \tilde{\Delta}_{q'} f)}_{R_q^{4,2}}$$

For the first term, we merely have (by virtue of (3.14)),

$$2^{q\sigma} \|R_q^{4,1}\|_{L^p} \leq C \|\nabla v\|_{L^\infty} \sum_{|q'-q|\leq 2} 2^{q'\sigma} \|\tilde{\Delta}_{q'} f\|_{L^p},$$

whence

$$(3.21) \quad 2^{q\sigma} \|R_q^{4,1}\|_{L^p} \leq C c_q \|\nabla v\|_{L^\infty} \|f\|_{B_{p,r}^\sigma}.$$

For $R_q^{4,2}$, we proceed differently according to the value of $1/p + 1/p_1$. If $1/p + 1/p_1 \leq 1$, we define p_2 by the relation $1/p_2 := 1/p + 1/p_1$. Then under condition $\sigma > -1 - N/p_1$, proposition 1.4.2 combined with the embedding $B_{p_2,r}^{\sigma+\frac{N}{p_1}} \hookrightarrow B_{p,r}^\sigma$ insures that we have

$$(3.22) \quad 2^{q\sigma} \|R_q^{4,2}\|_{L^p} \leq C c_q \|\tilde{v}\|_{B_{p_1,\infty}^{\frac{N}{p_1}+1}} \|f\|_{B_{p,r}^\sigma}.$$

Now, if $1/p + 1/p_1 > 1$, the above argument has to be applied with p' instead of p_2 and one still ends up with (3.22) provided that $\sigma > -1 - N/p'$.

Putting (3.21), (3.22) together and appealing to (3.13), one gets

$$(3.23) \quad 2^{q\sigma} \|R_q^4\|_{L^p} \leq C c_q \|\nabla v\|_{B_{p_1,\infty}^{\frac{N}{p_1}}} \|f\|_{B_{p,r}^\sigma}.$$

Bounds for $2^{q\sigma} \|R_q^5\|_{L^p}$:

The arguments are the same as for R_q^4 . Under condition $\sigma > -N \min(1/p_1, 1/p')$, one gets

$$(3.24) \quad 2^{q\sigma} \|R_q^5\|_{L^p} \leq C c_q \|\nabla v\|_{B_{p_1, \infty}^{\frac{N}{p_1}}} \|f\|_{B_{p, r}^{\sigma}}.$$

Bounds for $2^{q\sigma} \|R_q^6\|_{L^p}$:

As $R_q^6 = \sum_{|q'-q| \leq 1} [\Delta_{q'}, \Delta_{-1} v] \cdot \nabla \Delta_{q'} f$, the first order Taylor formula yields

$$(3.25) \quad \begin{aligned} 2^{q\sigma} \|R_q^6\|_{L^p} &\leq C \sum_{|q'-q| \leq 1} \|\nabla \Delta_{-1} v\|_{L^\infty} 2^{q'\sigma} \|\Delta_{q'} f\|_{L^p}, \\ &\leq C c_q \|\nabla v\|_{L^\infty} \|f\|_{B_{p, r}^{\sigma}}. \end{aligned}$$

Combining inequalities (3.15), (3.16), (3.19) or (3.20), (3.23), (3.24) and (3.25) yields (3.8) and (3.9). The proof of inequality (3.10) is left to the reader (see exercise (3.2)).

3.3 Solving the transport equation in Besov spaces

Let us now state our existence result for the transport equation with data in Besov spaces:

Theorem 3.3.1. *Let p, p_1, r and σ be as in the statement of proposition 3.2.1. Let $f_0 \in B_{p, r}^s$ and $g \in L^1(0, T; B_{p, r}^s)$. Let v be a time dependent vectorfield with coefficients in $L^\rho(0, T; B_{\infty, \infty}^{-M})$ for some $\rho > 1$ and $M > 0$, and such that $Dv \in L^1(0, T; B_{p_1, \infty}^{\frac{N}{p_1}} \cap L^\infty)$ if $\sigma < 1 + \frac{N}{p_1}$, and $Dv \in L^1(0, T; B_{p_1, r}^{\sigma-1})$ if $\sigma > 1 + \frac{N}{p_1}$ or $\sigma = 1 + \frac{N}{p_1}$ and $r = 1$.*

Then equation (T) has a unique solution $f \in L^\infty(0, T; B_{p, r}^\sigma) \cap (\cap_{\sigma' < \sigma} \mathcal{C}([0, T]; B_{p, 1}^{\sigma'}))$ and the inequalities of proposition 3.2.1 hold true.

If moreover r is finite then we have $f \in \mathcal{C}([0, T]; B_{p, r}^\sigma)$.

Proof: Uniqueness readily stems from proposition 3.2.1 so let us tackle directly the proof of the existence. For the sake of conciseness, we treat only the case $\sigma < 1 + \frac{N}{p_1}$.

We first smooth out the data and the velocity field v by setting

$$f_0^n := S_n f_0, \quad g^n = \rho_n *_t S_n g \quad \text{and} \quad v^n = \rho_n *_t S_n v$$

where $\rho_n := \rho_n(t)$ stands for a sequence of mollifiers with respect to the time variable⁴.

We clearly have $f_0^n \in B_{p, r}^\infty$, $g \in \mathcal{C}([0, T]; B_{p, r}^\infty)$, $v^n \in \mathcal{C}([0, T] \times \mathbb{R}^N)$ and $Dv^n \in \mathcal{C}([0, T]; B_{p, r}^\infty)$ with $B_{p, r}^\infty := \bigcap_{s \in \mathbb{R}} B_{p, r}^s$. Moreover f_0^n is uniformly bounded in $B_{p, r}^\sigma$, g^n is uniformly bounded in $L^1(0, T; B_{p, r}^\sigma)$, v^n is uniformly bounded in $L^\rho(0, T; B_{\infty, \infty}^{-M})$ and Dv^n is uniformly bounded in $L^1(0, T; B_{p_1, \infty}^{\frac{N}{p_1}} \cap L^\infty)$.

Let f^n be the solution to

$$\partial_t f^n + v^n \cdot \nabla f^n = g^n, \quad f^n|_{t=0} = f_0^n.$$

Of course f^n is smooth and, according to proposition 3.2.1, we have

$$(3.26) \quad \|f^n(t)\|_{B_{p, r}^\sigma} \leq e^{C \int_0^t Z^n(\tau) d\tau} \left(\|f_0^n\|_{B_{p, r}^\sigma} + \int_0^t e^{-C \int_0^\tau Z^n(\tau') d\tau'} \|g^n(\tau)\|_{B_{p, r}^\sigma} d\tau \right)$$

⁴with no loss of generality, one can assume that v and g are defined on $\mathbb{R} \times \mathbb{R}^N$.

with $Z^n(t) := \|\nabla v^n(t)\|_{B_{p_1,\infty}^{\frac{N}{p_1}} \cap L^\infty}$.

Thus, according to the uniform bounds for f_0^n , g^n and v^n , one can conclude that sequence $(f^n)_{n \in \mathbb{N}}$ is uniformly bounded in $\mathcal{C}([0, T]; B_{p,r}^\sigma)$.

There are several ways to prove the convergence of $(f^n)_{n \in \mathbb{N}}$, all of them being quite cumbersome if written into details. For instance, one can show that $(f^n)_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathcal{C}([0, T]; B_{p,r}^{\sigma-1})$ (if σ is not too negative), or use a duality method or appeal to compactness arguments.

In the present notes, we are going to use compactness. Let us first observe that $(\partial_t f^n - g^n)_{n \in \mathbb{N}}$ is uniformly bounded in $L^\alpha(0, T; B_{p,\infty}^{-m})$ for some $\alpha > 1$, and some $m > 0$. This may be proved by observing that

$$(3.27) \quad \partial_t f^n - g^n = \Delta_{-1} v^n \cdot \nabla f^n + (\text{Id} - \Delta_{-1}) v^n \cdot \nabla f^n.$$

Indeed, because ∇f^n is uniformly bounded in $L^\infty(0, T; B_{p,r}^{s-1})$ and $\Delta_{-1} v^n$, uniformly bounded in $L^\rho(0, T; \mathcal{C}_b^\infty)$ (where $\mathcal{C}_b^\infty$ stands for the set of smooth bounded functions with bounded derivatives⁵), the first term in the right-hand side of (3.27) is uniformly bounded in $L^\rho(0, T; B_{p,r}^{s-1})$. Now, as for small enough ε the function $(\text{Id} - \Delta_{-1}) v^n$ is uniformly bounded in $L^\alpha(0, T; B_{p_1+\varepsilon,r}^{\frac{N}{p_1}+1-\varepsilon})$ for some $\alpha > 1$ (interpolate between the uniform bounds in $L^1(0, T; B_{p_1,\infty}^{\frac{N}{p_1}+1})$ and in $L^\rho(0, T; B_{\infty,\infty}^{-M})$ for $(\text{Id} - \Delta_{-1}) v^n$), one can conclude by appealing to propositions 1.4.1 and 1.4.2 that the last term in (3.27) is uniformly bounded in $L^\alpha(0, T; B_{p,\infty}^{-m})$ for some large enough $m > 0$.

Integrating in time and denoting $\bar{f}^n(t) := f^n - \int_0^t g^n(\tau) d\tau$, we thus gather that there exists some $\beta > 0$ such that the sequence $(\bar{f}^n)_{n \in \mathbb{N}}$ is uniformly bounded in $\mathcal{C}^\beta([0, T]; B_{p,\infty}^{-m})$, hence uniformly equicontinuous with values in $B_{p,\infty}^{-m}$.

Next, assuming that m is large enough, proposition 1.4.6 guarantees that for all $\varphi \in \mathcal{C}_c^\infty$ the map $u \mapsto \varphi u$ is compact from $B_{p,r}^s$ to $B_{p,\infty}^{-m}$. Combining Ascoli theorem and Cantor diagonal process thus insures that, up to a subsequence, sequence $(\bar{f}^n)_{n \in \mathbb{N}}$ converges in $\mathcal{S}'$ to some distribution $\bar{f}$ such that $\varphi \bar{f}$ belongs to $\mathcal{C}([0, T]; B_{p,\infty}^{-m})$ for all $\varphi \in \mathcal{C}_c^\infty$.

Finally, by taking advantage once again of the uniform bounds in $L^\infty(0, T; B_{p,r}^\sigma)$ and of the Fatou property for Besov spaces, we get $\bar{f} \in L^\infty(0, T; B_{p,r}^\sigma)$ and, by interpolation, better results of convergence so that it becomes easy to check that the function $f := \bar{f} + \int_0^t g(\tau) d\tau$ is a solution to (T) (that the data f_0^n , g^n and v^n converge to f , g and v may be easily deduced from their definition).

We still have to prove that $f \in \mathcal{C}([0, T]; B_{p,r}^\sigma)$ in the case where r is finite. Just by looking at the equation (T), it is easy to get $\partial_t f \in L^1(0, T; B_{p,\infty}^{-M'})$ for some large enough M' . Hence f belongs to $\mathcal{C}([0, T]; B_{p,\infty}^{-M'})$. Therefore $\Delta_q f \in \mathcal{C}([0, T]; L^p)$ for any $q \geq -1$, whence $S_q f \in \mathcal{C}([0, T]; B_{p,r}^\sigma)$ for all $q \in \mathbb{N}$.

We claim that the sequence of continuous $B_{p,r}^s$ valued functions $(S_q f)_{q \in \mathbb{N}}$ converges uniformly on $[0, T]$. Indeed, according to (1.1),

$$\Delta_{q'}(f - S_q f) = \sum_{\substack{|q''-q'| \geq 1 \\ q'' \geq q}} \Delta_{q'} \Delta_{q''} f$$

⁵Here comes the assumption that $v \in L^\rho(0, T; B_{\infty,\infty}^{-M})$.

whence

$$(3.28) \quad \|f - S_q f\|_{B_{p,r}^\sigma} \leq C \left(\sum_{q' \geq q-1} \left(2^{q'\sigma} \|\Delta_{q'} f\|_{L^p} \right)^r \right)^{\frac{1}{r}}.$$

Using inequality (3.7) to bound the right-hand side of (3.28), we gather

$$\begin{aligned} \|f - S_q f\|_{L_T^\infty(B_{p,r}^\sigma)} &\lesssim \left(\left(\sum_{q' \geq q-1} \left(2^{q'\sigma} \|\Delta_{q'} f_0\|_{L^p} \right)^r \right)^{\frac{1}{r}} + \int_0^T \left(\sum_{q' \geq q-1} \left(2^{q'\sigma} \|\Delta_{q'} g(t)\|_{L^p} \right)^r \right)^{\frac{1}{r}} dt \right. \\ &\quad \left. + \|f\|_{L_T^\infty(B_{p,r}^\sigma)} \int_0^T \left(\sum_{q' \geq q-1} c_{q'}(t)^r \right)^{\frac{1}{r}} Z(t) dt \right), \end{aligned}$$

with $\sum_{q' \geq -1} c_{q'}(t) \leq 1$ for all $t \in [0, T]$.

The first term clearly tends to zero when q tends to infinity. The terms in the integrals also tend to zero for almost every t . Lebesgue dominated convergence theorem enables us to conclude that $\|f - S_q f\|_{L_T^\infty(B_{p,r}^\sigma)}$ tends to zero when q tends to infinity. This achieves to proving that $u \in \mathcal{C}([0, T]; B_{p,r}^\sigma)$ in the case $r < +\infty$.

When $r = +\infty$, we use that for any $\sigma' < \sigma$ we have the embedding $B_{p,\infty}^\sigma \hookrightarrow B_{p,1}^{\sigma'}$ so that the above argument may be repeated in the space $B_{p,1}^{\sigma'}$. This yields $f \in \mathcal{C}([0, T]; B_{p,1}^{\sigma'})$. ■

3.4 On the Cauchy problem for a shallow water equation

Let us first point out that theorem 3.3.1 is a good starting point for the study of general (possibly N -dimensional) equations of the type

$$\partial_t u + u \cdot \nabla u = f(u, \nabla u, \dots),$$

including the incompressible Euler equation, provided the nonlinearity is of first order at most.

In the present section, we show how it may be used to solve a nonlinear one-dimensional shallow water equation which has been much studied recently: the so-called Camassa-Holm equation.

3.4.1 About Camassa-Holm equation

The one-dimensional model we are interested in reads:

$$(3.29) \quad \partial_t v - \partial_{xxx}^3 v + 2\kappa \partial_x v + 3v \partial_x v = 2\partial_x v \partial_{xx}^2 v + v \partial_{xxx}^3 v.$$

Above, the scalar function $u = u(t, x)$ stands for the fluid velocity at time $t \geq 0$ in the x direction and κ is a non-negative parameter. For simplicity, we assume that x belongs to $\mathbb{R}$. Similar results are true if x belongs to the circle, though.

Equation (3.29) has been derived independently by A. Fokas and B. Fuchssteiner in [27], and by R. Camassa and D. Holm in [6]. Its systematic mathematical study has been initiated in a series of papers by A. Constantin and J. Escher (see e.g [16]).

We shall concentrate on the case $\kappa = 0$ (which actually is not restrictive since the change of variables $u(t, x) = v(t, x - \kappa t) + \kappa$ leads to (3.29) with $\kappa = 0$). It turns out that equation (3.29) with $\kappa = 0$ has many remarkable structural properties. In particular, it has infinitely many conservation laws, the most obvious ones being the conservation of the average over $\mathbb{R}$ and of the H^1 norm for smooth solutions with sufficient decay at infinity. By taking advantage of this

latter property, Z. Xin and P. Zhang proved that (3.29) has global weak solutions for any data in H^1 (see [39]).

In the present chapter, we address the question of existence and uniqueness for the initial value problem. For simplicity, we restrict ourselves to the evolution for positive times. Of course, one would get similar results for negative times: this is just a matter of changing the initial condition u_0 into $-u_0$.

At this point, one can wonder which regularity assumptions are relevant for u_0 so that the initial value problem be well-posed in the sense of Hadamard (i.e. (CH) has a unique local solution in a suitable functional setting, and continuity with respect to the initial data holds true).

In order to go further into the quest for a good functional framework, let us observe that one can get rid very easily of the third order terms. Indeed: applying the pseudo-differential operator $(1 - \partial_x^2)^{-1}$ to (3.29), we discover that (CH) is equivalent to

$$(CH) \quad \begin{cases} \partial_t u + u \partial_x u = P(D) \left(u^2 + \frac{1}{2} (\partial_x u)^2 \right), \\ u|_{t=0} = u_0, \end{cases}$$

with $P(D) = -\partial_x(1 - \partial_x^2)^{-1}$.

Hence Camassa-Holm equation is nothing but a generalized Burgers equation with a nonlocal nonlinearity of order 0. In light of proposition 3.3.1, we can thus expect that a necessary condition for well-posedness for data in some functional space E is that E be embedded in the space Lip of continuous bounded functions with bounded derivatives.

Moreover, as the solution u is expected to be only in $\mathcal{C}([0, T]; E)$ (a gain of regularity cannot be expected in a Burgers like equation), the application

$$G : u \mapsto P(D) \left(u^2 + \frac{1}{2} (\partial_x u)^2 \right)$$

must map E to E continuously.

3.4.2 A well-posedness result and a blow-up criterion

If we restrict ourselves to the framework of nonhomogeneous Besov space $B_{p,r}^s$, the condition $E \subset \text{Lip}$ is equivalent to $s > 1 + 1/p$ (or $s \geq 1 + 1/p$ if $r = 1$), and no further restrictions are needed for the continuity of the map G (up to the endpoint $r = 1$, $s = 1$, $p = +\infty$ which has to be avoided). We shall see however that for proving uniqueness, our method requires that in addition we have $s > \max(1 + 1/p, 3/2)$. Let us mention in passing that in the periodic framework, by using the Lagrangian formulation for Camassa-Holm equation, it is actually possible to weaken the condition on s (see [26]).

For stating our local existence result, the following functional spaces are needed:

$$\begin{aligned} E_{p,r}^s(T) &:= \mathcal{C}([0, T]; B_{p,r}^s) \cap \mathcal{C}^1([0, T]; B_{p,r}^{s-1}) \quad \text{if } r < \infty, \\ E_{p,\infty}^s(T) &:= L^\infty(0, T; B_{p,\infty}^s) \cap \text{Lip}([0, T]; B_{p,\infty}^{s-1}) \end{aligned}$$

with $T > 0$, $s \in \mathbb{R}$ and $1 \leq p, r \leq \infty$.

Theorem 3.4.1. *Let $1 \leq p, r \leq \infty$ and $s > \max(3/2, 1 + 1/p)$. Let $u_0 \in B_{p,r}^s$. There exists a time $T > 0$ such that (CH) has a unique solution u in $E_{p,r}^s(T)$.*

We can also state a result of conservation of energy for smooth solutions:

Theorem 3.4.2. *Let s, p, r be as in theorem 3.4.1. Let $u \in E_{p,r}^s(T)$ be a solution of (CH) on $[0, T] \times \mathbb{R}$ with data $u_0 \in B_{p,r}^s \cap H^1$. Then the solution u to (CH) satisfies*

$$(3.30) \quad \forall t \in [0, T], \quad \|u(t)\|_{H^1} = \|u_0\|_{H^1}.$$

Before stating blow-up criteria, let us give the definition of the lifespan of solutions with data in $B_{p,r}^s$.

Definition. Let $u_0 \in B_{p,r}^s$. We define the lifespan $T_{u_0}^*$ of the solutions of (CH) with initial data u_0 as the supremum of positive times T such that (CH) has a solution $u \in E_{p,r}^s(T)$ on $[0, T] \times \mathbb{R}$.

Our main blow-up criterion reads:

Theorem 3.4.3. *Let u_0 be as in theorem 3.4.1 and u be the corresponding solution. Then*

$$T_{u_0}^* < \infty \implies \int_0^{T_{u_0}^*} \|\partial_x u(\tau)\|_{L^\infty} d\tau = \infty.$$

3.4.3 Uniqueness

Let us start this section with a lemma:

Lemma 3.4.4. *Let $1 \leq p, r \leq \infty$, and $(\sigma_1, \sigma_2) \in \mathbb{R}^2$ be such that*

$$B_{p,r}^{\sigma_2} \hookrightarrow \text{Lip}, \quad \sigma_1 \leq \sigma_2 \quad \text{and} \quad \sigma_1 + \sigma_2 > 2 + \max\left(0, \frac{2}{p} - 1\right).$$

Then the function $\mathcal{B} : (f, g) \mapsto P(D)(fg + \frac{1}{2}\partial_x f \partial_x g)$ maps $B_{p,r}^{\sigma_1} \times B_{p,r}^{\sigma_2}$ in $B_{p,r}^{\sigma_1}$.

Proof: We notice that $P(D)$ is a multiplier of degree -1 . Hence, according to proposition 1.3.7, it suffices to prove that the functional

$$H : (f, g) \mapsto fg + \frac{1}{2}\partial_x f \partial_x g$$

maps $B_{p,r}^{\sigma_1} \times B_{p,r}^{\sigma_2}$ in $B_{p,r}^{\sigma_1-1}$.

The term fg is easy to handle so we focus on the study of $\partial_x f \partial_x g$. By virtue of Bony's decomposition, we have

$$\partial_x f \partial_x g = T_{\partial_x f} \partial_x g + T_{\partial_x g} \partial_x f + R(\partial_x f, \partial_x g).$$

Proposition 1.4.1 insures that the application $(f, g) \mapsto T_{\partial_x f} \partial_x g$ is continuous from $B_{p,r}^{\sigma_1} \times B_{p,r}^{\sigma_2}$ to

- the space $B_{p,r}^{\sigma_1+\sigma_2-2-\frac{1}{p}}$ if $\sigma_1 < 1 + \frac{1}{p}$,
- the space $B_{p,r}^{\sigma_2-1-\varepsilon}$ for all $\varepsilon > 0$, if $\sigma_1 = 1 + \frac{1}{p}$ and $r > 1$,
- the space $B_{p,r}^{\sigma_2-1}$ if $\sigma_1 = 1 + \frac{1}{p}$ and $r = 1$, or $\sigma_1 > 1 + \frac{1}{p}$.

According to our assumptions on σ_1 , σ_2 , p and r , we thus can conclude that $(f, g) \mapsto T_{\partial_x f} \partial_x g$ maps $B_{p,r}^{\sigma_1} \times B_{p,r}^{\sigma_2}$ in $B_{p,r}^{\sigma_1-1}$.

Since $B_{p,r}^{\sigma_2-1} \hookrightarrow L^\infty$, proposition 1.4.1 readily yields the continuity of $(f, g) \mapsto T_{\partial_x g} \partial_x f$ from $B_{p,r}^{\sigma_1} \times B_{p,r}^{\sigma_2}$ to $B_{p,r}^{\sigma_1-1}$.

Finally, the remainder term maps $B_{p,r}^{\sigma_1} \times B_{p,r}^{\sigma_2}$ in $B_{p,r}^{\sigma_1+\sigma_2-2-\frac{1}{p}}$ (and thus in $B_{p,r}^{\sigma_1-1}$) provided that $\sigma_1 + \sigma_2 > 2 + \max(0, \frac{2}{p} - 1)$. ■

Uniqueness in theorem 3.4.1 is a straightforward corollary of the following proposition.

Proposition 3.4.5. *Let $1 \leq p, r \leq +\infty$ and $s > \max(1 + 1/p, 3/2)$. Suppose that we are given $(u, v) \in (L^\infty(0, T; B_{p,r}^s) \cap \mathcal{C}([0, T]; B_{p,r}^{s-1}))^2$ two solutions of (CH) with initial data $u_0, v_0 \in B_{p,r}^s$. Then we have for every $t \in [0, T]$:*

$$\|u(t) - v(t)\|_{B_{p,r}^{s-1}} \leq \|u_0 - v_0\|_{B_{p,r}^{s-1}} e^{C \int_0^t (\|u(\tau)\|_{B_{p,r}^s} + \|v(\tau)\|_{B_{p,r}^s}) d\tau}.$$

Proof: It is obvious that $w := v - u$ solves the transport equation:

$$\partial_t w + u \partial_x w = -w \partial_x v + P(D) \left(w(u+v) + \frac{1}{2} \partial_x w \partial_x (u+v) \right).$$

According to propositions 3.2.1 and 1.3.7, the following inequality holds true:

$$(3.31) \quad \|w(t)\|_{B_{p,r}^{s-1}} \leq \|w_0\|_{B_{p,r}^{s-1}} e^{C \int_0^t \|\partial_x u\|_{B_{p,r}^{s-1}} d\tau'} + C \int_0^t e^{C \int_\tau^t \|\partial_x u\|_{B_{p,r}^{s-1}} d\tau'} \times \left(\|w \partial_x v\|_{B_{p,r}^{s-1}} + \|\mathcal{B}(w, u+v)\|_{B_{p,r}^{s-1}} \right) d\tau,$$

where $\mathcal{B}$ stands for the symmetric bilinear function introduced in lemma 3.4.4.

Since $s > \max(\frac{3}{2}, 1 + \frac{1}{p})$, we thus have

$$\|\mathcal{B}(w, u+v)\|_{B_{p,r}^{s-1}} \leq C \|w\|_{B_{p,r}^{s-1}} (\|u\|_{B_{p,r}^s} + \|v\|_{B_{p,r}^s}).$$

Plugging this last inequality in (3.31) and applying Gronwall lemma completes the proof. $\blacksquare$

Remark. For proving uniqueness, we are led to estimate in $B_{p,r}^{s-1}$ the difference between two solutions although both solutions are in $B_{p,r}^s$. Owing to the term $(\partial_x u)^2$, the additional condition $s > \max(\frac{3}{2}, 1 + \frac{1}{p})$ is thus required. In fact, uniqueness is also in true in $B_{2,1}^{\frac{3}{2}}$, see [24]. It has been discovered recently in [26] that if using the Lagrangian formulation, one can directly prove uniqueness in a space with the regularity index s and forget about the restriction $s > \max(\frac{3}{2}, 1 + \frac{1}{p})$.

3.4.4 The proof of existence

Let us first enumerate the main steps of the proof of theorem 3.4.1.

- (i) Construction of approximate solutions of (CH) which are smooth solutions of some *linear* transport equation.
- (ii) Find a positive T for which these approximate solutions are uniformly bounded in $E_{p,r}^s(T)$.
- (iii) Prove that the sequence of approximate solutions is a Cauchy sequence in a larger space.
- (iv) Check that the limit has indeed the required regularity.

First step: approximate solution

We use a standard iterative process to build a solution. Starting from $u^0 := 0$, we then define by induction a sequence of smooth functions $(u^n)_{n \in \mathbb{N}}$ by solving the following linear transport equation:

$$(T_n) \quad \begin{cases} (\partial_t + u^n \partial_x) u^{n+1} = P(D) \left((u^n)^2 + \frac{1}{2} (\partial_x u^n)^2 \right), \\ u^{n+1}|_{t=0} = u_0^{n+1} := S_{n+1} \star u_0. \end{cases}$$

Since all the data belong to $B_{p,r}^\infty$, theorem 3.3.1 enables us to show by induction that for all $n \in \mathbb{N}$, the above equation has a global solution which belongs to $\mathcal{C}(\mathbb{R}^+; B_{p,r}^\infty)$.

Second step: uniform bounds

According to theorem 3.3.1 and to lemma 3.4.4, we have the following inequality for all $n \in \mathbb{N}$:

$$(3.32) \quad \|u^{n+1}(t)\|_{B_{p,r}^s} \leq C e^{CU^n(t)} \left(\|u_0\|_{B_{p,r}^s} + \int_0^t e^{-CU^n(\tau)} \|u^n(\tau)\|_{B_{p,r}^s}^2 d\tau \right)$$

with $U^n := \int_0^t \|u^n(\tau)\|_{B_{p,r}^s} d\tau$.

Let us fix a $T > 0$ such that $2C^2\|u_0\|_{B_{p,r}^s}T < 1$ and suppose that

$$(3.33) \quad \forall t \in [0, T], \quad \|u^n(t)\|_{B_{p,r}^s} \leq \frac{C\|u_0\|_{B_{p,r}^s}}{1 - 2C^2\|u_0\|_{B_{p,r}^s}t}.$$

Plugging (3.33) in (3.32) yields

$$\begin{aligned} \|u^{n+1}(t)\|_{B_{p,r}^s} &\leq \frac{1}{\sqrt{1-2C^2\|u_0\|_{B_{p,r}^s}t}} \left(\|u_0\|_{B_{p,r}^s} + C^2\|u_0\|_{B_{p,r}^s}^2 \int_0^t \frac{d\tau}{(1-2C^2\|u_0\|_{B_{p,r}^s}\tau)^{\frac{3}{2}}} \right), \\ &\leq \frac{C\|u_0\|_{B_{p,r}^s}}{1-2C^2\|u_0\|_{B_{p,r}^s}t}. \end{aligned}$$

Therefore, $(u^n)_{n \in \mathbb{N}}$ is uniformly bounded in $\mathcal{C}([0, T]; B_{p,r}^s)$. This clearly entails that $u^n \partial_x u^n$ is uniformly bounded in $\mathcal{C}([0, T]; B_{p,r}^{s-1})$. As the right-hand side of (T_n) has been shown to be uniformly bounded in $\mathcal{C}([0, T]; B_{p,r}^s)$, one can conclude that the sequence $(u^n)_{n \in \mathbb{N}}$ is uniformly bounded in $E_{p,r}^s(T)$.

Third step: convergence

We are going to show that $(u^n)_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathcal{C}([0, T]; B_{p,r}^{s-1})$.

We remark that for all $(m, n) \in \mathbb{N}^2$, we have

$$(\partial_t + u^{n+m} \partial_x)(u^{n+m+1} - u^{n+1}) = (u^n - u^{n+m}) \partial_x u^{n+1} + B(u^{n+m} - u^n, u^{n+m} + u^n).$$

Applying theorem 3.3.1 and lemma 3.4.4, and using that $B_{p,r}^{s-1}$ is an algebra yields

$$\begin{aligned} \forall t \in [0, T], \quad \|(u^{n+m+1} - u^{n+1})(t)\|_{B_{p,r}^{s-1}} &\leq e^{CU^{n+m}(t)} \left(\|u_0^{n+m+1} - u_0^{n+1}\|_{B_{p,r}^{s-1}} \right. \\ &\quad \left. + C \int_0^t e^{-CU^{n+m}(\tau)} \|u^{n+m} - u^n\|_{B_{p,r}^{s-1}} (\|u^n\|_{B_{p,r}^s} + \|u^{n+1}\|_{B_{p,r}^s} + \|u^{n+m}\|_{B_{p,r}^s}) d\tau \right). \end{aligned}$$

Since $(u^n)_{n \in \mathbb{N}}$ is uniformly bounded in $E_{p,r}^s(T)$ and

$$u_0^{n+m+1} - u_0^{n+1} = \sum_{q=n+2}^{n+m+1} \Delta_q u_0,$$

we finally get a constant C_T independent of n, m and such that for all $t \in [0, T]$, we have

$$\|(u^{n+m+1} - u^{n+1})(t)\|_{B_{p,r}^{s-1}} \leq C_T \left(2^{-n} + \int_0^t \|(u^{n+m} - u^n)(\tau)\|_{B_{p,r}^{s-1}} d\tau \right).$$

Arguing by induction, one can easily prove that

$$\|u^{n+m+1} - u^{n+1}\|_{L_T^\infty(B_{p,r}^{s-1})} \leq \frac{(TC_T)^{n+1}}{(n+1)!} \|u^m\|_{L_T^\infty(B_{p,r}^s)} + C_T \sum_{k=0}^n 2^{-(n-k)} \frac{(TC_T)^k}{k!}.$$

As $\|u^m\|_{L_T^\infty(B_{p,r}^s)}$ may be bounded independently of m , we can conclude to the existence of some new constant C'_T such that

$$\|u^{n+m+1} - u^{n+1}\|_{L_T^\infty(B_{p,r}^{s-1})} \leq C'_T 2^{-n}.$$

Hence $(u^n)_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathcal{C}([0, T]; B_{p,r}^{s-1})$, whence it converges to some limit function $u \in \mathcal{C}([0, T]; B_{p,r}^{s-1})$.

last step: conclusion

We now have to check that u belongs to $E_{p,r}^s(T)$ and satisfies (CH). Since $(u^n)_{n \in \mathbb{N}}$ is uniformly bounded in $L^\infty(0, T; B_{p,r}^s)$, Fatou property for Besov spaces guarantees that u also belongs to $L^\infty(0, T; B_{p,r}^s)$.

On the other hand, as $(u^n)_{n \in \mathbb{N}}$ converges to u in $\mathcal{C}([0, T]; B_{p,r}^{s-1})$, an interpolation argument insures that convergence actually holds true in $\mathcal{C}([0, T]; B_{p,r}^{s'})$ for any $s' < s$. It is then easy to pass to the limit in (T_n) and to conclude that u is indeed a solution to (CH).

Now, because u belongs to $L^\infty(0, T; B_{p,r}^s)$, the right-hand side of the equation

$$\partial_t u + u \partial_x u = P(D)(u^2 + \frac{1}{2}(\partial_x u)^2)$$

also belongs to $L^\infty(0, T; B_{p,r}^s)$. In the case $r < \infty$, theorem 3.3.1 enables us to conclude that $u \in \mathcal{C}([0, T]; B_{p,r}^s)$. Finally, using again the equation, we see that $\partial_t u$ is in $\mathcal{C}([0, T]; B_{p,r}^{s-1})$ if r is finite, and in $L^\infty(0, T; B_{p,r}^{s-1})$ otherwise. So finally, u belongs to $E_{p,r}^s(T)$.

Remark. If v_0 is in a small neighborhood of u_0 in $B_{p,r}^s$, the arguments above give the existence of a solution $v \in E_{p,r}^s(T)$ to (CH) with initial datum v_0 . Proposition 3.4.5 combined with an obvious interpolation ensures continuity with respect to the initial data in $\mathcal{C}([0, T]; B_{p,r}^{s'}) \cap \mathcal{C}^1([0, T]; B_{p,r}^{s'-1})$ for any $s' < s$.

The fact that continuity also holds in $\mathcal{C}([0, T]; B_{p,r}^s) \cap \mathcal{C}^1([0, T]; B_{p,r}^{s-1})$ when $r < +\infty$ is not obvious but belongs to the mathematical folklore. It may be proved through the use of a sequence of approximate solutions $(u_\epsilon)_{\epsilon > 0}$ for (CH) which converges uniformly in $\mathcal{C}([0, T]; B_{p,r}^s) \cap \mathcal{C}^1([0, T]; B_{p,r}^{s-1})$. A viscosity approximation gives the desired property of convergence.

3.4.5 Blow-up criterion and energy conservation

This section is devoted to the proof of theorems 3.4.3 and 3.4.2. Both theorems are based on the following lemma:

Lemma 3.4.6. *Let $1 \leq p, r \leq \infty$ and $s > 1$. Let $u \in L^\infty(0, T; B_{p,r}^s)$ solving (CH) on $[0, T) \times \mathbb{R}$ with $u_0 \in B_{p,r}^s$ as an initial datum. There exist a constant C depending only on s and p , and a universal constant C' such that for all $t \in [0, T)$, we have*

$$(3.34) \quad \|u(t)\|_{B_{p,r}^s} \leq \|u_0\|_{B_{p,r}^s} e^{C \int_0^t \|u(\tau)\|_{\text{Lip}} d\tau},$$

$$(3.35) \quad \|u(t)\|_{\text{Lip}} \leq \|u_0\|_{\text{Lip}} e^{C' \int_0^t \|\partial_x u(\tau)\|_{L^\infty} d\tau}.$$

Proof: Applying the last part of proposition 3.2.1 to (CH) and using the fact that $P(D)$ is a multiplier of order -1 yields

$$e^{-C \int_0^t \|\partial_x u\|_{L^\infty} d\tau} \|u(t)\|_{B_{p,r}^s} \leq \|u_0\|_{B_{p,r}^s} + C \int_0^t e^{-C \int_0^\tau \|\partial_x u\|_{L^\infty} d\tau'} \left(\|u^2\|_{B_{p,r}^{s-1}} + \|(\partial_x u)^2\|_{B_{p,r}^{s-1}} \right) d\tau.$$

As $s - 1 > 0$, we have, according to proposition 1.4.3,

$$\|u^2\|_{B_{p,r}^{s-1}} + \|(\partial_x u)^2\|_{B_{p,r}^{s-1}} \leq C \|u\|_{\text{Lip}} \|u\|_{B_{p,r}^s}.$$

Therefore

$$e^{-C \int_0^t \|\partial_x u\|_{L^\infty} d\tau} \|u(t)\|_{B_{p,r}^s} \leq \|u_0\|_{B_{p,r}^s} + C \int_0^t e^{-C \int_0^\tau \|\partial_x u\|_{L^\infty} d\tau'} \|u\|_{B_{p,r}^s} \|u\|_{\text{Lip}} d\tau.$$

Applying Gronwall lemma completes the proof of (3.34).

By differentiating once equation (CH) with respect to x , and applying the L^∞ estimate for transport equations, we easily prove that

$$e^{-\int_0^t \|\partial_x u\|_{L^\infty} d\tau} \|u(t)\|_{\text{Lip}} \leq \|u_0\|_{\text{Lip}} + \int_0^t e^{-\int_0^\tau \|\partial_x u\|_{L^\infty} d\tau} \|P(D)(u^2 + \frac{1}{2}(\partial_x u)^2)\|_{\text{Lip}} d\tau.$$

Now, exercise 3.3 guarantees that

$$\|P(D)(u^2 + \frac{1}{2}(\partial_x u)^2)\|_{\text{Lip}} \leq C' \|u\|_{\text{Lip}} \|\partial_x u\|_{L^\infty}$$

for some universal constant C' . Hence Gronwall lemma gives inequality (3.35). $\blacksquare$

We can now prove theorem 3.4.3. Let $u \in \bigcap_{T < T^*} E_{p,r}^s(T)$ be such that $\int_0^{T^*} \|\partial_x u(\tau)\|_{L^\infty} d\tau$ be finite. According to inequality (3.35), $\int_0^{T^*} \|u(\tau)\|_{\text{Lip}} d\tau$ is also finite. Hence, (3.34) insures that

$$(3.36) \quad \forall t \in [0, T^*), \|u(t)\|_{B_{p,r}^s} \leq M_{T^*} := \|u_0\|_{B_{p,r}^s} e^{C \int_0^{T^*} \|u(\tau)\|_{\text{Lip}} d\tau} < \infty.$$

Let $\epsilon > 0$ be such that $2C^2\epsilon M_{T^*} < 1$ where C stands for the constant used in the proof of theorem 3.3.1. We then have a solution $\tilde{u} \in E_{p,r}^s(\epsilon)$ to (CH) with initial datum $u(T^* - \epsilon/2)$. For the sake of uniqueness, $\tilde{u}(t) = u(t + T^* - \epsilon/2)$ on $[0, \epsilon/2)$ so that $\tilde{u}$ extends the solution u beyond T^* . We conclude that $T^* < T_{u_0}^*$ and theorem 3.4.3 is proved.

Let us now prove theorem 3.4.2. Assume that u_0 belongs to $H^1 \cap B_{p,r}^s$ with $s > \max(3/2, 1 + 1/p)$, and that the corresponding solution u is defined on $[0, T]$ and belongs to $E_{p,r}^s(T)$. We want to prove that the H^1 norm of u is conserved. Clearly, this cannot be done by using the approximation scheme of theorem 2.7 since it does not conserve the H^1 norm.

An alternative method is to mollify the initial datum u_0 and to define u^n as the maximal solution of (CH) corresponding to u_0^n then pass to the limit. This time, we set $u_0^n := \rho^n \star u_0$ where $(\rho^n)_{n \in \mathbb{N}}$ stands for a sequence of nonnegative mollifiers.

Owing to the nonnegativity of the mollifiers, one can easily check that

$$\|u_0^n\|_{B_{p,r}^s} \leq \|u_0\|_{B_{p,r}^s} \quad \text{and} \quad \|u_0^n\|_{H^1} \leq \|u_0\|_{H^1}.$$

Following the second step of the proof of theorem 3.4.1, we discover that there exists a constant C such that u^n is a solution of (CH) on $[0, \bar{T}] \times \mathbb{R}$ with

$$\bar{T} := C / \|u\|_{L_T^\infty(B_{p,r}^s)}$$

and $u^n \in E_{p,r}^s(\bar{T})$ uniformly.

Now, we also have $u_0^n \in H^4$ so that, there exists some $T^n > 0$ such that (CH) with data u_0^n has a solution $\tilde{u}^n \in \mathcal{C}([0, T^n]; H^4)$. By virtue of uniqueness, we actually have $\tilde{u}^n \equiv u^n$ on $[0, \min(\bar{T}, T^n)]$, and according to lemma (3.4.6) and to Besov embeddings,

$$\forall t \in [0, \min(\bar{T}, T^n)], \|u^n(t)\|_{H^4} \leq \|u_0^n\|_{H^4} e^{C \int_0^t \|u^n(\tau)\|_{B_{p,r}^s} d\tau}.$$

Note that the right-hand side may be bounded independently of n and of $t \in [0, \bar{T}]$. Therefore, arguing as in the proof of the first part of theorem 3.4.3, one can conclude that T^n may be chosen greater than $\bar{T}$.

Now, the smoothness of u^n enables us to derive directly from (3.29) that

$$(3.37) \quad \forall t \in [0, \bar{T}], \|u^n(t)\|_{H^1} = \|u_0^n\|_{H^1} \leq \|u_0\|_{H^1}.$$

Therefore, passing to the limit (use e.g. proposition 3.4.5) and using Fatou property for H^1 , one eventually gets $\|u(t)\|_{H^1} \leq \|u_0\|_{H^1}$ for $t \in [0, \bar{T}]$.

For proving the reverse inequality, one can solve the equation backward, starting from $u(\bar{T})$. Then, arguing as above and using uniqueness, one can assert that $\|u(\bar{T} - t)\|_{H^1} \leq \|u(\bar{T})\|_{H^1}$ for $t \leq C/\|u\|_{L_{\bar{T}}^\infty(B_{p,r}^s)}$. Repeating the argument several times, we finally get $\|u(t)\|_{H^1} = \|u_0\|_{H^1}$ for all $t \in [0, \bar{T}]$.

It is now easy to get equality on $[0, T]$. Indeed, arguing as above yields equality on $[\bar{T}, 2\bar{T}]$, $[2\bar{T}, 3\bar{T}]$, etc. until the whole interval $[0, T]$ is exhausted.

Let us end this chapter with the statement of a slightly weaker blow-up criterion in the spirit of the celebrated Beale-Kato-Majda criterion for Euler equations (see for example [3] or [10]):

Proposition 3.4.7. *Under the assumption that $u_0 \in B_{p,r}^s$ with $s > \max(3/2, 1 + 1/p)$, we have:*

$$(3.38) \quad T_{u_0}^* < +\infty \implies \int_0^{T_{u_0}^*} \|u(\tau)\|_{B_{\infty,\infty}^1} d\tau = +\infty.$$

Proof: This merely stems from the following logarithmic interpolation inequality:

$$\|u\|_{\text{Lip}} \leq C \left(1 + \|u\|_{B_{\infty,\infty}^1} \log(e + \|u\|_{B_{p,r}^s}) \right)$$

which holds true whenever $s > 1 + 1/p$ and may be deduced from proposition 1.3.6 and the embeddings $\dot{B}_{\infty,1}^1 \cap L^\infty \hookrightarrow \text{Lip}$ and $B_{p,1}^{s-1} \hookrightarrow B_{\infty,\infty}^{s-1-\frac{1}{p}}$.

Plugging it into (3.36), we get

$$\|u(t)\|_{B_{p,r}^s} \leq \|u_0\|_{B_{p,r}^s} e^{Ct} e^{C \int_0^t \|u\|_{B_{\infty,\infty}^1} \log(e + \|u\|_{B_{p,r}^s}) d\tau}.$$

Therefore, easy calculations lead to

$$\log(e + \|u(t)\|_{B_{p,r}^s}) \leq \log(e + \|u_0\|_{B_{p,r}^s}) + Ct + C \int_0^t \|u\|_{B_{\infty,\infty}^1} \log(e + \|u\|_{B_{p,r}^s}) d\tau.$$

Gronwall lemma thus yields

$$\log(e + \|u(t)\|_{B_{p,r}^s}) \leq \left(\log(e + \|u_0\|_{B_{p,r}^s}) + Ct \right) e^{C \int_0^t \|u\|_{B_{\infty,\infty}^1} d\tau}.$$

One can now conclude that if $\int_0^T \|u\|_{B_{\infty,\infty}^1} dt < \infty$ then $u \in L^\infty(0, T; B_{p,r}^s)$. Arguing as in theorem 3.4.3 completes the proof of the proposition. $\blacksquare$

Remark. The fact that $\|\partial_x u(t)\|_{L^\infty}$ may be replaced with the weaker norm $\|\partial_x u(t)\|_{B_{\infty,\infty}^0}$ is actually not very sensitive to the structure of the equation. As pointed out above, a similar criterion may be stated (with the same method) for the incompressible Euler equations, and also for a number of quasilinear first order equations.

We want to conclude this section with a more precise blow-up statement which is specific to the Camassa-Holm equation:

Proposition 3.4.8. *Under the assumption that $u_0 \in B_{p,r}^s \cap H^1$ with $s > \max(3/2, 1 + 1/p)$, we have:*

$$(3.39) \quad T_{u_0}^* < +\infty \implies \int_0^{T_{u_0}^*} \inf_{x \in \mathbb{R}} u(\tau, x) d\tau = -\infty.$$

Proof: The reader is referred to [18]. $\blacksquare$

3.5 Exercises

Exercise 3.1. Let $v : \mathbb{R} \times \mathbb{R}^N \rightarrow \mathbb{R}^N$ be a smooth time-dependent vectorfield, globally Lipschitz with respect to the space variable, and $\psi_{t,s}$, the flow of v .

- 1) Check that $\psi_{t,s}$ is defined for all $(t, s) \in \mathbb{R}^2$.
- 2) Prove that $\forall (t, s) \in \mathbb{R}^2, \forall x \in \mathbb{R}^N, \det D\psi_{t,s}(x) = \exp\left(\int_s^t \operatorname{div} v(\tau, \psi_{\tau,s}(x)) d\tau\right)$.
- 3) Prove inequality (3.4) for smooth solutions which belong to $L^\infty(0, T; L^p)$.
- 4) Prove estimates in $W^{1,p}$ spaces for the solutions to the transport equation (T).

Exercise 3.2. Prove inequality (3.10).

Exercise 3.3. Throughout the exercise, the function u is defined over $\mathbb{R}$.

- 1) Prove that $(1 - \partial_{xx}^2)^{-1}u = \frac{1}{2} e^{-|\cdot|} * u$.

- 2) From the first question, deduce that

$$\begin{aligned} \|(1 - \partial_{xx}^2)^{-1}u\|_{L^\infty} &\leq \|u\|_{L^\infty}, & \|(1 - \partial_{xx}^2)^{-1}\partial_x u\|_{L^\infty} &\leq \|u\|_{L^\infty}, \\ \|(1 - \partial_{xx}^2)^{-1}\partial_x(u^2)\|_{L^\infty} &\leq 2\|u\|_{L^\infty}\|\partial_x u\|_{L^\infty}, & \|(1 - \partial_{xx}^2)^{-1}\partial_{xx}^2(u^2)\|_{L^\infty} &\leq 2\|u\|_{L^\infty}\|\partial_x u\|_{L^\infty}. \end{aligned}$$

- 3) Let $P(D) := \partial_x(1 - \partial_{xx}^2)^{-1}$. Prove the existence of a universal constant C such that

$$\|P(D)(u^2 + \frac{1}{2}(\partial_x u)^2)\|_{\text{Lip}} \leq C\|u\|_{\text{Lip}}\|\partial_x u\|_{L^\infty}.$$

Chapter 4

A short insight into compressible fluid mechanics

4.1 About the model

We first briefly explain how the system of equations for the flow of compressible fluids may be derived from basic physics.

4.1.1 Physical conservation laws

We assume that the fluid under consideration fills in a time independent open domain D of $\mathbb{R}^N$ and may be characterized for every material point $x \in D$ at time $t \in \mathbb{R}$, by

- its *velocity* field $u := u(t, x)$,
- its *density* $\rho := \rho(t, x)$,
- its *internal energy* $e := e(t, x)$,
- its *entropy* by unit mass $s := s(t, x)$.

For each subdomain Ω of D , one can define the following physical quantities:

- the *mass*: $M(\Omega) := \int_{\Omega} \rho \, dx$.
- the *momentum*: $P(\Omega) := \int_{\Omega} \rho u \, dx$.
- the *energy*: $E(\Omega) := \int_{\Omega} (\frac{1}{2} \rho |u|^2 + \rho e) \, dx$.
- the *entropy*: $S(\Omega) := \int_{\Omega} \rho s \, dx$.

Denoting by ψ_t the flow¹ of u and $\Omega_t := \psi_t(\Omega)$, and assuming that there is no production or loss of mass, we have the following conservation law:

$$(4.1) \quad \frac{d}{dt} M(\Omega_t) = \frac{d}{dt} \int_{\Omega_t} \rho \, dx = 0.$$

For the momentum, we have

$$(4.2) \quad \frac{d}{dt} P(\Omega_t) = \frac{d}{dt} \int_{\Omega_t} \rho u \, dx = \int_{\Omega_t} \rho f \, dx + \int_{\partial \Omega_t} (\sigma \cdot n) \, d\Sigma$$

¹See the definition in chapter 3.

where the first term on the right-hand side represents external volume forces with density f (such as gravity for instance) and the second term, surface forces.

In the full generality, the strain about $x \in \partial\Omega_t$ is some vector valued function $\mathcal{T}$ depending on t , x and on the unit exterior normal vector n at x . Under the assumption of small strains, this function may be linearized so that it reduces to $\mathcal{T}(t, x, n) = \sigma(t, x) \cdot n$ for some *strain tensor* σ . In the absence of mass couples, an assumption that we are going to do from now on, the *angular momentum* $\int_{\Omega_t} x \wedge (\rho u)(t, x) dx$ is also conserved and one can prove that σ is a symmetric tensor.

As regards the energy conservation, we have

$$(4.3) \quad \frac{d}{dt} E(\Omega_t) = \frac{d}{dt} \int_{\Omega_t} \rho \left(e + \frac{|u|^2}{2} \right) dx = \int_{\Omega_t} \rho f \cdot u dx + \int_{\partial\Omega_t} (\sigma \cdot n) \cdot n d\Sigma - \int_{\partial\Omega_t} q \cdot n d\Sigma$$

where q is the *heat flux* vector.

Finally, introducing the temperature T , the entropy balance writes:

$$(4.4) \quad \frac{d}{dt} S(\Omega_t) = \frac{d}{dt} \int_{\Omega_t} \rho s dx \geq - \int_{\partial\Omega_t} \left(\frac{q \cdot n}{T} \right) d\Sigma.$$

4.1.2 The full model

From the global conservation laws (4.1), (4.2), (4.3) and (4.4), one can obtain a system of PDE's involving ρ , u , e , and s . This is a mere consequence of the following (formal) lemma.

Lemma 4.1.1. *Let Ω be an open subdomain of D , ψ be the flow of u and $\Omega_t := \psi_t(\Omega)$. The following equalities hold true for all scalar function b :*

$$\begin{aligned} \frac{d}{dt} \int_{\Omega_t} b dx &= \int_{\Omega_t} (\partial_t b + \operatorname{div} bu) dx, \\ &= \int_{\Omega_t} \partial_t b dx + \int_{\partial\Omega_t} (b u \cdot n) d\Sigma. \end{aligned}$$

Combining lemma 4.1.1 with relations (4.1), (4.2), (4.3) and (4.4), we obtain the following equations for the evolution of compressible flows:

$$(4.5) \quad \partial_t \rho + \operatorname{div}(\rho u) = 0,$$

$$(4.6) \quad \partial_t \rho u + \operatorname{div}(\rho u \otimes u) = \rho f + \operatorname{div} \sigma,$$

$$(4.7) \quad \partial_t \left(\rho \left(e + \frac{|u|^2}{2} \right) \right) + \operatorname{div} \left(\rho \left(e + \frac{|u|^2}{2} \right) u \right) = \rho f \cdot u + \operatorname{div}(\sigma \cdot u) - \operatorname{div} q,$$

$$(4.8) \quad \partial_t \rho s + \operatorname{div}(\rho s u) \geq - \operatorname{div} \left(\frac{q}{T} \right).$$

From a mathematical viewpoint, the above equations are too general to be handled. In the next section, we shall restrict ourselves to particular fluids.

4.1.3 Simplifying assumptions

We shall first assume that the fluid is Newtonian, namely

- the tensor σ is a linear function of Du , invariant under rigid transforms,
- the fluid is isotropic (i.e the physical quantities depend only on (t, x)).

As a consequence, it may be shown that σ has the following form:

$$\sigma = (\lambda \operatorname{div} u - p) \operatorname{Id} + 2\mu D(u)$$

where the scalar function $p = p(t, x)$ is the *pressure*, λ and μ are the *viscosity coefficients* and $D(u) := \frac{1}{2}(Du + {}^tDu)$ is the *deformation tensor*. One can also introduce the *viscous stress tensor* :

$$\tau := \lambda \operatorname{div} u \operatorname{Id} + 2\mu D(u).$$

Finally, we shall assume that *Fourier law* is satisfied, namely $q = -k\nabla T$.

In full generality, the coefficients k , λ and μ may depend on ρ and T . For the sake of simplicity however, we shall assume from now on that they are constant functions. We thus obtain the following system of equations:

$$(4.9) \quad \partial_t \rho + \operatorname{div}(\rho u) = 0,$$

$$(4.10) \quad \partial_t \rho u + \operatorname{div}(\rho u \otimes u) - \mu \Delta u - (\lambda + \mu) \nabla \operatorname{div} u + \nabla p = \rho f,$$

$$(4.11) \quad \partial_t \left(\rho \left(e + \frac{|u|^2}{2} \right) \right) + \operatorname{div} \left(\rho \left(e + \frac{|u|^2}{2} \right) u \right) + \operatorname{div} p u - k \Delta T = \rho f \cdot u + \operatorname{div}(\tau \cdot u) - \operatorname{div} q,$$

$$(4.12) \quad \partial_t \rho s + \operatorname{div}(\rho s u) \geq k \operatorname{div} \left(\frac{\nabla T}{T} \right).$$

In dimension N , the above system has $N+2$ equations (and an inequation) for $N+5$ unknown scalar functions (namely ρ , $u_1, \dots, u_N$, p , e , T and s). In order to close the system, three additional equations are needed. As regards the entropy s , we assume that the so-called *Gibbs relation* is satisfied:

$$Tds = de + p d\left(\frac{1}{\rho}\right).$$

We still need two other equalities (often called *state relations*) involving p , ρ , e , s and T . One can assume for example that $p = P(\rho, T)$ and $e = \varepsilon(\rho, T)$ for some given functions P and ε which may be determined from further physical considerations depending on the nature of the fluid. Combining Gibbs relation with the mass, momentum and energy equations, we get

$$\partial_t \rho s + \operatorname{div}(\rho s u) = k \operatorname{div} \left(\frac{\nabla T}{T} \right) + \frac{\tau : D(u)}{T} + k \frac{|\nabla T|^2}{T^2}.$$

Hence, according to the entropy inequality, we must have

$$\tau : D(u) + k \frac{|\nabla T|^2}{T} \geq 0.$$

This yields the following additional constraints on λ , μ and k :

$$(4.13) \quad k \geq 0, \quad \mu \geq 0 \quad \text{and} \quad 2\mu + N\lambda \geq 0.$$

Let us give some important examples:

- Monoatomic gases in dimension $N = 3$ satisfy $2\mu + 3\lambda = 0$. For more general fluids, $2\mu + 3\lambda$ may be positive.
- Inviscid fluids are such that $\mu = \lambda = 0$.
- By definition, a non conducting fluid is such that $k = 0$.

4.1.4 Barotropic fluids

From now, we shall consider a simplified model for compressible fluids, the so-called *barotropic Navier-Stokes equations*:

$$(4.14) \quad \begin{cases} \partial_t \rho + \operatorname{div} \rho u = 0, \\ \partial_t \rho u + \operatorname{div}(\rho u \otimes u) - \mu \Delta u - (\lambda + \mu) \nabla \operatorname{div} u + \nabla p = \rho f \end{cases}$$

with $p = P(\rho)$ for some suitably smooth function P .

The above system may be derived from the general model (4.10), (4.11), (4.12) under the assumption that s is a constant and that the fluid is non conducting. Note that in the viscous case (that we are going to consider in the next sections) the assumption of constant entropy is somewhat inconsistent with physics since the term $\tau : D(u)$ appearing in the entropy balance may be positive. As a matter of fact, it turns out that, from a mathematical viewpoint, the barotropic (or isentropic) model keeps most of the features of the full model.

4.2 Local well-posedness in critical spaces

From now on, we restrict ourselves to viscous fluids (i.e $\mu > 0$) and we further assume (for the sake of simplicity only) that $\lambda + \mu = 0^2$. We focus on flows whose density is a small perturbation of a positive constant, say 1 to simplify the notation.

Denoting $a = \rho - 1$, the barotropic system for smooth enough solutions with positive density thus reduces to

$$(NSC) \quad \begin{cases} \partial_t a + u \cdot \nabla a + (1 + a) \operatorname{div} u = 0, \\ \partial_t u - \frac{\mu}{1+a} \Delta u + u \cdot \nabla u + \nabla g = f, \end{cases}$$

where $g = G(a)$ stands for the chemical potential expressed in term of a . The function G is assumed to be conveniently smooth and, with no loss of generality, vanishes at 0.

The conducting thread for finding an appropriate framework is the same as in the incompressible case (see chapter 2): using functional spaces which have the same scaling invariance as the system, if any.

For the velocity, one can expect the functional framework to be the same as the one used in chapter 2. Now, one can notice that (NSC) is invariant for all $\lambda > 0$ by the rescaling $(a, u) \mapsto (a_\lambda, u_\lambda)$ with

$$a_\lambda(t, x) = a(\lambda^2 t, \lambda x) \quad \text{and} \quad u_\lambda(t, x) = \lambda u(\lambda^2 t, \lambda x)$$

provided that the chemical potential has been changed into $\lambda^2 g$.

This motivates us to solve system (NSC) in a functional space whose norm is invariant for all λ (up to a an irrelevant constant) by the above transform. If we restrict to homogeneous Besov spaces, this suggests us to consider $a_0 \in \dot{B}_{p_1, r_1}^{\frac{N}{p_1}}$ and $u_0 \in \dot{B}_{p_2, r_2}^{\frac{N}{p_2}-1}$ for some $1 \leq p_1, p_2, r_1, r_2 \leq \infty$. In order to avoid the appearance of vacuum however, a L^∞ bound on the density is needed. Hence, we shall assume that $r_1 = 1$ which, in view of Besov embeddings, provides a L^∞ control for free. For technical reasons, we shall further assume that $r_2 = 1$ and that $p_1 = p_2 = p$.

Finally, we thus expect to get existence in the following functional space:

$$E_T^p := \left\{ (a, u) \in \tilde{C}_T(\dot{B}_{p,1}^{\frac{N}{p}}) \times (\tilde{C}_T(\dot{B}_{p,1}^{\frac{N}{p}-1}))^N \mid \|u\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}+1})} < \infty \right\}$$

where $\tilde{C}_T(\dot{B}_{p,1}^s) := \mathcal{C}([0, T]; \dot{B}_{p,1}^s) \cap \tilde{L}_T^\infty(\dot{B}_{p,1}^s)$ for $s \in \mathbb{R}$.

²Actually, the very same results may be proved under the weaker assumption that $\mu > 0$ and $\lambda + 2\mu > 0$ (see [20])

Let us now state our well-posedness result:

Theorem 4.2.1. *Assume that the space dimension is $N \geq 2$ and that $1 \leq p < 2N$. There exists a positive constant ε such that for all $u_0 \in \dot{B}_{p,1}^{\frac{N}{p}-1}$, $f \in L_{loc}^1(\mathbb{R}^+; \dot{B}_{p,1}^{\frac{N}{p}-1})$ and $a_0 \in \dot{B}_{p,1}^{\frac{N}{p}}$ such that $\|a_0\|_{\dot{B}_{p,1}^{\frac{N}{p}}} \leq \varepsilon$ then there exists a positive time T such that (NSC) has a solution (a, u) on $[0, T] \times \mathbb{R}^N$ which belongs to E_T^p .*

Besides there exists a positive constant $\eta > 0$ such that uniqueness holds true in E_T^p whenever $\|a\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} \leq \eta$ if $N \geq 3$ and $p < N$, or $\|a\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} \leq \eta$ if $N = 2$ or $p = N$.

4.2.1 The existence proof

Existence in theorem 4.2.1 may be obtained by mean of the following form of Schauder-Tychonoff fixed point theorem (stated by M. Hukuhara in [29]):

Theorem (Hukuhara). *Let K be a convex subset of a locally convex topological linear space E , and Φ be a continuous self-mapping of K . If besides $\Phi(K)$ is contained in a compact subset of K then Φ has a fixed point in K .*

Let us briefly enumerate the main steps of the proof.

In the first step, we show that our existence problem amounts to finding a fixed point for some map Φ . In the next two steps, we state various a priori estimates for Φ which will help us to find an appropriate functional framework for applying Hukuhara's theorem. In the fourth step, we show that Hukuhara's theorem indeed applies. This yields existence in a space slightly larger than E_T^p (time continuity is missing). In the fifth step, we show that the constructed solution actually belongs to E_T^p . The last step is devoted to uniqueness (which actually relies on independent arguments).

First step: construction of the functional Φ

Introducing $\tilde{a} := a/(1+a)$ and u_L the solution to

$$\partial_t u_L - \mu \Delta u_L = f, \quad u_L|_{t=0} = u_0,$$

we have to solve

$$\begin{cases} \partial_t a + u \cdot \nabla a = -(1+a) \operatorname{div} u, \\ \partial_t \bar{u} - \mu \Delta \bar{u} = -u \cdot \nabla u - \mu \tilde{a} \Delta u - \nabla g \end{cases}$$

with initial data $a|_{t=0} = a_0$ and $u|_{t=0} = 0$, and $u := u_L + \bar{u}$.

This motivates us to introduce the functional $\Phi : (b, \bar{v}) \mapsto (\Phi_1(b, \bar{v}), \Phi_2(b, \bar{v}))$ defined by

$$\Phi_1(b, \bar{v})(t) := a_0 \circ \psi_{0,t}^v - \int_0^t ((1+b) \operatorname{div} v) \circ \psi_{\tau,t}^v d\tau \quad \text{with} \quad v := u_L + \bar{v},$$

where $\psi_{s,s'}^v$ stands for the flow of v , and

$$\Phi_2(b, \bar{v})(t) := - \int_0^t e^{\mu(t-\tau)\Delta} (v \cdot \nabla v + \mu \tilde{b} \Delta v + \nabla g) d\tau$$

with $\tilde{b} := b/(1+b)$ and $g := G(b)$.

Second step: a priori estimates

We shall prove that for suitably small T and a_0 , the functional Φ has a fixed point in the Banach space

$$\tilde{E}_T^p := \tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}}) \times (\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1}) \cap L_T^1(\dot{B}_{p,1}^{\frac{N}{p}+1}))^N.$$

Let $(R, \eta) \in (0, 1)^2$ to be fixed hereafter. We denote by $B_T^{R, \eta}$ the set of $(b, \bar{v})$ in $\tilde{E}_T^p$ such that

$$\|b\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} \leq R \quad \text{and} \quad \|\bar{v}\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} + \mu \|\bar{v}\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}+1})} \leq \eta.$$

In what follows, we assume that R has been chosen so small as to satisfy³

$$\|b\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} \leq R \implies |b(t, x)| \leq \frac{1}{2} \quad \text{for all } (t, x) \in [0, T] \times \mathbb{R}^N.$$

We claim that if $R_0 := \|a_0\|_{\dot{B}_{p,1}^{\frac{N}{p}}}$, R , η and T are sufficiently small then the function Φ maps $B_T^{R, \eta}$ to $B_T^{R, \eta}$.

Indeed, taking advantage of proposition 3.2.1, we get

$$\|\Phi_1(b, \bar{v})\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} \leq e^{CV(T)} \left(\|a_0\|_{\dot{B}_{p,1}^{\frac{N}{p}}} + \int_0^T e^{-CV(t)} \|(1+b) \operatorname{div} v\|_{\dot{B}_{p,1}^{\frac{N}{p}}} dt \right)$$

with $V(t) := \exp(\int_0^t \|Dv\|_{\dot{B}_{p,1}^{\frac{N}{p}}} d\tau)$.

Hence, using proposition 1.4.3,

$$\|\Phi_1(b, \bar{v})\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} \leq e^{CV(T)} \left(\|a_0\|_{\dot{B}_{p,1}^{\frac{N}{p}}} + \int_0^T e^{-CV(t)} (1 + \|b\|_{\dot{B}_{p,1}^{\frac{N}{p}}}) \|\operatorname{div} v\|_{\dot{B}_{p,1}^{\frac{N}{p}}} dt \right).$$

Using the definition of R_0 and V , and the fact that $\|b\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} \leq R$, we end up with

$$(4.15) \quad \|\Phi_1(b, \bar{v})\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} \leq e^{CV(T)} R_0 + (e^{CV(T)} - 1)(1 + R).$$

Next, according to theorem 2.2.3, we have the following inequality for $\Phi_2(b, \bar{v})$:

$$\|\Phi_2(b, \bar{v})\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} + \mu \|\Phi_2(b, \bar{v})\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}+1})} \leq C \int_0^T \left(\|v \cdot \nabla v\|_{\dot{B}_{p,1}^{\frac{N}{p}-1}} + \mu \|\tilde{b} \Delta v\|_{\dot{B}_{p,1}^{\frac{N}{p}-1}} + \|G(b)\|_{\dot{B}_{p,1}^{\frac{N}{p}}} \right) dt.$$

Appealing to propositions 1.5.11, 1.5.12 and 1.5.13, we thus get⁴

$$\|\Phi_2(b, \bar{v})\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} + \mu \|\Phi_2(b, \bar{v})\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}+1})} \leq C \int_0^T \left(\|v\|_{\dot{B}_{p,1}^{\frac{N}{p}}}^2 + \|b\|_{\dot{B}_{p,1}^{\frac{N}{p}}} (1 + \mu \|\Delta v\|_{\dot{B}_{p,1}^{\frac{N}{p}-1}}) \right) dt,$$

whence

$$(4.16) \quad \begin{aligned} & \|\Phi_2(b, \bar{v})\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} + \mu \|\Phi_2(b, \bar{v})\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}+1})} \\ & \leq C \left(\|u_L\|_{L_T^2(\dot{B}_{p,1}^{\frac{N}{p}})}^2 + \|\bar{v}\|_{L_T^2(\dot{B}_{p,1}^{\frac{N}{p}})}^2 + R(T + \mu V(T)) \right). \end{aligned}$$

³Remind that $\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}}) \hookrightarrow L^\infty(0, T; \dot{B}_{p,1}^{\frac{N}{p}}) \hookrightarrow L^\infty(0, T \times \mathbb{R}^N)$.

⁴Here, $N \geq 2$ and $p < 2N$ are needed

Let us assume that T is so small as to satisfy

$$T + \mu \|u_L\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}+1})} \leq \eta \quad \text{and} \quad \mu^{\frac{1}{2}} \|u_L\|_{L_T^2(\dot{B}_{p,1}^{\frac{N}{p}})} \leq \eta.$$

As $(b, \bar{v})$ is in $B_T^{R,\eta}$, we also easily get by interpolation that

$$\mu^{\frac{1}{2}} \|\bar{v}\|_{L_T^2(\dot{B}_{p,1}^{\frac{N}{p}})} \leq \eta.$$

Hence, plugging all these inequalities in (4.15) and (4.16), we end up (up to a change of C) with

$$\begin{aligned} \|\Phi_1(b, \bar{v})\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} &\leq e^{\frac{C\eta}{\mu}} R_0 + 2(e^{\frac{C\eta}{\mu}} - 1), \\ \|\Phi_2(b, \bar{v})\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} + \mu \|\Phi_2(b, \bar{v})\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}+1})} &\leq C\left(\frac{\eta^2}{\mu} + \eta R\right). \end{aligned}$$

Now, if we fix a positive $R \leq \min(1, (2C)^{-1})$, and assume that $R_0 \leq R/4$ and that η is so small as to satisfy

$$\frac{C\eta}{\mu} \leq \frac{1}{2} \quad \text{and} \quad 4(e^{\frac{C\eta}{\mu}} - 1) \leq R,$$

then the function Φ maps $B_T^{R,\eta}$ in $B_T^{R,\eta}$.

Third step: time derivatives

The compactness of the map Φ will be supplied by the following lemma:

Lemma 4.2.2. *Let $(b, \bar{v})$ be in $B_T^{R,\eta}$ with T , R and η chosen according to the previous step. Denote $a := \Phi_1(b, \bar{v})$ and $\bar{u} := \Phi_2(b, \bar{v})$.*

Then $\partial_t a \in L^2(0, T; \dot{B}_{p,1}^{\frac{N}{p}-1})$ and $\partial_t \bar{u} \in L^{\frac{2}{1+\alpha}}(0, T; \dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}-2+\alpha})$ for any $\alpha \in [-1, 1]$ such that $\alpha > \max\left(2\left(1 - \frac{N}{p}\right), 2 - N\right)$. Besides, there exists a constant $C_{T,\alpha}^{R,\eta}$ depending only on R , η , T , α , N , p , μ and G and such that

$$\|\partial_t a\|_{L_T^2(\dot{B}_{p,1}^{\frac{N}{p}-1})} + \|\partial_t \bar{u}\|_{L_T^{\frac{2}{1+\alpha}}(\dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}-2+\alpha})} \leq C_{T,\alpha}^{R,\eta}.$$

Proof: Let us first focus on $\partial_t a$. By definition of a , we have

$$(4.17) \quad \partial_t a = -v \cdot \nabla a - (1 + b) \operatorname{div} v \quad \text{with} \quad v := u_L + \bar{v}.$$

Because v belongs to $L^2(0, T; \dot{B}_{p,1}^{\frac{N}{p}})$, a, b are in $L^\infty(0, T; \dot{B}_{p,1}^{\frac{N}{p}})$ and $p < 2N$, propositions 1.5.11 and 1.5.12 insure that

$$\|\partial_t a\|_{L_T^2(\dot{B}_{p,1}^{\frac{N}{p}-1})} \leq C \left(1 + \|a\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} + \|b\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} \right) \|v\|_{L_T^2(\dot{B}_{p,1}^{\frac{N}{p}})},$$

whence, since $(a, \bar{u})$ and $(b, \bar{v})$ are in $B_T^{R,\eta}$,

$$\|\partial_t a\|_{L_T^2(\dot{B}_{p,1}^{\frac{N}{p}-1})} \leq C(1 + 2\eta)\eta\mu^{-\frac{1}{2}}.$$

Let us now consider $\partial_t \bar{u}$. We use that

$$(4.18) \quad \partial_t \bar{u} = \mu \Delta \bar{u} - \mu \tilde{b} \Delta v - v \cdot \nabla v - \nabla g.$$

As $\bar{u}$ and v are in $L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}+1}) \cap L^\infty(0, T; \dot{B}_{p,1}^{\frac{N}{p}-1})$, interpolation yields

$$\Delta \bar{u}, \Delta v \in L^{\frac{2}{1+\alpha}}(0, T; \dot{B}_{p,1}^{\frac{N}{p}-2+\alpha}) \quad \text{for any } \alpha \in [-1, 1].$$

Because b and thus $\tilde{b}$ are in $L^\infty(0, T; \dot{B}_{p,1}^{\frac{N}{p}})$ (use proposition 1.5.13), we gather that the first two terms in the right-hand side of (4.18) are in $L^{\frac{2}{1+\alpha}}(0, T; \dot{B}_{p,1}^{\frac{N}{p}-2+\alpha})$ (provided that $\alpha > \max(2 - N, 2 - \frac{2N}{p})$ for the second one) with a bound depending only on R, η, μ, N, α and p .

Next, since $v \in L^2(0, T; \dot{B}_{p,1}^{\frac{N}{p}})$ and $\nabla v \in L^{\frac{2}{\alpha}}(0, T; \dot{B}_{p,1}^{\frac{N}{p}-2+\alpha})$, propositions 1.5.11 and 1.5.12 insure that $v \cdot \nabla v$ belongs to $L^{\frac{2}{1+\alpha}}(0, T; \dot{B}_{p,1}^{\frac{N}{p}-2+\alpha})$ under the same condition on α . Finally, using proposition 1.5.13, it is easy to see that the last term in the right-hand side of (4.18) is in $L^\infty(0, T; \dot{B}_{p,1}^{\frac{N}{p}-1})$ and may be bounded in terms of R, μ, T, N, p, α and G . ■

Fourth step: the fixed point argument

Let us introduce the functional space

$$Y_T^p := \tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}}) \times \left(\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1}) \cap \mathcal{M}_T(\dot{B}_{p,1}^{\frac{N}{p}+1}) \right)^N$$

where $\mathcal{M}_T(\dot{B}_{p,1}^{\frac{N}{p}+1})$ stands for the space of bounded measures on $[0, T]$ with values in $\dot{B}_{p,1}^{\frac{N}{p}+1}$.

The space Y_T^p endowed with the norm

$$\|(b, \bar{v})\|_{Y_T^p} := \|b\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} + \|b\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} + \int_0^T d\|v(t)\|_{\dot{B}_{p,1}^{\frac{N}{p}+1}}$$

is a Banach space. Besides, we notice that Y_T^p is the dual space of

$$X_T^p := \tilde{L}_T^1(\dot{B}_{p',\infty}^{-\frac{N}{p}}) \times \left(\tilde{L}_T^1(\dot{B}_{p',\infty}^{1-\frac{N}{p}}) + \mathcal{C}([0, T]; \dot{B}_{p',\infty}^{-\frac{N}{p}-1}) \right)^N$$

where $\tilde{L}_T^1(\dot{B}_{q,\infty}^s)$ stands for the completion of $\mathcal{S}([0, T] \times \mathbb{R}^n)$ for the norm of $\tilde{L}_T^1(\dot{B}_{q,\infty}^s)$, and that X_T^p is a separable Banach space ⁵ (see exercise 4.2).

We claim that for R, T and η chosen according to the previous steps, Hukuhara's theorem applies with the functional space Y_T^p endowed with the weak star topology and the function Φ restricted to the set

$$B_{T,\alpha}^{R,\eta} := \left\{ (b, \bar{v}) \in B_T^{R,\eta}, \|\partial_t b\|_{L_T^2(\dot{B}_{p,1}^{\frac{N}{p}-1})} + \|\partial_t \bar{v}\|_{L_T^{\frac{2}{1+\alpha}}(\dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}-2+\alpha})} \leq C_{T,\alpha}^{R,\eta} \right\}$$

with $\max(2 - N, 2(1 - \frac{N}{p}), -1) < \alpha < 1$ and $C_{T,\alpha}^{R,\eta}$ defined in lemma 4.2.2.

(i) Properties of $B_{T,\alpha}^{R,\eta}$ and Y_T^p .

On one hand, since Y_T^p is the dual space of a Banach space, we gather that Y_T^p endowed with the weak star topology is a locally convex topological linear space.

On the other hand, $B_{T,\alpha}^{R,\eta}$ is obviously a convex subset of Y_T^p .

Finally, according to lemma 4.2.2, we have $\Phi(B_{T,\alpha}^{R,\eta}) \subset B_{T,\alpha}^{R,\eta}$. Since $B_{T,\alpha}^{R,\eta} \subset B_T^{R,\eta}$, it is clear that Φ is a self-map of $B_{T,\alpha}^{R,\eta}$.

⁵Having the property that the space Y_T^p is the dual of some separable Banach space turns out to be crucial for applying Hukuhara's theorem, this is the reason why we take $\mathcal{M}_T(\dot{B}_{p,1}^{\frac{N}{p}+1})$ rather than $L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}+1})$ above.

(ii) Continuity of Φ .

Since $B_{T,\alpha}^{R,\eta}$ is a bounded subset of Y_T^p and Y_T^p is the dual space of a separable Banach space, it suffices to state the sequential continuity of Φ for the weak star topology.

So let us assume that $(b^n, \bar{v}^n)_{n \in \mathbb{N}} \in \left(B_{T,\alpha}^{R,\eta}\right)^\mathbb{N}$ satisfies $(b^n, \bar{v}^n) \rightharpoonup (b, \bar{v})$ weak $*$.

Let $(a^n, \bar{u}^n) := \Phi(b^n, \bar{v}^n)$. Since $(a^n, \bar{u}^n)$ belongs to $B_T^{R,\eta}$ for all $n \in \mathbb{N}$, there exists a distribution $(a, \bar{u})$ verifying

$$(4.19) \quad \|a\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} \leq R \quad \text{and} \quad \|\bar{u}\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} + \mu \|\bar{u}\|_{\mathcal{M}_T(\dot{B}_{p,1}^{\frac{N}{p}+1})} \leq \eta$$

and such that, up to a subsequence, we have $(a^n, \bar{u}^n) \rightharpoonup (a, \bar{u})$ weak $*$.

Hence it is only a matter of showing that $(a, \bar{u}) = \Phi(b, \bar{v})$.

The result will be obtained by “passing to the limit” in the equation satisfied by $(a^n, \bar{u}^n)$. To make this possible, we first have to prove that sequences $(a^n, \bar{u}^n)_{n \in \mathbb{N}}$ and $(b^n, \bar{v}^n)_{n \in \mathbb{N}}$ converge in a stronger sense.

Let us first focus on $(b^n, \bar{v}^n)_{n \in \mathbb{N}}$.

Because $(b^n, \bar{v}^n)_{n \in \mathbb{N}}$ belongs to $B_{T,\alpha}^{R,\eta}$, Hölder inequality insures that sequence $(b^n)_{n \in \mathbb{N}}$ (resp. $(\bar{v}^n)_{n \in \mathbb{N}}$) is bounded in the space $\mathcal{C}^{\frac{1}{2}}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}})$ (resp. $\mathcal{C}^{\frac{1-\alpha}{2}}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}-2+\alpha})$). The sequence $(b^n)_{n \in \mathbb{N}}$ of $\left(\dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}}\right)$ valued functions is thus uniformly equicontinuous on $[0, T]$. A slight generalization of proposition 1.4.6 to homogeneous spaces (see exercise 1.19) guarantees that the map $z \mapsto \phi z$ is compact from $\dot{B}_{p,1}^{\frac{N}{p}}$ to $\left(\dot{B}_{p,1}^{\frac{N}{p}} + \dot{B}_{p,1}^{\frac{N}{p}-1}\right)$ for all $\phi \in \mathcal{C}_c^\infty$. Combining Ascoli’s theorem and Cantor diagonal process, we gather that $(b^n)_{n \in \mathbb{N}}$ has a convergent subsequence in $\mathcal{C}([0, T]; \left(\dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}}\right)_{loc})$: there exists a subsequence of $(b^n)_{n \in \mathbb{N}}$ and some $b' \in \mathcal{S}'_h$ such that for all $\phi \in \mathcal{C}_c^\infty$, $(\phi b^n)_{n \in \mathbb{N}}$ tends to $\phi b'$ in $\mathcal{C}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}})$.

A similar argument provides us with a subsequence of $(\bar{v}^n)_{n \in \mathbb{N}}$ and some $\bar{v}' \in \mathcal{S}'_h$ such that for all $\phi \in \mathcal{C}_c^\infty$, sequence $(\phi \bar{v}^n)_{n \in \mathbb{N}}$ tends to $\phi \bar{v}'$ in $\mathcal{C}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}-2+\alpha})$.

Of course, since by assumption we have $(b^n, \bar{v}^n) \rightharpoonup (b, \bar{v})$ weak $*$, we actually have $b' = b$ and $\bar{v}' = \bar{v}$. Arguing by interpolation, we conclude that for all $\phi \in \mathcal{C}_c^\infty$ and $\varepsilon \in (0, 1)$, the whole sequence $(\phi b^n, \phi \bar{v}^n)_{n \in \mathbb{N}}$ tends to $(\phi b, \phi \bar{v})$ in

$$\mathcal{C}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}-\varepsilon} + \dot{B}_{p,1}^{\frac{N}{p}}) \times \left(\mathcal{C}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}-1-\varepsilon}) \cap L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}+1-\varepsilon})\right)^N.$$

Let us now focus on the convergence of $(a^n, \bar{u}^n)_{n \in \mathbb{N}}$. Since Φ is a self-map of $B_{T,\alpha}^{R,\eta}$, the sequence $(a^n, \bar{u}^n)_{n \in \mathbb{N}}$ is also uniformly bounded in the space

$$\mathcal{C}^{\frac{1}{2}}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}}) \times \left(\mathcal{C}^{\frac{1-\alpha}{2}}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}-2+\alpha})\right)^N.$$

Arguing as above, we thus gather that, up to extraction, we have $(a^n, \bar{u}^n) \rightarrow (a, u)$ in $\mathcal{C}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}-\varepsilon} + \dot{B}_{p,1}^{\frac{N}{p}}) \times \left(\mathcal{C}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}-1-\varepsilon}) \cap L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}+1-\varepsilon})\right)^N$ for all $\varepsilon \in (0, 1)$.

Now, denoting $u^n := \bar{u}^n + u_L$ and $v^n := \bar{v}^n + u_L$, we have for all $n \in \mathbb{N}$,

$$(4.20) \quad \begin{cases} \partial_t a^n + v^n \cdot \nabla a^n = -(1 + b^n) \operatorname{div} v^n, \\ \partial_t \bar{u}^n - \mu \Delta \bar{u}^n = -(v^n \cdot \nabla v^n + \mu \frac{b^n}{1+b^n} \Delta v^n + \nabla(G(b^n))), \\ a^n|_{t=0} = a_0, \quad \bar{u}^n|_{t=0} = 0. \end{cases}$$

The properties of strong convergence which have been proved above enable us to pass to the limit in the system. We get

$$(4.21) \quad \begin{cases} \partial_t a + v \cdot \nabla a = -(1+b) \operatorname{div} v, \\ \partial_t \bar{u} - \mu \Delta \bar{u} = -(v \cdot \nabla v + \mu \frac{b}{1+b} \Delta v + \nabla(G(b))), \\ a|_{t=0} = a_0, \quad \bar{u}|_{t=0} = 0. \end{cases}$$

Since $(b, \bar{v})$ belongs to $B_T^{R,\eta}$, uniqueness in theorems 3.3.1 and 2.2.3 ensures that $(a, \bar{u}) = \Phi(b, \bar{v})$ as wanted.

(iii) Compactness of Φ .

It suffices to state that any sequence $(a^n, \bar{u}^n) := \Phi(b^n, \bar{v}^n)$ with $(b^n, \bar{v}^n) \in B_{T,\alpha}^{R,\eta}$ has a convergent subsequence in $B_{T,\alpha}^{R,\eta}$.

Since in particular $(a^n, \bar{u}^n)$ belongs to $B_T^{R,\eta}$ for all n , there exists some distribution $(a, \bar{u})$ satisfying (4.19) and such that $(a^n, \bar{u}^n) \rightharpoonup (a, \bar{u})$ weak $*$. Proving that $\bar{u}$ actually belongs to $L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}+1})$ is the main difficulty. For that, we use the fact that for all $n \in \mathbb{N}$, we have

$$\partial_t \bar{u}^n - \mu \Delta \bar{u}^n = -(v^n \cdot \nabla v^n + \mu \tilde{b}^n \Delta v^n + \nabla G(b^n)).$$

Since $(b^n, \bar{v}^n) \in B_T^{R,\eta}$, it is easy to show that the right-hand side is uniformly bounded in $L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}-1})$. Hence, according to theorem 2.2.3, there exists some constant C such that for all $n \in \mathbb{N}$, we have

$$\|\partial_t \bar{u}^n - \mu \Delta \bar{u}^n\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}-1})} + \|\bar{u}^n\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} \leq C.$$

Taking advantage of inequality (2.5) and of the definition of the Besov space $\dot{B}_{p,1}^{\frac{N}{p}-1}$ we thus get, up to a change of C a nonnegative sequence $(c_q)_{q \in \mathbb{Z}}$ satisfying $\sum c_q \leq 1$ and such that

$$\|\bar{u}_n\|_{L^1(T_1, T_2; \dot{B}_{p,1}^{\frac{N}{p}+1})} \leq C \sum_q c_q \left(\frac{1 - e^{-\kappa \mu (T_2 - T_1) 2^{2q}}}{\kappa \mu} \right)$$

whenever $0 \leq T_1 \leq T_2 \leq T$.

In other words, the sequence $(\bar{u}_n)_{n \in \mathbb{N}}$ of $\dot{B}_{p,1}^{\frac{N}{p}+1}$ valued functions is equiintegrable on $[0, T]$. Hence, according to Dunford-Pettis theorem, the limit function $\bar{u}$ belongs to $L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}+1})$.

Finally, by making use of the uniform bounds in $L^2(0, T; \dot{B}_{p,1}^{\frac{N}{p}-1} \times \left(L_T^{\frac{2}{1+\alpha}}(\dot{B}_{p,1}^{\frac{N}{p}-1} + \dot{B}_{p,1}^{\frac{N}{p}-2+\alpha}) \right)^N$ for $(\partial_t a^n, \partial_t \bar{u}^n)$ and taking one more extraction if needed, we conclude that $(a, \bar{u}) \in B_{T,\alpha}^{R,\eta}$.

Now, Hukuhara's theorem applies. Denoting by $(a, \bar{u})$ a fixed point of Φ in $B_{T,\alpha}^{R,\eta}$ and setting $u := u_L + \bar{u}$, we thus have proved the existence of a solution (a, u) in $\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1}) \times (\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1}) \cap L_T^1(\dot{B}_{p,1}^{\frac{N}{p}+1}))^N$ for (NSC).

Fifth step: time continuity

In order to complete the existence part in theorem 4.2.1, we still have to prove that $a \in \mathcal{C}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}})$ and that $u \in \mathcal{C}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}-1})$.

As $a \in \tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})$ and $u \in \tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1}) \cap L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}+1})$, we readily check that the right-hand side of the density equation in (NSC) belongs to $L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}})$. Therefore, theorem 3.3.1 insures that $a \in \mathcal{C}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}})$.

Similarly, the right-hand side of the velocity equation is in $L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}-1})$, so theorem 2.2.3 yields $u \in \mathcal{C}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}-1})$.

Last step: uniqueness

Uniqueness in theorem 4.2.1 stems from the following proposition.

Proposition 4.2.3. *Assume that $N \geq 2$ and $1 \leq p \leq N$. Let (a^1, u^1) and (a^2, u^2) be two solutions in E_T^p of (NSC) with the same initial data $(a_0, u_0) \in \dot{B}_{p,1}^{\frac{N}{p}} \times (\dot{B}_{p,1}^{\frac{N}{p}-1})^N$ and external force $f \in L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}-1})$. There exists a constant $\eta \in (0, 1)$ such that if*

$$(4.22) \quad \|a^1\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} \leq \eta \quad \text{in the case } N \geq 3 \quad \text{and } p < N,$$

$$(4.23) \quad \|a^1\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} \leq \eta \quad \text{in the case } N = 2 \quad \text{or } p = N,$$

then $(a^2, u^2) \equiv (a^1, u^1)$ on $[0, T]$.

Proof: Throughout the proof, we denote $\tilde{a}^i := a^i / (1 + a^i)$.

Let us write a system of equations for $(\delta a, \delta u) := (a^2 - a^1, u^2 - u^1)$. We have

$$(4.24) \quad \begin{cases} \partial_t \delta a + u^2 \cdot \nabla \delta a = \sum_{i=1}^3 \delta F_i, \\ \partial_t \delta u - \mu \Delta \delta u = \sum_{i=1}^5 \delta G_i, \end{cases}$$

$$\begin{aligned} \text{with} \quad \delta F_1 &:= -\delta u \cdot \nabla a^1, & \delta F_2 &:= -\delta a \operatorname{div} u^2, & \delta F_3 &:= -(1 + a^1) \operatorname{div} \delta u, \\ \delta G_1 &:= -u^2 \cdot \nabla \delta u, & \delta G_2 &:= -\delta u \cdot \nabla u^1, \\ \delta G_3 &:= -\mu (\tilde{a}^2 - \tilde{a}^1) \Delta u^2, & \delta G_4 &:= -\mu \tilde{a}^1 \Delta \delta u, & \delta G_5 &:= -\nabla (G(a^2) - G(a^1)). \end{aligned}$$

Owing to the term δF_1 in the first equation of (4.24), one can expect to be able to bound δa in $L^\infty(0, T; \dot{B}_{p,1}^{\frac{N}{p}-1})$ rather than in $L^\infty(0, T; \dot{B}_{p,1}^{\frac{N}{p}})$. This loss of one derivative will induce also a loss of one derivative when bounding δu and then one will be in troubles for bounding δG if $p > N$. This is the reason why we need a stronger condition on p for uniqueness than for existence.

Let us first treat the case $N \geq 3$ and $1 \leq p < N$ which is easier⁶. Uniqueness is going to be proved in the following functional space (defined below for $\bar{T} > 0$, $N \geq 3$ and $1 \leq p < N$):

$$F_{\bar{T}}^p := L_{\bar{T}}^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1}) \times (L_{\bar{T}}^\infty(\dot{B}_{p,1}^{\frac{N}{p}-2}) \cap L_{\bar{T}}^1(\dot{B}_{p,1}^{\frac{N}{p}}))^N.$$

Of course, we first have to check that $(\delta a, \delta u)$ belongs to $F_{\bar{T}}^p$. For δa , this is easy because, according to lemma 4.2.2, we have $\partial_t a^i \in L^2(0, T; \dot{B}_{p,1}^{\frac{N}{p}-1})$. Hence $a^i - a_0$ is in $\mathcal{C}([0, T]; \dot{B}_{p,1}^{\frac{N}{p}-1})$.

⁶Of course, in view of Besov embeddings, we have $E_p^T \hookrightarrow E_{p'}^T$ for $p' \geq p$ hence, if one makes the slightly stronger assumption that (4.22) holds with $\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})$ instead of $L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})$ then the general case is a consequence of the endpoint case $p = N$.

Dealing with δu is slightly more involved. Let $\bar{u}^i := u^i - \bar{u}_L$ with $\bar{u}_L$ the solution to the following linear heat equation:

$$(4.25) \quad \partial_t \bar{u}_L - \mu \Delta \bar{u}_L = f - \nabla(G(a_0)), \quad \bar{u}_L|_{t=0} = u_0.$$

We obviously have $\bar{u}^i(0) = 0$ and

$$\partial_t \bar{u}^i - \mu \Delta \bar{u}^i = -\mu \tilde{a}^i \Delta u^i - u^i \cdot \nabla u^i - \nabla(G(a^i) - G(a_0)).$$

Since $p < N$ and $N \geq 3$, and $\bar{a}^i \in L^\infty(0, T; \dot{B}_{p,1}^{\frac{N}{p}-1})$, it is now easy to check that the right-hand side belongs to $L^2(0, T; \dot{B}_{p,1}^{\frac{N}{p}-2})$. Hence $\bar{u}^i$ belongs to $\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-2}) \cap \tilde{L}_T^1(\dot{B}_{p,1}^{\frac{N}{p}})$ according to theorem 2.2.3, and one can now conclude that $(\delta a, \delta u)$ belongs to F_T^p .

Next, applying proposition 3.2.1 to the first equation of (4.24), we get for $\bar{T} \leq T$:

$$\|\delta a\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} \leq e^{C\|u^2\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}+1})}} \|\delta F\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}-1})}.$$

Propositions 1.5.11 and 1.5.12 yield

$$\begin{aligned} \|\delta F_1\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}-1})} &\lesssim \|\nabla a^1\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} \|\delta u\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}})}, \\ \|\delta F_2\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}-1})} &\lesssim \|\delta a\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} \|\operatorname{div} u^2\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}})}, \\ \|\delta F_3\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}-1})} &\lesssim (1 + \|a^1\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})}) \|\operatorname{div} \delta u\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}-1})}. \end{aligned}$$

Hence, as $u^2 \in L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}+1})$, there exists some constant C_T independent of $\bar{T}$ such that

$$(4.26) \quad \|\delta a\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} \leq X(\bar{T}) \|\delta a\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} + C_T \|\delta u\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}})},$$

with $X(\bar{T}) := C_T \|u^2\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}})}$.

Next, applying theorem 2.2.3 to the second equation of (4.24) yields

$$\|\delta u\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-2})} + \|\delta u\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}})} \lesssim \sum_{i=1}^5 \|\delta G_i\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}-2})}.$$

We notice that, as $\dot{B}_{p,1}^{\frac{N}{p}}(\mathbb{R}^N) \hookrightarrow \mathcal{C}(\mathbb{R}^N)$, we have $a^i \in \mathcal{C}([0, T] \times \mathbb{R}^N)$. Hence if η has been chosen small enough in (4.22) so that $|a^1(t, x)| \leq \frac{1}{4}$ on $[0, T]$ then we have, for small enough $\bar{T}$,

$$(4.27) \quad \|a^i(t)\|_{L^\infty} \leq \frac{1}{2} \quad \text{for } i = 1, 2 \quad \text{and } t \in [0, \bar{T}].$$

Therefore applying propositions 1.5.11, 1.5.12 and 1.5.13 yields (remind that $p < N$ and $N \geq 3$ has been assumed)

$$\begin{aligned} \|\delta G_1\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}-2})} &\lesssim \|u^2\|_{L_T^2(\dot{B}_{p,1}^{\frac{N}{p}})} \|\nabla \delta u\|_{L_T^2(\dot{B}_{p,1}^{\frac{N}{p}-2})}, \\ \|\delta G_2\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-2})} &\lesssim \|\nabla u^1\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}})} \|\delta u\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-2})}, \\ \|\delta G_3\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-2})} &\lesssim (1 + \|a^1\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} + \|a^2\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})}) \|\Delta u^2\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}-1})} \|\delta a\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})}, \\ \|\delta G_4\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-2})} &\lesssim \|a^1\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} \|\Delta \delta u\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}-2})}, \\ \|\delta G_5\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-2})} &\lesssim T(1 + \|a^1\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} + \|a^2\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})}) \|\delta a\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})}. \end{aligned}$$

Assuming that $\bar{T}$ has been chosen so small as to satisfy $X(\bar{T}) \leq 1/2$, inequality (4.26) implies that

$$\|\delta a\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})} \leq 2C_T \|\delta u\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}})}.$$

Hence, we finally obtain

$$(4.28) \quad \|\delta u\|_{\tilde{L}_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-2})} + \|\delta u\|_{\tilde{L}_T^1(\dot{B}_{p,1}^{\frac{N}{p}})} \leq Y(\bar{T}) \|\delta u\|_{\tilde{L}_T^1(\dot{B}_{p,1}^{\frac{N}{p}})}$$

with, up to a change of C_T ,

$$Y(\bar{T}) := C \|a^1\|_{L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}})} + C_T (\bar{T} + \|u^1\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}+1})} + \|u^2\|_{L_T^1(\dot{B}_{p,1}^{\frac{N}{p}+1})} + \|u^2\|_{L_T^2(\dot{B}_{p,1}^{\frac{N}{p}})}).$$

We stress that C depends only on N and p and that

$$\limsup_{\bar{T} \rightarrow 0^+} Y(\bar{T}) \leq C\eta.$$

Hence assuming $C\eta < 1$ entails $(\delta a, \delta u) \equiv 0$ on a suitably small time interval $[0, \bar{T}]$. Standard connectivity arguments then lead to $\bar{T} = T$.

Let us now treat the limit case⁷ where $p = N$. Obviously, the above proof fails because some terms in the right-hand side of the equation of δu (such as $\delta a \Delta u^2$ for instance) cannot be estimated in $L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}-2})$ in terms of norms of δa in $L_T^\infty(\dot{B}_{p,1}^{\frac{N}{p}-1})$ and u^2 in $L^1(0, T; \dot{B}_{p,1}^{\frac{N}{p}+1})$. On the other hand, according to exercise 1.22, those terms may be bounded in the slightly larger space $L_T^1(\dot{B}_{p,\infty}^{\frac{N}{p}-2})$.

At this point, we could try to get bounds for δu in $L^\infty(0, T; \dot{B}_{N,\infty}^{-1}) \cap \tilde{L}_T^1(\dot{B}_{N,\infty}^1)$, but we then have to face the lack of control on δu in $L^1(0, T; L^\infty)$ (because, in contrast with $\dot{B}_{N,1}^1$, the space $\dot{B}_{N,\infty}^1$ is not embedded in L^∞) so that we run into troubles when estimating δF_1 . The key to that difficulty relies on the following logarithmic interpolation inequality which is a straightforward generalization of proposition 1.5.3 to $\tilde{L}_T^\rho(\dot{B}_{p,r}^s)$ spaces:

$$(4.29) \quad \|w\|_{L_T^1(\dot{B}_{N,1}^1)} \lesssim \|w\|_{\tilde{L}_T^1(\dot{B}_{N,\infty}^1)} \log \left(e + \frac{\|w\|_{\tilde{L}_T^1(\dot{B}_{N,\infty}^0)} + \|w\|_{\tilde{L}_T^1(\dot{B}_{N,\infty}^2)}}{\|w\|_{\tilde{L}_T^1(\dot{B}_{N,\infty}^1)}} \right).$$

It is then possible to conclude to uniqueness by taking advantage of a generalization of Gronwall lemma: the so called Osgood lemma (see e.g. [13]).

To begin with, let us introduce the functional space in which we are going to prove uniqueness:

$$F_{\bar{T}} := L^\infty(0, \bar{T}; \dot{B}_{N,\infty}^0) \times \left(L^\infty(0, \bar{T}; \dot{B}_{N,\infty}^{-1}) \cap \tilde{L}_{\bar{T}}^1(\dot{B}_{N,\infty}^1) \right)^N.$$

Let us first state that $(\delta a, \delta u) \in F_{\bar{T}}$. For δa , the proof given in the non critical case still works. As $\dot{B}_{N,1}^0 \hookrightarrow \dot{B}_{N,\infty}^0$, we get that $\delta a \in L^\infty(0, T; \dot{B}_{N,\infty}^0)$.

For δu , we just have to check that the right-hand side of (4.25) belongs to $\tilde{L}_T^1(\dot{B}_{N,\infty}^{-1})$. This is due to propositions 1.5.11, 1.5.12 and exercise 1.22 generalized to $\tilde{L}_T^\rho(\dot{B}_{p,r}^s)$ spaces.

⁷If $N = 2$, the general case is a consequence of the case $p = 2$ by virtue of Besov embeddings.

Let us now bound δa . Combining theorem 3.3.1 with propositions 1.5.11 and 1.5.12 and the embedding $\dot{B}_{N,1}^1 \hookrightarrow \dot{B}_{N,\infty}^1 \cap L^\infty$ gives:

$$\|\delta a\|_{L_t^\infty(\dot{B}_{N,\infty}^0)} \leq C e^{C \int_0^t \|u^2\|_{\dot{B}_{N,1}^2} d\tau} \int_0^t \left(\|\delta a\|_{\dot{B}_{N,\infty}^0} \|\operatorname{div} u^2\|_{\dot{B}_{N,1}^1} + \|\delta u\|_{\dot{B}_{N,1}^1} \left(1 + \|a^1\|_{\dot{B}_{N,1}^1}\right) \right) d\tau,$$

whence, according to Gronwall inequality,

$$\|\delta a\|_{L_t^\infty(\dot{B}_{N,\infty}^0)} \leq C e^{C \int_0^t \|u^2\|_{\dot{B}_{N,1}^2} d\tau} (1 + \|a^1\|_{L_t^\infty(\dot{B}_{N,1}^1)}) \|\delta u\|_{L_t^1(\dot{B}_{N,1}^1)}.$$

Making use of inequality (4.29) with $w = \delta u$, we end up with

$$\|\delta a\|_{L_t^\infty(\dot{B}_{N,\infty}^0)} \leq C_T \|\delta u\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^1)} \log \left(e + \frac{\|\delta u\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^0)} + \|\delta u\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^2)}}{\|\delta u\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^1)}} \right)$$

for some constant C_T depending only on the bounds for a^1 and u^2 on $[0, T]$.

Remark that since $\tilde{L}_t^\infty(\dot{B}_{N,1}^0) \hookrightarrow L_t^1(\dot{B}_{N,1}^0)$ for finite t , we have

$$\forall t \in [0, T], \|\delta u\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^0)} + \|\delta u\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^2)} \leq V(t) := V^1(t) + V^2(t) < \infty$$

with

$$V^i(t) := \int_0^t \left(\|u^i(\tau)\|_{\dot{B}_{N,1}^0} + \|u^i(\tau)\|_{\dot{B}_{N,1}^2} \right) d\tau.$$

Therefore,

$$(4.30) \quad \|\delta a\|_{L_t^\infty(\dot{B}_{N,\infty}^0)} \leq C_T \|\delta u\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^1)} \log \left(e + \frac{V(t)}{\|\delta u\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^1)}} \right)$$

with V a non-decreasing function of $t \in [0, T]$ with bounded first derivative.

Let us now turn to the proof of estimates for δu . According to theorem 2.2.3, we have

$$\|\delta u\|_{L_t^\infty(\dot{B}_{N,\infty}^{-1})} + \|\delta u\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^1)} \lesssim \sum_{i=1}^5 \|\delta G_i\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^{-1})}.$$

Using proposition 1.5.11, exercise 1.22 and that $L^1(0, t; \dot{B}_{N,\infty}^{-1}) \hookrightarrow \tilde{L}_t^1(\dot{B}_{N,\infty}^{-1})$, we get for $t \leq T$:

$$\begin{aligned} \|\delta u\|_{L_t^\infty(\dot{B}_{N,\infty}^{-1})} + \|\delta u\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^1)} &\lesssim \|u^2\|_{\tilde{L}_t^2(\dot{B}_{N,1}^1)} \|\delta u\|_{\tilde{L}_t^2(\dot{B}_{N,\infty}^0)} + \|\delta u\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^1)} \|a^1\|_{\tilde{L}_t^\infty(\dot{B}_{N,1}^1)} \\ &\quad + \int_0^t \left[\|u^1\|_{\dot{B}_{N,1}^2} \|\delta u\|_{\dot{B}_{N,\infty}^{-1}} + \left(1 + \|a^1\|_{\dot{B}_{N,1}^1} + \|a^2\|_{\dot{B}_{N,1}^1}\right) \|\delta a\|_{\dot{B}_{N,\infty}^0} \left(1 + \|u^2\|_{\dot{B}_{N,1}^2}\right) \right] d\tau. \end{aligned}$$

Let us assume that the constant η in inequality (4.23) is small enough so that the second term in the right-hand side may be absorbed by the left-hand side. Next, remark that $\|u^2\|_{\tilde{L}_t^2(\dot{B}_{N,1}^1)}$ tends to 0 when t goes to 0 hence there exists a positive $\bar{T}$ such that the first term may also be absorbed⁸ for all $t \in [0, \bar{T}]$.

We end up with the following inequality:

$$\|\delta u\|_{L_t^\infty(\dot{B}_{N,\infty}^{-1})} + \|\delta u\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^1)} \lesssim \int_0^t \left(\|u^1\|_{\dot{B}_{N,1}^2} \|\delta u\|_{\dot{B}_{N,\infty}^{-1}} + (1 + \|u^2\|_{\dot{B}_{N,1}^2}) \|\delta a\|_{\dot{B}_{N,\infty}^0} \right) d\tau.$$

⁸By interpolation, one easily gets $\|\delta u\|_{\tilde{L}_t^2(\dot{B}_{N,\infty}^0)} \leq \|\delta u\|_{\tilde{L}_t^2(\dot{B}_{N,\infty}^1)}^{\frac{1}{2}} \|\delta u\|_{\tilde{L}_t^\infty(\dot{B}_{N,\infty}^{-1})}^{\frac{1}{2}}$.

One can now plug (4.30) in the above inequality. Denoting $X(t) := \|\delta u\|_{L_t^\infty(\dot{B}_{N,\infty}^{-1})} + \|\delta u\|_{\tilde{L}_t^1(\dot{B}_{N,\infty}^1)}$, we eventually get

$$\forall t \in [0, \bar{T}], \quad X(t) \leq C \int_0^t (1+V'(\tau))X(\tau) \log\left(e + \frac{V(T)}{X(\tau)}\right) d\tau.$$

As

$$V' \in L^1(0, T) \quad \text{and} \quad \int_0^1 \frac{dr}{r \log\left(e + \frac{V(T)}{r}\right)} = +\infty,$$

Osgood lemma (see e.g. lemma 3.1 in [13]) entails that $X \equiv 0$ on $[0, \bar{T}]$. This means that (a^1, u^1) and (a^2, u^2) coincide on $[0, \bar{T}]$. Then appealing to the usual connectivity argument completes the proof. $\blacksquare$

Remark. Let us stress the fact that in the statement of proposition 4.2.3, the space E_T^p may be replaced by its counterpart with $L^p(0, T; \dot{B}_{p,r}^s)$ spaces instead of $\tilde{L}_T^p(\dot{B}_{p,r}^s)$. This is due to the fact that theorems 2.2.3 and 3.3.1 provide missing tildes for free.

The critical case is the only case where keeping a tilde in condition (4.23) is fundamental. Note however that this assumption involves only the size of a^1 , not the regularity. We do not know how to prove uniqueness in theorem 4.2.1 in the case $N = 2$ or $p = N$ in a functional framework with no $\tilde{L}_T^p(\dot{B}_{p,r}^s)$ spaces.

4.3 Further results

Let us first point out that as a by-product of our proof of existence and of inequality (2.5), we get the following lower bound for the existence time:

$$T \geq \sup\{t > 0, \quad f_1(t) \leq \eta \quad \text{and} \quad f_2(t) \leq \eta\}$$

with

$$\begin{aligned} f_1(t) &:= \sum_q 2^{q(\frac{N}{p}-1)} (\|\dot{\Delta}_q u_0\|_{L^p} + \|\dot{\Delta}_q f\|_{L_t^1(L^p)}) \left(\frac{1-e^{-\kappa\mu t 2^{2q}}}{\kappa\mu 2^{2q}} \right), \\ f_2(t) &:= \sum_q 2^{q(\frac{N}{p}-1)} (\|\dot{\Delta}_q u_0\|_{L^p} + \|\dot{\Delta}_q f\|_{L_t^1(L^p)}) \left(\frac{1-e^{-2\kappa\mu t 2^{2q}}}{2\kappa\mu 2^{2q}} \right)^{\frac{1}{2}} \end{aligned}$$

and η such that

$$2C\eta \leq \mu \quad \text{and} \quad 16(e^{\frac{C\eta}{\mu}} - 1) \leq \|a_0\|_{\dot{B}_{p,1}^{\frac{N}{p}}}.$$

Above C is the constant appearing in the proof of existence.

This lower bound is rather inexplicit unless u_0 and f have more regularity.

Let us finally mention that theorem 4.2.1 may be extended or improved in many ways:

- One can show a small data global existence and uniqueness result in critical spaces under the condition that

$$\|\rho_0 - \bar{\rho}\|_{\dot{B}_{2,1}^{\frac{N}{2}} \cap \dot{B}_{2,1}^{\frac{N}{2}-1}} + \|u_0\|_{\dot{B}_{2,1}^{\frac{N}{2}-1}} + \|f\|_{L^1(\dot{B}_{2,1}^{\frac{N}{2}})} \ll 1$$

for some constant positive density $\bar{\rho}$ such that $P'(\bar{\rho}) > 0$ (see [17] for more details).

- The condition that the density is a small perturbation of a constant state may be relaxed provided some extra regularity is available (see [20]). It is however fundamental for the initial density to be bounded away from 0.

In a recent paper however, it has been stated that, for suitably smooth data, existence *and* uniqueness results may be proved in some cases where vacuum occurs (see [15]).

- Similar results may be proved for *polytropic* heat conducting compressible fluids (see [20] and [23]).
- After a suitable rescaling, one can state results of convergence for the barotropic Navier-Stokes equations to the incompressible Navier-Stokes equations when the Mach number ε tends to 0 (see [22] and [21]).

Finally, we want to stress that combining a priori estimates in Besov spaces for the heat equation and for the transport equation may give local well-posedness results for many other evolution PDE's coming from fluid mechanics or other fields of physics. One can mention for example that this method leads to such results for the system of density-dependent incompressible Navier-Stokes equations (see [19]) and for visco-elastic fluids (see [14]).

4.4 Exercises

Exercise 4.1. Prove lemma 4.1.1 for smooth u , b , and Ω .

Exercise 4.2. Let X_T^p and Y_T^p be defined according to page 76. Prove that both X_T^p and Y_T^p are Banach spaces, that X_T^p is separable and that Y_T^p is the dual of X_T^p .

Exercise 4.3. Prove the passage to the limit from system (4.20) to system (4.21).

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List of symbols

$[r]$, 12

$\approx$, 7

$\lesssim$, 7

$\mathcal{B}(0, R)$, 7

$\overline{\mathcal{B}}(0, R)$, 7

$B_{p,r}^s$, 13

$\dot{B}_{p,r}^s$, 31

$\mathcal{C}(0, R_1, R_2)$, 7

$\mathcal{C}_0$, 16

$\mathcal{C}_b^\infty$, 59

$\mathcal{C}_c^\infty$, 15

C^r , 12

Δ_q , 9

$\dot{\Delta}_q$, 29

Δ_q^{per} , 10

$\mathcal{F}$, 7

$F_{p,r}^s$, 15

H^s , 11

Q_a^N , 10

$R(u, v)$, 21

$\dot{R}(u, v)$, 35

$\mathcal{S}$, 7

S^m , 20

$\mathcal{S}'$, 7

$\mathcal{S}'_h$, 29

S_q , 9

$\dot{S}_q$, 29

S_q^{per} , 10

$\mathbb{T}_a^N$, 10

$T_u v$, 21

$\dot{T}_u v$, 35

$\tilde{\mathbb{Z}}_a^N$, 10

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